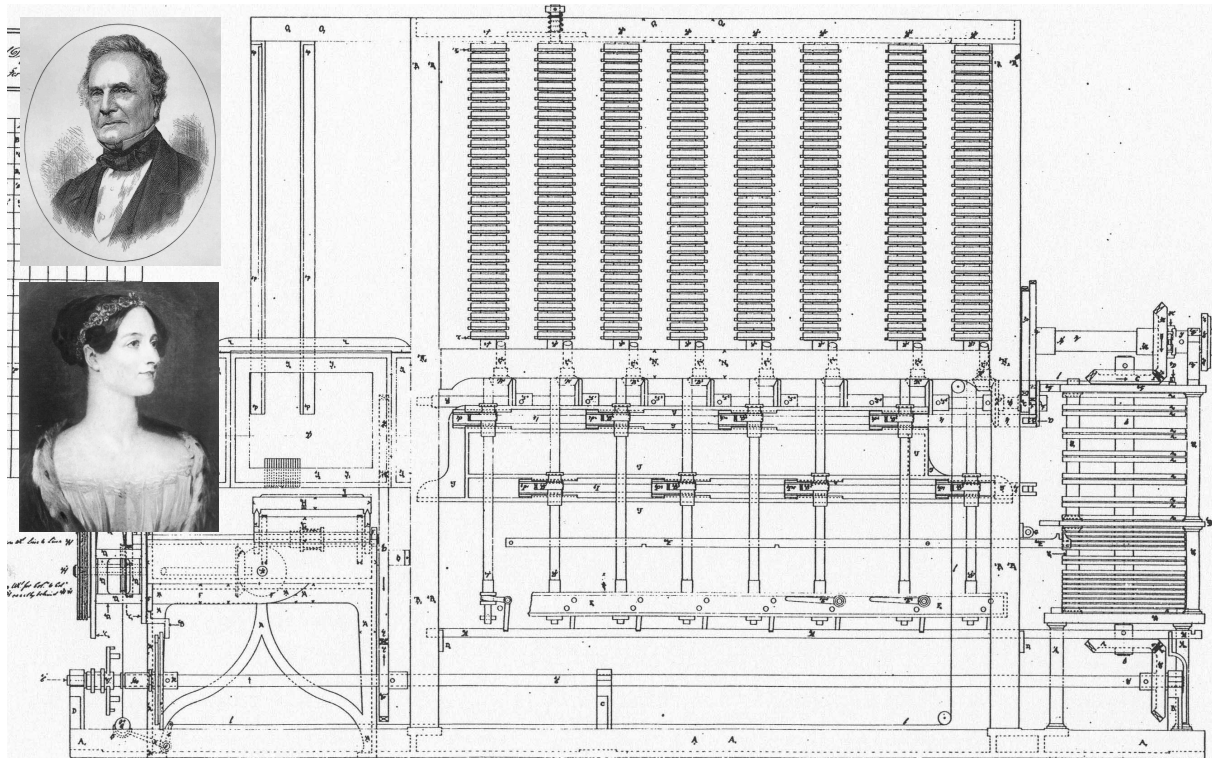


Charles Babbage, Ada Lovelace, and the Dawn of Computing

Instructor's Materials

Version 3.1



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Mark M. Meysenburg

Mark M. Meysenburg, 2016

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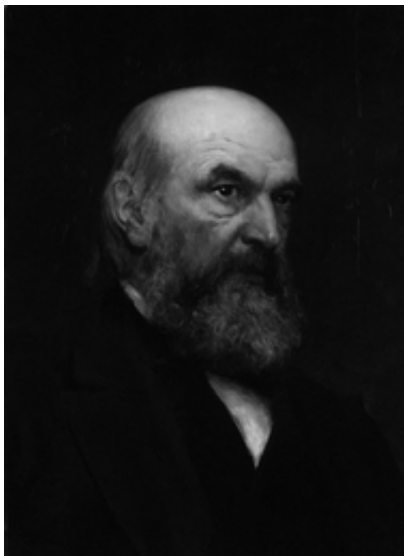
Chapter 1

Individual role sheets

John Couch Adams

Astronomer, co-discoverer of Neptune

Pro-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are John Couch (pronounced “cooch”) Adams, co-discoverer in 1846 of the planet Neptune. As of 1839, you are a Fellow at St. John’s College, Cambridge.

You were born in 1819 on a farm near Launceston, Cornwall, in the far southwest of England. Your mother’s uncle (and your namesake), John Couch, left his library to her. Through this library, you were exposed to your life’s passion, astronomy. In your primary schooling you learned the classics and excelled in Greek, but you had to teach yourself mathematics. You were apparently successful in this venture, for you were able to perform the calculations to predict an annular eclipse (where the moon blocks out the sun, but not completely, so that a bright ring appears around the moon) in 1836. You then traveled to the predicted location and observed the occurrence.

Thanks to a small inheritance to you mother, and to your work tutoring students, you were able to afford to attend St. John’s College, Cambridge, starting in 1839. Four years later you did so well on your final exams that you graduated as Senior Wrangler, the top mathematics undergraduate at Cambridge. In 1843 you became a Fellow at St. John’s

(meaning you obtained a paid research position there – similar to a “post-doc” position in modern American academia).

Your most important discovery for the purposes of this game began in 1841, when you decided to investigate

... the irregularities of the motion of Uranus ... in order to find out whether they may be attributed to the action of an undiscovered planet beyond it. [OR05]

In 1845, through quite a lot of complex, arduous mathematics, you were able to come up with a prediction for where the new planet Neptune would be found if someone looked for it with a telescope. You gave this information to the directory of the Cambridge University Observatory, James Challis, and he, in turn, provided you with a letter of introduction to George Airy, the Astronomer Royal. You went to see Airy at Greenwich twice, but as you had not made an appointment, you were unable to see him. Failing to secure an in-person meeting, you left your data at Greenwich for Airy.

Airy apparently read your letter, but he had a question for you regarding your data, and sent the question to you in another letter. Airy wondered “...whether the errors of the radius vector would be explained by the same theory that explained the errors of longitude.” [LB86]. You felt that the question was about a trivial matter, and so you never replied. You assumed that Challis and / or Airy would have notified other astronomers in the United Kingdom about your predictions, so that someone would train their telescope to your prediction location and see if the new planet existed. However, neither Challis nor Airy told anyone, and therefore no astronomers in the UK were looking for the new planet.

In June 1846, the French Astronomer Urbain Le Verrier published a prediction of the planets location, and in September 1846 a German astronomer at the Berlin Observatory was the first to view Neptune.

In this game you are young – not yet out of undergraduate school until game session (GS) 6 – and emblematic of a new breed of “men of science.” Rather than being a landed gentleman with enough time, talent, and money to pursue science as a hobby, you represent a new generation of professional “scientists.” You are mild-mannered and deferential, almost to a fault, and so you must take care to exhibit these traits during your GS 6 debate with Airy regarding the priority of the discovery of the planet Neptune.

Objectives

You have two main objectives in this game. First, you want your pro-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government decides, both opportunities, to fund the development of Babbage’s engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Secondly, you want to convince the world that you should be considered a co-discoverer of Neptune, if not the sole discoverer. In concert with this objective, you should convince other players in the game that your calculations would have been completed even sooner if you had had access to one of Babbage's computing engines – i.e., argue for your faction even when you are talking about the priority of Neptune. Although you are not a fellow of the Royal Society, other members of your faction are. If you can convince a FRS to introduce a resolution to the Royal stating that you should be considered co-discoverer of Neptune, and if the Royal passes the resolution, you win the game, regardless of the performance of the pro-Babbage faction.

Secret: You should not let your faction colleagues know that you win if the resolution is introduced and passed.

Responsibilities

As John Couch Adams, you have responsibilities to your pro-Babbage faction, to defend your priority in the discovery of Neptune, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibilities to the pro-Babbage faction

As a member of the pro-Babbage faction, you should do whatever you can to help the faction win the game; this is also obviously in your best interest, since one of the ways for you to win the game is for your faction to triumph. Early in the game, you can help your faction by being a vocal proponent of Babbage, in the settings where this is appropriate (i.e., not during Royal Society meetings, since you are not a FRS). You can ask good questions or make brief podium speeches to help propel your faction to victory.

The primary way you can help your faction is to make an excellent speech during GS 6, convincing other players that you are a rightful discoverer (or at least co-discoverer) of Neptune, and that you would have discovered the new planet even earlier if one of Babbage's machines had been available to speed up the calculations.

Regarding the priority of Neptune and the usefulness of a calculating engine

During GS 6, you will make a speech asserting that you were the first to discover Neptune, through the calculations you delivered to Airy. Your calculations were completed before

those of Le Verrier. More importantly to one of the main ideas of this game, you should also make the case that, had one of Babbage's Analytical Engines been available, you would have been able to make the calculations even more quickly, thus ensuring that Neptune would be discovered by Britain and not by France. This argument will illustrate how Babbage's invention could be practical and also could bring scientific glory to Great Britain.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are not a FRS, you should plan on making a speech supporting your faction's position during one of the sessions that take place at the Mechanic's Institute, GS 2 or 5. Coordinate with your faction leader (Herschel) and the other members of your faction so you can make a speech with the maximum impact for your faction. You have been invited by the Royal Society to speak during GS 6, and so the prohibition from speaking during a RS session is relaxed for you during that game session.

Powers

Since you are not a member of the government, the nobility, or the Royal Society, your formal powers in the game are very limited. This does not mean you are not a valuable member of your faction, or that you cannot win the game. You can earn PCs for yourself and / or your faction, by making an excellent podium speech, by making an effective case regarding the discovery of the planet Neptune, and asking insightful questions of the opposing faction.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 6 speech, due before GS 6 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As a member of the men of science community, you have important relationships with the main ideas, texts, and other characters in the game.

With ideas

You are a very capable mathematician, and so you are able to easily grasp the concepts behind Babbage's computing engines, and you also can see their importance. Of course, the amount your calculations regarding Neptune would have been accelerated with one of the machines is an excellent example. Therefore, you support the idea of government funding for both of Babbage's engines.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally, like yourself). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You will not be able to vote in this election, but you should support the candidate who you think is most likely to favor those in the new generation of men of science, regardless of which faction the candidate belongs to.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

Since you are young in this game, you have not had much of an opportunity to become intimately involved with the core texts, nor have you published yet yourself.

With other roles

Although you have a mild and deferential manner, and have much respect for Airy (and even Le Verrier), you and Airy disagree about the priority of Neptune and about whether Babbage's Analytical Engine would have made any difference in the discovery of the planet.

Mary Somerville's 1842 publication, *The Connection of the Physical Sciences*, was influential on you. In this work, Somerville noted that the planet Uranus was orbiting differently than predicted, and she speculated that there might be another planet, further beyond Uranus, whose gravitational pull was causing the odd behavior. This observation helped inspire you to begin your calculations to predict where Neptune would be found. Thus, you have a natural ally during GS 6 in Somerville. However, she will not be able to speak during GS 6, and so you should work with Somerville, encouraging her to point out the irregularities in Uranus' orbit during GS 5, setting the stage for your GS 6 speech.

You also do not share much common ground – other than faction affiliation, where appropriate – with the older members of the MOS community. You respect their work, but it is hard for you to relate to them personally; your up-from-the-bootstraps background bears little resemblance to the life of leisure of the landed gentry MOS members. This puts you in conflict with Babbage, Crosse, Gilbert, Herschel, Lovelace, Sabine, and Somerville. On the other hand, you share a professional approach to science with Airy, Brunel, Buckland, Clement, Faraday, and Lardner.

With your faction

Although you are not a politician or a FRS, you are a valuable member of your faction. You have something the faction needs: a concrete example, showing harm to Britain's reputation arising from the lack of one of Babbage's computing engines. One of the engines would have sped up your calculations, moving forward the timetable for the discovery of Neptune, and therefore securing scientific glory for the nation. As it stands, the glory of the discovery of Neptune went to France instead. This is not a theoretical harm, not a ship that might have crashed due to faulty navigational tables, not a minor inconvenience caused by a few mistakes in published logarithm tables. You are the only person in this time frame that can

show a concrete, detrimental impact due to the lack of a working Difference or Analytical engine.

Strategy advice

While you have an excellent case to make for building Babbage's engines, make sure you do not talk about the discovery of Neptune until the proper year – that is, not until GS 6. Before that, you should be an active, but polite, questioner of the anti-Babbage faction, and make a podium speech or two in support of your faction.

When the time to discuss Neptune comes, you should emphasize the harm to Britain's scientific reputation due to France being first to view Neptune. Communicate how useful one of the engines would have been in your quest for the planet, making clear that Britain would have gotten there first had your calculations been sped up. Of course, you should also argue that you deserve discoverer (or at least co-discoverer) credit, since your calculations were started so early. Also, try to get one of your FRS faction members to introduce a resolution to the Royal Society in GS 6, stating this. If the Royal passes this resolution, you win the game regardless of whether your faction wins.

Secret: You should not let your faction colleagues know that you win if the resolution is introduced and passed.

To learn more

[Bal90] Sir Robert Stawell Ball. *The Story of the Heavens*. Cassell & Company, Ltd., London, 1890. Available on Google Books

[LB86] John E. Littlewood and Béla Bollobás. *Littlewood's Miscellany*. Cambridge University Press, 1986

[OR05] J. J. O'Connor and E. F. Robertson. Adams biography. <http://www-groups.dcs.st-and.ac.uk/~history/Biographies/Adams.html>, February 2005

Summary of your individual victory objectives

You win the game if either of the following takes place:

- Your pro-Babbage faction wins the game, by securing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction. Or,
- **Secret:** A resolution is passed by the Royal Society naming you as the rightful co-discoverer or discoverer of Neptune.

Portrait of John Couch Adams FRS, by Sir Hubert von Herkomer, from the National Portrait Gallery, St Martin's Place, London; <http://www.npg.org.uk/collections/search/person/mp00026/john-couch-adams?search=sas&sText=john+couch+adams>

Charles Babbage FRS

British polymath

Pro-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Charles Babbage, British polymath (a person whose expertise spans a significant number of different subject areas) and inventor of the Difference and the Analytical Engines. You are also a member of the Men of Science (MOS) community in the game. You are a man of science, or natural philosopher, of the old kind – a man whose wealth and social position have allowed you to pursue your mathematical, engineering, economic, and scientific interests. You have also been the leading proponent in England of applying mathematics and science to industry and commerce.

You are a member of the MOS community and of the pro-Babbage faction, although you are not the leader of the faction. In fact, your job in the faction could be considered sabotage; you are responsible for taking certain actions that will work against the best interests of your faction.

Secret: Do not tell the members of your faction – or anyone else, for that matter – of your counter-productive responsibilities.

You are “the child of two revolutions: industrial revolution in Britain, and political and social revolution in France.” [Hym85] You were born on December 26th, 1791, in Walworth,

Surrey, England. In the present day, Walworth is a part of greater London. Your father was a banker named Benjamin Babbage, and your mother was Betsy Plumleigh Teape. Although he was from a wealthy family, your father was proud that he had made most of his fortune on his own.

You were educated at a variety of private boarding schools, showing an early interest and immense talent in mathematics and the creation of mechanical devices. You attended Trinity College, Cambridge starting in October 1810.

You were hopeful of learning more advanced mathematics at Cambridge, but you were disappointed, partially because of the state of mathematics at the university and partially because of your own stubbornness. The professors recommended that you should study the mathematics that would be on the standard graduation examination and the Tripos exam, the examination taken by select students to determine who would graduate with honors. The mathematics questions on these exams had not been updated since 1747; in a sense, Cambridge was still resting on the laurels of Newton, while the world (particularly continental Europe) had moved forward. British mathematicians continued to use Newton's notation in calculus problems, while everyone else had adopted Leibniz's style, the notation familiar to calculus students today. In addition, Newtonian calculus approached problems visually, with graphs, lines, and charts, while the newer analytical calculus instead relied on the manipulation of symbolic equations to solve problems.

On your way to Cambridge, you purchased a French calculus textbook at considerable expense from a French bookshop, and you wanted to learn more about the analytical mathematics you read about in the book. At least one Cambridge scholar, Robert Woodhouse, had written extensively on this subject. But three professors – possibly including Woodhouse himself – recommended that you should study for the graduation exams rather than wasting time on more advanced topics. You did not take this advice.

Thankfully, you had access to mathematical works from the continent via the Cambridge library, and so were able to further your knowledge of analytical mathematics in that way.

At Cambridge, you, John Herschel, and others formed the Analytical Society. The society was dedicated to improving the state of mathematics and Cambridge, and then in the whole of Britain. Your primary goal – which the Analyticals succeeded in achieving – was to switch Britain from Newton's calculus notation to Leibniz's notation, and to promote the use of analytical mathematics in the U.K. Eventually the Society published a compilation of papers on analytical mathematics, written by yourself and Herschel, entitled *Memoirs of the Analytical Society*. Your mathematical approach to science, industry, and all things, was cemented during your time with the Analyticals.

You graduated from Cambridge in 1814, but just barely. You had not studied for the standard graduation exam, much less the Tripos honors examination. During your graduation exam, you were unable to answer question after question regarding Newtonian mathematics. Eventually, one of the examiners (perhaps Woodhouse) asked you a question from one of your own papers in the *Memoirs of the Analytical Society*, and with one correct mathematics answer, you passed the exam and earned your degree.

Thanks to your work on modernizing the state of mathematics in Britain, among other factors, you were elected as a Fellow of the Royal Society in 1816. Between 1815 and 1820 you performed a large volume of important work in pure mathematics, mostly in the theory of functions.

However, during this early post-Cambridge period you had a difficult time finding paid employment, for a variety of reasons. You applied for mathematical professorships first at Haileybury, the East India College, and then at the University of Edinburgh in Scotland. Although technically qualified, you did not receive either position, most likely because of your family's nonconformist past and your liberal political leanings. In 1824 you were invited by some acquaintances to partner with them in the formation of a new life insurance company, called the Protector. You produced some very good actuarial tables, but unfortunately the company never came to fruition. You were worried about the financial security of your family – having married Georgina Whitmore in 1814, following a secret engagement – until the death of your father in 1827. The inheritance you received from your father, approximately £100,000, allowed you to live quite comfortably, and also to pursue your mathematical and scientific interests. Unfortunately, later in 1827 your wife and your youngest son also died, leaving you devastated.

In 1828, while you were touring Europe as a way to take your mind off your losses, you were named as the Lucasian Professor of Mathematics at Cambridge. The Lucasian Chair is perhaps the most prestigious academic position that one can hold anywhere in the world; Isaac Newton held the chair for 33 years in the seventeenth century, and Stephen Hawking held it for 30 years in the twentieth. You replaced Sir George Biddell Airy, who had bested you for the chair just two years prior in 1826.

During the time span covered by this game, you and your close friends are politically liberal and perhaps even slightly radical. You twice ran for Parliament, first in 1832. You ran as a Whig, a candidate of reason and reform. You were not elected in either campaign, however.

Aside from your intellect, one of your greatest gifts is the ease with which you are able to mix, mingle, and converse with people from every class in society. You have “immense personal charm and boundless vitality,” and are “equally at ease with intelligent working men, country clergymen, men of science, at court, or at the dining tables of the aristocracy.” [Hym85] The parties you hold at your house at 1 Dorset Street, London, are legendary. John Herschel, William Whewell, Florence Nightingale, Lord Tennyson, Michael Faraday, Charles Darwin, Charles Dickens, Ada Lovelace, and the Duke of Wellington are just some of the luminaries that have attended your soirées. An invitation to one of your Saturday evening parties is one of the most coveted social achievements in London society. In fact, this game begins with one of your parties in GS 1.

You have done productive work in many different areas, but the consuming passion of your life has been your Difference, and later your Analytical, engines. The idea of creating a Difference Engine to mechanize the production of numerical tables first came to you in 1821 or 1822. The idea excited you so much that you became physically ill. In 1823 you received £1,700 from the British government to begin the construction of your first

Difference Engine. A fragment from this project, constructed by Joseph Clement, is the only tangible outcome of the Difference Engine project to date as our game begins in 1828. The Difference Engine fragment sat in your drawing room at 1 Dorset Street for many years; it was a popular attraction among the learned guests at your parties.

The construction of these machines is very, very expensive; certainly beyond your means. This, plus the benefits the nation would reap if they were constructed, have led you and those in your faction to seek out government funding for the project.

Even though you are able to converse with anyone, you also have a reputation among the artistically-inclined as something of a wet blanket. Consider this excerpt of a letter you wrote in 1851 to Lord Tennyson regarding one of his poems:

Sir: In your otherwise beautiful poem “The Vision of Sin” there is a verse which reads – “Every moment dies a man, Every moment one is born.” It must be manifest that if this were true, the population of the world would be at a standstill . . . I would suggest that in the next edition of your poem you have it read – “Every moment dies a man, Every moment $1\frac{1}{16}$ is born.” . . . The actual figure is so long I cannot get it onto a line, but I believe the figure $1\frac{1}{16}$ will be sufficiently accurate for poetry. I am, Sir, yours, etc., Charles Babbage.

Objectives

Your objectives in this game are somewhat complicated. You want your pro-Babbage faction to win the game, of course; if the faction wins, you win. Your faction wins if the British government decides, on both of the opportunities, to fund the development of one of your Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Secret: However, the actions you are required to take during the game could certainly be considered sabotage against your own faction, making it more difficult for your faction (and therefore yourself) to win the game. During GS 4, you will make a speech that is strongly critical of the Royal Society, and the content and tenor of this speech should come as a surprise to everyone, especially those in your faction. Your GS 4 speech reflects the 1830 publication of your work *Reflections on the Decline of Science in England, and on Some of its Causes* [Bab30], which was stridently critical of the political crony-ism in the Royal (amongst other complaints).

You obviously cannot tell anyone in the game of your actions beforehand, nor discuss your motivations for them at any time before the game post-mortem. You must come off as a gregarious, friendly person, an ardent supporter of your Engines, and as a person of great integrity.

Responsibilities

As Charles Babbage, you have responsibilities to preside over your party during GS 1, responsibilities to your pro-Babbage faction, a duty to point out the flaws you perceive in the Royal Society, the responsibility to nominate John Herschel for the next president of the Royal Society, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Conducting GS1, your Dorset Street party

You host a party at your Dorset Street residence to begin the game. As host, you are responsible for making sure that the activities and speeches scheduled for GS 1 all happen. The activities for GS 1, and their order, are:

- Speech by The Fury. The Fury will lecture all of the characters in the game regarding the proper etiquette and manner of behavior that should be observed throughout the game.
- Ice breaker. Players will spend a few minutes playing a Georgian-era party game to begin the game. The GM will provide a handout describing the game, how it should be played, and how players can win PCs in the party game.
- PM speech by the current PM. He will outline the priorities and goals his government will have while he is in office.
- Frend and Somerville will make a joint presentation, teaching those in attendance how to use a slide rule and mathematical tables to perform calculations.
- William King will make a presentation on the supremacy of reason over imagination.
- John Herschel will make a speech introducing the Difference Engine fragment. Herschel will describe its purpose, history, and so on.
- At the end of the session the GM will conduct the persuasiveness metric poll of the indeterminates.

During any non-scheduled time in the session, the *podium rule* is in effect. The podium rule states that any character wishing to speak should approach the podium and wait to be recognized by the presiding officer. You are the presiding officer for GS 1. You must make sure that you are fair to anyone who wishes to speak, even those that are not in your faction. Characters are encouraged to ask questions of presenters, at the discretion of the presiding officer, but they should not make speeches to the class without approaching the podium.

Speak with the GM before GS 1 about the party, so that you are well prepared for the conduct of the session.

Responsibilities to the Pro-Babbage Faction

As a member of the pro-Babbage faction, other than the “sabotage” element (described in the Objectives section above, and also below), you should do whatever you can to help the faction win the game. The main way you can help your faction is to work hard to understand how the Difference and Analytical Engines work, and especially work hard to be able to explain why constructing either machine would be beneficial. You should constrain your vision to the benefits of performing mathematical calculations more quickly and with more accuracy.

Regarding the flaws in the Royal Society

Secret: During GS 4, you will make a presentation to the Royal Society summarizing your criticisms of the Society you published in your 1830 in *Reflections on the Decline of Science in England, and on Some of its Causes* [Bab30]. You must be harshly critical of the current status of the society during this presentation. You should study this document to understand what your criticisms of the Royal should be.

Also, this presentation must be a surprise to everyone in the game, including those in your faction. *Accordingly, do not confer with anyone, other than the game-master, regarding your presentation before you make it during GS 4.*

Nominating John Herschel for Royal Society President

During GS 4, each faction will nominate a candidate to serve as Royal Society President after Davies Gilbert. You must nominate your faction’s candidate, John Herschel. Prepare a short nomination speech, in which you should try to make a convincing argument as to why Herschel, a dignified man of science, should lead the Royal. Contrast (in a positive manner, if possible) Herschel with the other faction’s candidate, the Duke of Sussex.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound

plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are a FRS, you should plan on making a speech supporting your faction’s position during one of the sessions that take place at the Royal Society, GS 3, 4, or 6. Coordinate with your faction leader (Herschel) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

Since you are not a member of the government, nor a member of the nobility, you have no formal power to make decisions regarding your machines – or anything else, for that matter. But, because of your connections and socializing with nobility and Whig politicians, you may be able to exert some informal influence on them.

You are an influential fellow of the Royal Society (FRS), and if the Royal is asked to take up the matter of your machines, you can help steer the society to recommend that the government provide more funding for development and construction. Of course, after your presentation in GS 4, you and your faction’s influence within the Royal Society may be diminished.

Your greatest power in the game – other than your secret, pseudo-sabotage responsibility – is to be an expert on the Difference and Analytical engines, and their value in terms of the way they could speed up the calculation of complex mathematical equations.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for your engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 4 speech, due before GS 4 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide

more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in British science during the early to mid-1800s, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding the idea of the government funding large-scale scientific and engineering projects, you are clearly in favor of the practice. You believe from the core of your being that your Difference and Analytical Engines will be useful for scientists and engineers in the Kingdom. At the same time, you know that their construction is well beyond your ability to pay; there is no way for you to privately fund the project. You believe that it is in the best interest of the British government to fund the development of the engines.

You are a member of the last generation of the old-school men of science community, i.e., a wealthy gentleman with enough time, talent, and money to pursue your scientific interests at your leisure. Although you do not have anything against him personally, this puts you somewhat at odds with younger men of science like John Couch Adams. Men like Adams do not have large sums of money to pursue scientific studies as a hobby, rather, they do science for a living – science is their profession as well as their passion. If you are among the last of the wealthy, amateur scientific gentlemen, Adams and his colleagues are among the first of the professional scientists (although he is not yet called a “scientist.”)

Fellow men of science of the old school are Crosse, Gilbert, Herschel, Lovelace, Sabine, and Somerville. Those representative of the new, professional science are Adams, Airy, Brunel, Buckland, Clement, Faraday, and Lardner. Other men of science of your generation might look down upon the new, professional scientists, begrudging them their social status. You, however, do not; you have a reputation as a man who can carry on a conversation with anyone from nobility to the educated working man.

As for the conflict between imagination and reason, you are firmly aligned with William

King on the supremacy of reason, and opposed to Byron's Ghost, the champion of imagination.

With texts

As a true polymath, you have studied widely and published in many different fields. Within the time period of this game, you have published on the actuarial tables used by life insurance companies (*A Comparative View of the Various Institutions for the Assurance of Lives*, 1826), on the problems in the Royal Society (*Reflections on the Decline of Science in England, and on Some of Its Causes*, 1830), on operations research (*On the Economy of Machinery and Manufactures*, 1835), on religion (*The Ninth Bridgewater Treatise*, 1835), and on mathematics (*Table of the Logarithms of the Natural Numbers from 1 to 108000*, 1841).

Interestingly, what you have not published on is your Difference or Analytical Engines. The only detailed work describing how one of your engines operates to be published contemporaneously is Menabrea's article in *Scientific Memoirs*, *Sketch of the Analytical Engine invented by Charles Babbage ... with notes by the translator* (1843), translated into English and annotated with voluminous notes by your dear friend, Ada Lovelace. The lack of published work about your Engines may make it more difficult for others in the MOS community to fully understand the importance of your ideas.

With other roles

You met John Herschel when you were both students at Cambridge. You, Herschel, and several others formed the Analytical Society there, and have been fast friends ever since.

You are also good friends with I. K. Brunel. An example of your friendship happened in 1838. You devised an ingenious method to test the smoothness of ride on Brunel's wide gauge Great Western Railway. At the time, the nascent railway system in Britain used tracks of two different widths, or gauges, making it difficult for people or goods to move across the kingdom via rail. The controversy was called the "war of the gauges." Babbage outfitted a special railway car, called a dynamometer car, filled with instrumentation. The car measured the speed of the car, the frequency and intensity of the car's shaking, and so on. (Think of a dynamometer car as the equivalent of a modern jetliner's black box.) Data from your dynamometer car helped convince the government to make Brunel's seven-foot railway gauge the standard in the southwest of England. Airy disagreed with you and Brunel, throwing his support to the narrow gauge option.

Dionysius Lardner has been a vocal supporter of your Difference Engine, publishing a review of the Engine in the *Edinburgh Review* in 1834. Lardner made a case that the error-free tables produced by the Difference Engine would prevent navigational errors, and thus the loss of ships at sea.

You have a special relationship with Augusta Ada King-Noel, Countess of Lovelace (a.k.a. Ada Lovelace). You first were introduced to Ada by Mary Somerville at one of your Dorset

Street parties. Ada was young and beautiful, but also she was intelligent and mathematically gifted. She understood the operating concept and purpose of your Difference Engine almost immediately after seeing the Engine fragment at your party. She gained more information about the Difference Engine from one of Dionysius Lardner's lectures. You arranged for Ada to take more advanced mathematical tutoring from Augustus De Morgan of the University of London. Later, she became a trusted collaborator on your Analytical Engine project. She translated one of the few published papers on the Analytical Engine from Italian into English, and added her own notes to the article in the process. Her voluminous notes – three times the length of the original article – showed that she had a vision of what the machine truly do, beyond mere calculation of mathematical formulae. You consider Ada to be the “high priestess of the Analytical Engine,” and your “enchanted math fairy.” It is also true that you value Ada's celebrity status as the only legitimate child of Lord Byron. If her fame helps you to achieve your goals regarding the calculating Engines, so much the better.

You do not have a good relationship with George Biddell Airy, who preceded you as the Lucasian Chair of Mathematics at Cambridge. Perhaps your relationship can be summed up in Airy's one-word opinion of your calculating engines: “worthless.”

You have been harshly critical of Edward Sabine in *Reflections on the Decline of Science in England, and on Some of Its Causes*, accusing him of falsifying data in his work on determining longitude.

You share political sympathies with two of the indeterminates, the Whig MPs Grey and Russell. You are politically opposed to the Tories, and thus are generally against the policies of the Tory indeterminates, Peel, Robinson, and Wellesley.

With your faction

The special – and secret – relationship with your faction is outlined in the Objectives and Responsibilities sections above. Remember that you are an important member of the faction, nonetheless. You are one of the few people in the game that really, truly understands the importance of your Difference and Analytical engines. As such, you are important to the faction as someone who can persuade the indeterminates that your engines are important and should be funded by the government.

Strategy advice

You have the most difficult role in this game. As a player acting out the role of Charles Babbage in the game, you know that the actions you are required to take in the game (harshly criticizing the Royal Society) are counter-productive to your faction. Of course, Babbage himself did not see his actions that way; he was simply stating what he saw as the correct take on the Royal Society.

Accordingly, even though you know as an actor that your actions are self-defeating in

terms of the game, you cannot allow that knowledge to seep through in to your portrayal of Babbage in the game. You must perform your duties in the game and be an enthusiastic advocate for your Difference and Analytical Engines even though you may be fighting a losing battle.

To learn more

[Bab30] Charles Babbage. *Reflections on the Decline of Science in England, and on Some of its Causes*. B. Fellowes and J. Booth, London, 1830. Available via Google Books

[Hym85] Anthony Hyman. *Charles Babbage, Pioneer of the Computer*. Princeton University Press, Princeton, NJ, 1985

Summary of your individual victory objectives

You win the game if your faction wins. Your faction wins if the British government decides to fund your Difference or Analytical Engines on both funding decision opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Isambard Kingdom (I.K.) Brunel FRS

British engineer

Pro-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Isambard Kingdom (I. K.) Brunel. Although small in stature, you are a giant figure in British engineering and industrialization. You are an accomplished mechanical and civil engineer in the U.K. – perhaps the most accomplished. Your work includes the Thames Tunnel, the Great Western Railway, and the S. S. Great Western, one of the first two steamships to successfully cross the Atlantic Ocean from the U.K. to New York.

You were born on April 9th, 1806, in Portsmouth, on the south coast of England. Your father, Marc Isambard Brunel, was another incredibly accomplished engineer and also a friend of Charles Babbage. Your mother, Sophia Kingdom, was the daughter of a Royal Navy contractor; she met your father while studying French in Rouen, France. Your father recommended you for entry to the French École Polytechnique engineering school – for no such schools yet existed in Marc Brunel’s adopted Britain – in 1822, but as a “foreigner” you were not accepted. Instead you apprenticed with Abraham-Louis Breguet in France and then returned to England.

Since your return, you have had a profound impact on the state of engineering, industrialization, and public transportation in Britain. You helped your father complete the Thames Tunnel project, in spite of almost dying during a collapse and subsequent flood. You were instrumental in the creation of the Great Western Railway, the first railway linking London to the southwest and west of England and most of Wales. You also built the S. S. Great Western, the second steamship to cross the Atlantic (and the first to do so with a comfortable reserve of coal in her bunkers), and the S. S. Great Eastern, an iron-hulled steamship that utilized a propeller in addition to her paddle-wheels. These accomplishments are joined by other, perhaps more mundane, designs and constructions, including the Clifton Suspension Bridge. You are elected as a Fellow of the Royal Society (FRS) in 1830.

Objectives

You have one main objective in the game. You want your pro-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government decides, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As I. K. Brunel, you have a responsibility to preside over the lectures at the Mechanic's Institute during GS 5, responsibilities to your pro-Babbage faction, a responsibility to show that government funding of engineering is worthwhile, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to preside over GS 5

You will preside over the second set of lectures at the Mechanic's Institute, which take place during GS 5. As the presiding character, you are responsible for making sure that the activities and speeches scheduled for GS 5 all happen. The activities for GS 5, and their order, are:

- Social time. This is a brief period of time for general communication between characters, and for the factions to confer and strategize.

- PM speech. The newly-elected PM, will outline the priorities and goals his government will have while he is in office.
- Joseph Clement will speak regarding his work with Babbage on the Difference Engine
- Your speech regarding the Thames Tunnel and the value of government spending for large scientific and engineering projects. See below for more details regarding your speech.
- Countess Lovelace will speak on the analytical engine.
- Crosse will speak on the concept of interchangeable parts in manufacturing, and how the concept may be applied to Babbage's machines.
- At the end of the session the gamemaster will conduct the persuasiveness metric poll of the indeterminates.

During any non-scheduled time in the session, the *podium rule* is in effect. The podium rule states that any character wishing to speak should approach the podium and wait to be recognized by the presiding officer. You are the presiding officer for GS 5. You must make sure that you are fair to anyone who wishes to speak, even those that are not in your faction. Characters are encouraged to ask questions of presenters, at the discretion of the presiding officer, but they should not make speeches to the class without approaching the podium.

Speak with the GM before GS 5 about the lectures at the Mechanic's Institute, so that you are well prepared for the conduct of the session.

Responsibilities to the Pro-Babbage Faction

As a member of the pro-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. Early in the game, you can help your faction by being a vocal proponent for Babbage and his calculating engines. You can ask good questions and make short podium speeches to help propel your faction to victory.

The second way you can help your faction win the game is to convince others that it is appropriate for the British government to provide substantial funding for large-scale scientific and engineering projects. Your primary opportunity to do this is your speech in GS 5, described in the next section.

Regarding the value of government funding for science and engineering projects

As the game starts in 1828, your father Marc's Thames Tunnel project, to construct an underwater tunnel under the Thames river in London, is in its third year. The project was initially privately funded via a group of investors who formed the Thames Tunnel Company. Your father's chief engineer became ill as a result of the putrid sewer gas leeching into the tunnel from the river, and you took over the project in 1826. In 1828, you barely survived the second flooding of the tunnel, and, beset by financial problems, the tunnel was walled

off and work halted. Tunneling resumed in 1835 and was complete in 1841. All told, the cost was £454,000 to dig the tunnel and £180,000 to fit it out. Of the total £634,000, £246,000 was a loan from the British Treasury.

During GS 5, you are to present a speech defending the Thames Tunnel as a good investment of Treasury money, with the implication that Treasury investment in one of Babbage's machines would likewise be a good investment.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are not a FRS until 1830, you should plan on making a speech supporting your faction's position during one of the sessions that take place at the Royal Society after you are made a FRS: GS 4 or 6. Coordinate with your faction leader (Herschel) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

Since you are not a member of the government, nor a member of the nobility, you have no formal power to make decisions regarding Babbage's machines – or anything else, for that matter. However, as of 1830, you are an influential fellow of the Royal Society (FRS), and if the Royal is asked to take up the matter of Babbage's machines, you can help steer the society to recommend that the government provide more funding for development and construction.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using nudges, you can influence the way that the PM votes regarding funding for Babbage’s engines, and even influence who will be PM during GS 6. By using the other type of cards, powers, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 5 speech, due before GS 5 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the “winning” PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in British science during the early to mid-1800s, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

You are a very capable engineer, and so you are able to easily understand the way Babbage’s computing engines work. However, the mathematics behind their workings and the true degree of their usefulness may elude you.

You are not, however, a typical MOS. You are young, and not a landed gentleman, and so you do not have large reserves of money to allow you to pursue engineering as a hobby. You earn your living by creating grand, useful things: tunnels, railways, steamships, bridges, and the like. Accordingly, you believe it is appropriate and proper to be a professional man of science, rather than simply a wealthy, talented amateur. In addition, since the Thames Tunnel relied on government funding, you are obviously in favor of the British government funding engineering and scientific endeavors, if they are likely to produce useful outcomes. Although you may not fully grasp Babbage’s mathematical vision for the Difference and

Analytical Engines, you take him at his word and are in favor of government support for the projects.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. This means that when the next president of the Royal Society is chosen in GS 4, you should vote for the candidate who you think is most likely to favor those in the new generation of men of science, regardless of which faction the candidate belongs to.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

Engineering was very much a working-man's occupation in the time frame of this game, and as such, engineers did not write papers or books. Unlike the mathematicians and scientists in your faction (Babbage, Herschel, Adams, and even Lovelace and Somerville), you have not written any publications on your work.

With other roles

Like your father, you are fast friends with Charles Babbage. An example of your friendship happened In 1838. Babbage devised an ingenious method to test the smoothness of ride on your wide gauge Great Western Railway. At the time, the nascent railway system in Britain used tracks of two different widths, or gauges, making it difficult for people or goods to move across the kingdom via rail. The controversy was called the "war of the gauges." Babbage outfitted a special railway car, called a dynamometer car, filled with instrumentation. The car measured the speed of the car, the frequency and intensity of the car's shaking, and so on. (Think of dynamometer car as the equivalent of a modern jetliner's "black box.") Data from Babbage's dynamometer car helped convince the government to make your seven-foot railway gauge the standard in the southwest of England. George Biddell Airy, who is something of a nemesis to Charles Babbage, was also asked for his input regarding the war of the gauges; he recommended that the competing narrow-gauge system be employed rather than your wide gauge.

Although you are on the same faction, and hold broadly the same opinions regarding the big ideas in the game, you have an adversarial relationship with the scientific populist Dionysius Lardner. On several occasions, Lardner has very publicly criticized your great engineering projects; worse, his criticisms have largely been scientifically unsound. For example, when you were supervising construction of the Box Tunnel for the Great Western Railway, which included a one-foot decrease in elevation for every 100 feet of length, Lardner performed

some (erroneous) mathematics and concluded that if a train lost its brakes in the tunnel, it would emerge from the other end traveling at a speed in excess of 120 miles per hour. You rightfully pointed out that Lardner had forgotten to factor in friction and air resistance; in fact, a runaway train would exit the tunnel at something like 40 miles per hour, scarcely faster than its normal speed. Lardner also said your S. S. Great Western, the second steamship to cross the Atlantic, was more likely to voyage to the moon than it was to make the crossing safely. The ship steamed into New York harbor in April 1838, with 200 tons of coal still in her bunkers, proving just how wrong Lardner's claims were.

In 1830, the president of the Royal Society, Davies Gilbert, was asked to evaluate designs for a proposed bridge over the River Avon in Bristol. He selected your design for a suspension bridge. Construction was started in 1831, but riots in Bristol caused the project to be abandoned. Work was not started again until 1864.

You have no love for the bureaucracy of the government; you feel the interests of Britain would best be served if the government did not interfere. Of course, your experience with the Thames Tunnel also has convinced you that, unfortunately, government funds are sometimes a necessary evil in order for progress to take place.

With your faction

Although you are not a member of the nobility or the government, you are a valuable member to your faction. First, you are a FRS as of 1830, and so you can work to convince the other Fellows to vote in favor of Babbage and his Engines when the subject arises.

Secondly, you have something the faction needs: a concrete example, showing the benefits of government funding of a large-scale engineering project, namely, the Thames Tunnel. You are one of the few people in this time frame that can show concrete benefits to the nation from a publicly funded, non-military engineering project. In your GS 5 speech, you should draw parallels between the success of the Thames Tunnel and the projected success of the Analytical Engines if its construction is funded by the government.

Strategy advice

To keep with the timeline of the game, you should refrain from playing up the success of the Thames Tunnel project until your speech in GS 5. The Tunnel was not complete until 1843, and before that time, the project was something of a laughing stock with the public and a sore point with you. Before GS 5, you should be a solid friend and supporter of Babbage; you can speak to the engineering feasibility of the project, based on your mechanical engineering expertise. You can be an active questioner of the anti-Babbage faction, and make a podium speech or two in support of your faction.

To learn more

[Das12] Mike Dash. The Epic Struggle to Tunnel Under the Thames. <http://www.smithsonianmag.com/history/the-epic-struggle-to-tunnel-under-the-thames-14638810/>, 2012

[Ken16] Maev Kennedy. Brunel’s Thames Tunnel (and Accidental Brothel) Becomes New Arts Space. <https://www.theguardian.com/stage/2016/apr/15/brunel-thames-tunnel-arts-space-grand-entrance-hall>, 2016

[Mus16] The Brunel Museum. The Thames Tunnel. <http://www.brunel-museum.org.uk/history/the-thames-tunnel/>, 2016

Summary of your individual victory objectives

You win the game if your faction wins. Your pro-Babbage faction wins the game, by securing government funding for the development of Babbage’s engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Image of I. K. Brunel in front of the launching chains for the S. S. Great Eastern in 1857, from Wikimedia Commons. URL: <https://commons.wikimedia.org/wiki/File:IKBrunelChains.jpg>

Andrew Crosse

British man of science

Pro-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Andrew Crosse, an amateur man of science best known for your experiments with electricity. Your experiments with very large voltaic piles, which discharged with exceptionally large reports, led to you being known in your neighborhood as the “thunder and lightning man.” Mary Shelly, the author of the novel *Frankenstein*, is an acquaintance, and her husband Percy Bysshe Shelley attended one of your lectures on atmospheric electricity in 1814; this may or may not have helped to influence later editions of the famous novel.

You were born on July 17, 1784, at your father’s estate, Fyne Court, in Sedgemoor, southwest of Bristol. Your parents were Richard Crosse and Susannah Mary Porter; your father was the sheriff of your home county, Somersetshire. Your father was also interested in the study of science, counting Benjamin Franklin and Joseph Priestley amongst his friends. So, when you started to show interest in the study of natural phenomena, especially electricity, both he and your mother encouraged your studies.

You attended Brasenose College, Oxford, starting in 1802 and graduating in 1805. You intended to study law and enter the bar, but after returning home to Fyne Court, you

devoted yourself to your experiments with electricity and electro-chemistry full time. You converted the music hall of your home into a laboratory for your experiments. Soon you had wires strung on the trees around the estate and a 50-jar voltaic pile. (A voltaic pile is an electrical battery, the first battery capable of producing a continuous current through a circuit.) Many of your experiments involved charging and then rapidly discharging the pile, resulting in a crack “as loud as those of a cannon.” These experiments led you to be known as the “thunder and lightening man” to your neighbors.

Your experiments have not been without controversy. In 1836, during some electrocrystallization experiments, you observed some tiny mites appearing within the experimental apparatus, in conditions which you judged inhospitable to life. A newspaper heard of the incident and published a story about it, dubbing the mites *Acarus crossii*. More newspapers reprinted the story, and some were under the impression that you claimed to have created the mites. This did not sit well; you received letters accusing you of blasphemy for claiming to have created life, and some of the letters included death threats. To make the controversy worse, other scientists claimed to have conducted the same experiment and repeated the results! In fact, you never did claim to have created the mites. You assumed that mite eggs were in the potash solution before you passed the electrical current through it, and somehow managed to survive the experience to hatch.

Objectives

You have one main objective in the game. You want your pro-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government decides, on both of the opportunities, to fund the development of Babbage’s Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As Andrew Crosse, you have responsibilities to your pro-Babbage faction, a responsibility to show how interchangeable, manufactured parts could make construction of Babbage’s Analytical Engine more likely, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to the pro-Babbage faction

As a member of the pro-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. Before and after your GS 5 speech, you can help your faction by being a vocal proponent for Babbage and his calculating engines. You can ask good questions and make short podium speeches to help propel your faction to victory.

Responsibility to speak regarding interchangeable parts

During GS 5, you will present a speech at the Mechanic's Institute regarding the concept of interchangeable parts, and how the concept could make construction of Babbage's Analytical Engine more likely. Interchangeable parts, first popularized by Eli Whitney in the U.S. around the turn of the 19th century, would make initial construction of the engine easier, since presumably many of the parts of the engine which serve similar purposes (e.g., the hundreds of geared wheels representing digits) could be made to identical specifications. In addition, interchangeable parts would ease in the maintenance of the engine, and make it easier to construct further engines. With your speech, you should try to counterbalance arguments from the anti-Babbage faction about how difficult construction of the engine will be.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are not a FRS, you should plan on making a speech supporting your

faction's position during the first session that takes place at the Mechanic's Institute, GS 2. Coordinate with your faction leader (Herschel) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

Since you are not a member of the government, the nobility, or the Royal Society, your formal powers in the game are very limited. This does not mean that you are not a valuable member of your faction, or that you cannot win the game. By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 5 speech, due before GS 5 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the "winning" PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in British science during the early to mid-1800s, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding the idea of the government funding large-scale scientific and engineering projects, you are in favor of the practice. If there is a precedent set of funding scientific research,

you might be able to convince the government to fund your work as well. So, if only for selfish reasons, you support the funding of Babbage's computing engines.

You are a member of the last generation of old-school men of science community, i.e., a wealthy gentleman with enough time, talent, and money to pursue your scientific interests at your leisure. Although you do not have anything against them personally, this puts you somewhat at odds with younger men of science like John Couch Adams. Men like Adams do not have large sums of money to pursue scientific studies as a hobby; rather, they do science for a living – science is their profession as well as their passion. If you are among the last of the wealthy, amateur scientific gentlemen, Adams and his colleagues are among the first of the professional scientists. This tension will be played out in GS 4, when the next president of the Royal Society is elected. You should support the candidate who you think best represents the values of old-school men of science, and who you think has the most political power.

As for the conflict between imagination and reason, you are more aligned with Byron's Ghost, the champion of imagination, and opposed to William King, who argues for the supremacy of reason.

With texts

Although there have been many popular press articles written about you (stemming from the 1836 misunderstanding), you have not yet published any of your own scientific work.

With other roles

Your status as an old-school man of science gives you a natural affinity for others from a similar background: Babbage, Gilbert, Herschel, Lovelace, Sabine, and Somerville. Similarly, you are not of the same class as those who pursue science professionally: Adams, Airy, Brunel, Buckland, Clement, Faraday, and Lardner.

You have a connection with Countess Lovelace for two very different reasons. First, Lovelace sought you out when she was thinking about the "calculus of the nervous system." Her desire was to create a mathematical system to explain human behavior, and she thought that electricity – recently shown to cause movements in the muscles of dead animals and people – might be involved. She initially sought the help of Michael Faraday, but he had to decline due to illness. So, she sought you out instead. She stayed at your estate for an extended period of time, but came away no further along in her work and disappointed in the chaotic nature of life there.

The other connection to Lovelace is more salacious. Beginning around 1844, your son John Crosse and Ada had an extramarital affair. Do not mention this publicly! A gentleman such as yourself would never think of airing the dirty laundry of another member of the nobility.

With your faction

Although you are not a member of the nobility or the government, you are a valuable member to your faction. Before and after your GS 5 speech, you can earn PCs for your faction by paying careful attention to the anti-Babbage arguments, asking questions that expose weaknesses, and making effective podium speeches to promote your faction.

Strategy advice

To prepare for your GS 5 speech, you should research about Eli Whitney and the circumstances of his interchangeable parts demonstration before the United States Congress. Also look in to other early pioneers of interchangeable parts, including Honoré LeBlanc and Samuel Bentham.

To learn more

[Cro57] Mrs. Cornelia A. H. Crosse. *Memorials, Scientific and Literary, of Andrew Crosse, the Electrician*. Longman, Brown, Green, Longmans, & Roberts, London, 1857. Available via Google Books

Summary of your individual victory objectives

You win the game if your faction wins. Your pro-Babbage faction wins the game, by securing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Michael Faraday FRS

English man of science

Pro-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Michael Faraday, an extremely influential English man of science. Your studies have focused on chemistry and electricity, specifically the fields of electromagnetism and electrochemistry. You are an excellent experimentalist and have a knack for explaining complicated scientific concepts in easy-to-understand language. As one measure of your influence on modern science and technology, the unit used to measure electrical capacitance, the farad, is named in your honor. You are the first scientific celebrity in Great Britain, and probably the leading man of science of your generation.

You were born on September 22, 1791, in Newington Butts, Surrey, near London, to working-class, nonconformist parents. Your father, James, was a blacksmith. Children of your class received only basic education, as they were expected to find work in the factories or take an apprenticeship and become craftsmen. Therefore, an intelligent and curious child such as yourself had to educate himself. You were able to do this largely during your apprenticeship with a bookbinder. You were able to read widely at the bookbinder's, developing a special interest in science and electricity in particular.

In 1812 you managed to acquire tickets to attend lectures by the renowned chemist Sir Humphry Davy. You took copious notes, compiled them into a 300-page book, and sent

the book to Davy, who was impressed and supportive of your budding scientific ambitions. In 1813, when Davy's eyesight was damaged while working with a volatile chemical, he hired you on as his assistant. It has been said that Davy's greatest discovery was Faraday.

While Davy respected your scientific abilities and valued your contributions to his work, he was a member of the aristocracy are you are not. To illustrate: when you accompanied Davy on a continental European tour between 1813 and 1815, you traveled as Davy's valet. Davy's wife, in particular, did not consider you an equal. She forced you to ride outside their coach, made you eat with the other servants, and generally made your life during the tour miserable.

Your scientific accomplishments by the time our game starts are already very impressive. You performed an experiment showing how magnetic fields effect electric current in 1821, liquefied chlorine in 1823, isolated benzene in 1825, and made important discoveries regarding alloys of iron in 1826. In 1827, you succeeded Davy as a professor at the Royal Institution of Great Britain. Your achievements were so numerous and so important that you were named Fellow of the Royal Society on August 1st, 1824. You will win the Royal's most prestigious award, the Copley Medal, twice: first in 1832 and then again in 1838.

You have worked so diligently on your studies – particularly trying to determine what electricity is and how it flows through certain materials – that your body seemingly gave out in 1839. For six years, you performed virtually no creative science. But, starting in 1845, you were able to resume your work.

Throughout your life, your nonconformist Christian faith as a member of the Sandemanian denomination has been very influential. Your childhood home served as the third Sade-manian church in London, as the denomination spread from Scotland to England in the 1760s. Your church teaches that the accumulation of personal wealth does not fit within the teachings of the Bible. For religious reasons, you turned down a knighthood, twice refused to be considered for president of the Royal Society, and refused to help with the production of chemical weapons for the Crimean war. Before your death in 1867, you also turned down the opportunity to be interred in Westminster Abbey with the likes of Isaac Newton; instead, your body was buried in the nonconformist section of a cemetery in north London.

Your strong faith has at least one practical implication in terms of the game: if other players at any time lead the group as a whole or in some part in prayer, you must quietly refuse to participate. The Sademanian faith holds that praying with someone from another denomination is unlawful.

Objectives

You have one main objective in the game. You want your pro-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government decides, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If

both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As Michael Faraday, you have a responsibility to introduce a resolution to the Royal Society recommending that the government fund construction of Babbage's Difference Engine, responsibilities to your pro-Babbage faction, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Regarding a pro-Babbage resolution

During GS 3, the Royal Society will debate whether or not to recommend that the government fund the development of Babbage's Difference Engine. You are to introduce a resolution saying that the Royal recommends that the government fund Babbage. Your speech should explain why you think this is appropriate; work with your faction (in particular with your faction leader, John Herschel) to make sure you include all of your faction's arguments for funding Babbage's device.

Responsibilities to the pro-Babbage faction

As a member of the pro-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. Early in the game, you can help your faction by being a vocal proponent for Babbage and his calculating engines. You can ask good questions and make short podium speeches to help propel your faction to victory.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret

term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are a FRS, you should plan on making a speech supporting your faction’s position during one of the other sessions that take place at the Royal Society: GS 4 or 6. Coordinate with your faction leader (Herschel) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

Since you are not a member of the government, nor a member of the nobility, you have no formal power to make decisions regarding Babbage’s machines – or anything else, for that matter. However, as of 1824, you are an influential fellow of the Royal Society (FRS), and if the Royal is asked to take up the matter of Babbage’s machines, you can help steer the society to recommend that the government provide more funding for development and construction.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage’s engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 3 speech, due before GS 3 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in British science during the early to mid-1800s, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

You are a very capable and distinguished man of science, and so you are able to understand the broad strokes of how Babbage’s computing engines could be of value. However, you are not a distinguished mathematician, and so the mathematics behind their workings and the true degree of their usefulness may elude you. Regarding the idea of the government funding large-scale scientific and engineering projects, you are in favor of the practice. If there is a precedent set of funding scientific research, you might be able to convince the government to fund your work as well. So, if only for selfish reasons, you support the funding of Babbage’s computing engines.

You are not, however, a typical MOS. You are not a landed gentleman, and you do not have large reserves of money to allow you to pursue science as a hobby. Your “commoner” status was only confirmed by the behavior of Davy’s wife during your European tour of 1813 – 1815. You have, through your God-given talent, worked your way up in the scientific world. Accordingly, you believe it is appropriate and proper to be a professional man of science, rather than simply a wealthy, talented amateur. Your religion’s take on personal wealth also causes you to consider those MOS members who are independently wealthy with some suspicion. This tension will be played out in GS 4, when the next president of the Royal Society is elected. When the next president of the Royal Society is chosen in GS 4, you should vote for the candidate who you think is most likely to favor those in the new generation of men of science, regardless of which faction the candidate belongs to.

As for the conflict between imagination and reason, you are firmly aligned with William King on the supremacy of reason, and opposed to Byron’s Ghost, the champion of imagination.

With texts

As the pre-eminent man of science of the time, you have published several books. During the time frame of the game, your works include the following:

- *Chemical Manipulation, Being Instructions to Students in Chemistry*, 1827.
- *Experimental Researches in Electricity, Vol I*, 1837.

- *Experimental Researches in Electricity, Vol II, 1844.*

With other roles

Your status as a commoner, a professional man of science, puts you somewhat at odds with the more old-school men of science: Babbage, Crosse, Gilbert, Herschel, Lovelace, Sabine, and Somerville. But, for the same reason, you have a natural affinity with others pursuing science as a profession: Adams, Airy, Brunel, Buckland, and Clement.

You have a connection with Countess Lovelace. She sought you out when she was thinking about the “calculus of the nervous system.” Her desire was to create a mathematical system to explain human behavior, and she thought that electricity – recently shown to cause movements in the muscles of dead animals and people – might be involved. She approached you regarding a collaboration, and you would have been happy to oblige, but you had to decline due to illness. You recommended that she contact Andrew Crosse, and so she approached him regarding the project.

With your faction

You have a key role in the game, from your faction’s point of view: you make the final, summary argument as to why the government should fund the Difference Engine. Despite your class and religious differences, work to be cordial and professional with your faction mates, so you can effectively coordinate with them to make an effective GS 3 resolution and speech.

Strategy advice

Pay careful attention to the arguments that your faction makes in favor of the Difference Engine during the first two game sessions, and also the arguments that the anti-Babbage faction makes. Try to gauge which arguments have made the most impact on the indeterminates. Then, in your resolution and speech during GS 3, try to amplify your faction’s best arguments, while refuting the most effective arguments of the anti-Babbage faction.

Summary of your individual victory objectives

You win the game if your faction wins. Your pro-Babbage faction wins the game, by securing government funding for the development of Babbage’s engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Portrait of Michael Faraday FRS, by Thomas Phillips, from the National Portrait Gallery, St Martin's Place, London; <http://www.npg.org.uk/collections/search/portrait/mw02170/Michael-Faraday?LinkID=mp01529&search=sas&sText=michael+faraday&role=sit&rNo=1>

Sir John Frederick William Herschel, 1st Baronet KH FRS

British polymath

Pro-Babbage faction leader and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Sir John Frederick William Herschel, British polymath (a person whose expertise spans a significant number of different subject areas). You are a member of the men of science (MOS) community in the game. You are a man of science, or natural philosopher, of the old kind – a man whose wealth and social position have allowed you to pursue your mathematical, astronomic, chemical, photographic, biological, and other scientific interests. You are a fellow of the Royal Society (FRS), and as of 1831 a First Baronet of the Royal Guelphic Order.

Since you are named a Baronet of the Royal Guelphic Order in 1831, you are entitled to the honorific “Sir” in the 1846 game sessions. Before that, you are simply John Herschel.

You were born in Slough, Buckinghamshire (a few miles west of London), on March 7, 1792, to the astronomer William Herschel and his wife Caroline. You studied at Eton and then at St. John’s College, Cambridge. You excelled at mathematics (and other subjects) at Cambridge, graduating as Senior Wrangler, the top mathematics undergraduate, in 1813. After Cambridge, you worked with your father, building a powerful telescope and making

several important astronomical observations. You were elected as a Fellow of the Royal Society (FRS) in 1813, and your work in astronomy earned you the Copley Medal from the Society in 1821. You were made a Knight of the Royal Guelphic Order in 1831.

At Cambridge, you, Charles Babbage, and others formed the Analytical Society. The society was dedicated to improving the state of mathematics and Cambridge, and then in the whole of Britain. Your primary goal – which the Analyticals succeeded in achieving – was to switch Britain from Newton’s calculus notation to Leibniz’s notation, which was in use in France and the rest of Europe.

Objectives

You have one main objective in the game. You want your pro-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government decides, on both of the opportunities, to fund the development of Babbage’s Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Secret: You also win the game if you are elected as Royal Society president during GS 4.

Responsibilities

As John Herschel, you have responsibilities as the leader of the pro-Babbage faction, to introduce the Difference Engine fragment during GS 1, the responsibility to run for the presidency of the Royal Society in GS 4, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibilities to the pro-Babbage faction

The members of your faction each have a responsibility to bring forward one or more arguments for why the British Government should fund construction of Babbage’s Difference or Analytical Engine. Your job, as leader of the pro-Babbage faction, is to make sure that these arguments are made in the most effective way. You should help your faction members recognize and exploit the areas where the anti-Babbage faction’s arguments are weakest. If the anti-Babbage faction seems to be persuading the indeterminates, you should rally your faction members to mount effective counter-arguments in their podium speeches. In

addition, you should work closely with Faraday to develop the arguments he will present in GS 3, in support of his resolution that the Royal Society should recommend funding for the Difference Engine.

Introduction of the Difference Engine fragment

During GS 1, after the introductory party games and the PM's speech are complete, you are to give a speech that introduces the Difference Engine fragment that Babbage keeps in his home. You should cover what it is intended to do, briefly sketch how it works, and then talk about some of the ways in which the engine would be beneficial.

Responsibility to run for the Royal Society presidency

During GS 4, the Royal Society will elect a new president, to replace Davies Gilbert. You will be your faction's nominee for the position. During GS 4, you will be nominated for the position by Charles Babbage; you should work with him to come up with good reasons why the society should elect you, a respected man of science, to the position. You should pursue this position vigorously, since winning the presidency allows you to win the game regardless of how your faction performs.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are a FRS, you should plan on making a speech supporting your faction's position during one of the sessions that take place at the Royal Society, GS 3, 4,

pr 6. Coordinate with your faction members so you can make a speech with the maximum impact for your faction.

Powers

Since you are not a member of the government, nor a member of the aristocracy by heredity, you have no formal power to make decisions regarding Babbage's machines – or anything else, for that matter. But, because of your connections and socializing with nobility and politicians, you may be able to exert some informal influence on them.

You are an influential fellow of the Royal Society (FRS), and if the Royal is asked to take up the matter of Babbage's machines, you can help steer the society to recommend that the government provide more funding for development and construction.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 1 speech, due before GS 1 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in British science during the early to mid-1800s, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding the idea of the government funding large-scale scientific and engineering projects, you are in favor of the practice. If there is a precedent set of funding scientific research, you might be able to convince the government to fund your work as well. So, if only for selfish reasons, you support the funding of Babbage's computing engines.

You are a member of the last generation of the old-school men of science community, i.e., a wealthy gentleman with enough time, talent, and money to pursue your scientific interests at your leisure. Although you do not have anything against them personally, this puts you somewhat at odds with younger men of science like John Couch Adams. Men like Adams do not have large sums of money to pursue scientific studies as a hobby, rather, they do science for a living – science is their profession as well as their passion. If you are among the last of the wealthy, amateur scientific gentlemen, Adams and his colleagues are among the first of the professional scientists.

As for the conflict between imagination and reason, you are firmly aligned with William King on the supremacy of reason, and opposed to Byron's Ghost, the champion of imagination.

With texts

You have published several respected works of science, both in academic and popular publications. Selected works include:

- *On the Hyposulphurous Acid and its Compounds*, in The Edinburgh Philosophical Journal, 1821.
- *On the Aberration of Compound Lenses and Object-Glasses*, 1821
- Articles on light, sound, and astronomy in *Encyclopædia Metropolitana*, first published in 1817.

With other roles

You met Charles Babbage when you were both students at Cambridge. You, Babbage, and several others formed the Analytical Society there, and have been fast friends ever since.

One of your most important works, *A preliminary discourse on the study of natural philosophy*, was published in 1831 in Dionysius Lardner's *Cabinet Cyclopædia*, and so you have a close relationship with Lardner.

As an old-school man of science, you have a natural affinity for others in the same class: Babbage, Crosse, Gilbert, Lovelace, Sabine, and Somerville. Likewise, you do not have much in common with the new breed of men of science, who study science as a profession: Adams, Airy, Brunel, Buckland, Clement, Faraday, and Lardner.

With your faction

As an organizer and strategist, you are very important to your faction. All of your faction members look up to you for your intelligence and scientific accomplishments. You are very committed to Babbage and his engines, so you should work diligently to try to procure funding for the projects.

Strategy advice

It would be wise to have out-of-class meetings with your faction members – either in person or via some electronic means – before each game session, to coordinate everyone’s efforts. If the gamemaster requires the text of speeches to be published well in advance of the session they will be presented in, you should make sure your faction members read the speeches of the anti-Babbage faction in advance, and then prepare counter arguments that can be presented via podium speeches.

To learn more

[Her30] John Frederick William Herschel. *Preliminary Discourse on the Study of Natural Philosophy*. Longman, Bees, Orme, Brown, & Green, London, 1830. Available via Google Books

Summary of your individual victory objectives

You win the game if your faction wins. Your pro-Babbage faction wins the game, by securing government funding for the development of Babbage’s engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Your second way to win the game is to be elected as the president of the Royal Society during GS 4.

Portrait of Sir John Frederick William Herschel, 1st Bt, by Julia Margaret Cameron, from the National Portrait Gallery, St Martin’s Place, London; URL: <http://www.npg.org.uk/collections/search/portrait/mw07487/Sir-John-Frederick-William-Herschel-1st-Bt?LinkID=mp02165&search=sas&sText=john+herschel&role=sit&rNo=5>

Dionysius Lardner FRS

Irish man of science

Pro-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Dionysius Lardner, a member of the Royal Society and a professor at University College in London. You are one of the first people to be called a scientific journalist, and editor of a 133-volume encyclopedia of science, the *Cabinet Cyclopædia*. You are a dynamic lecturer and a very effective popularizer of scientific topics, and in this game you are responsible for lecturing on Babbage's Difference Engine, among other things.

You were born in 1793 in Dublin, Ireland, the son of solicitor (a legal practitioner qualified to give general legal advice and conduct legal proceedings, as opposed to a barrister, a lawyer who works in higher-level courts). Your father wanted you to follow in his legal career footsteps, but you instead entered Trinity College at the University of Dublin in 1812. After your graduation in 1819 you began to write and lecture on mathematical and scientific subjects. In the year the game begins (1828), you have just been named professor of natural philosophy and mathematics at University College in London. You have also just been selected as a Fellow of the Royal Society (FRS). You have a remarkable knack for explaining complicated mathematical and scientific concepts in terms that ordinary people can understand, which has made your books and lectures quite popular.

On an unfortunate personal note, you have something of a penchant for having affairs with married women. Your first occurred during the 1820s, and a second affair with the wife of a Royal Army captain will cause you to flee England, along with your paramour Mary Spicer Heaviside, for the relative safety of Paris in 1840. For the purposes of this game, you will not leave for France until after the last game session, which takes place in 1846.

Objectives

You win the game in one of two ways. First, you win if your faction wins. Your faction wins if the British government decides, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

In addition, you have a second victory condition that is not connected to the success of your faction. If you are able to make at least two bad arguments during the game (see below), and document them for the gamemaster, without getting caught by the other players in the game, you win.

Responsibilities

As Dionysius Lardner, you have the responsibility to preside over the lectures at the Mechanic's Institute during GS 2, responsibilities to your faction, the responsibility to explain the Difference Engine and its importance, a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game, and a critical thinking skills responsibility that is yours alone. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to preside over GS 2

You will preside over the first set of lectures at the Mechanics Institute, which take place during GS 2. As the presiding character, you are responsible for making sure that the activities and speeches scheduled for GS 2 all happen. The activities for GS 2, and their order, are:

- Social time. This is a brief period of time for general communication between characters, and for the factions to confer and strategize.
- PM speech by the current PM. The newly-elected PM will outline the priorities and goals his government will have while he is in office.

- The Duke of Sussex will speak on the virtues of laissez-faire economic policies.
- Sabine will speak on the mathematics of navigation. You will provide the tables that he will use to teach others how to solve his problem. **NOTE: You MUST speak with the gamemaster before GS 2, so you can get the tables and understand what is going to happen with them.**
- Your speech regarding the dangers of erroneous mathematical tables and the value of the Difference Engine. See below for more details regarding your speech.
- Captain Swing will speak on the plight of farmers and the dangers of automation.
- At the end of the session the gamemaster will conduct the persuasiveness metric poll of the indeterminates.

During any non-scheduled time in the session, the *podium rule* is in effect. The podium rule states that any character wishing to speak should approach the podium and wait to be recognized by the presiding officer (you). You must make sure that you are fair to anyone who wishes to speak, even those that are not in your faction. Characters are encouraged to ask questions of presenters, at the discretion of the presiding officer, but they should not make speeches to the class without approaching the podium.

Speak with the GM before GS 2 about the lectures at the Mechanics Institute, so that you are well prepared for the conduct of the session.

Responsibilities to the pro-Babbage faction

As a member of the pro-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. You can do this by making an excellent, persuasive case for the Difference Engine in GS 2. Later in the game, you can help your faction by being a vocal proponent for Babbage and his calculating engines. You can ask good questions and make short podium speeches to help propel your faction to victory.

Regarding the Difference Engine

You are responsible for presenting a lecture outlining the workings of Babbage's proposed Difference Engine during GS 2. You should briefly explain how the engine would work, and then argue why the machine is so vital to Great Britain. For this, you should especially emphasize the importance of accurate navigational tables to the British Empire.

One way you can do this is by providing the mathematical tables for tangents and square roots that will be required to compute the answers to the problem that Sabine will speak on. When you visit with the GM before GS 2, the GM will provide you with copies of two different versions of each table. One version of each table will have the correct values, while the other version will have errors that will cause wrong answers for Sabine's problem. It is up to you and your faction to determine how to distribute the tables during GS 2. I.e.,

do you want the anti-Babbage faction to have the erroneous tables while the pro-Babbage faction has the correct ones? What about the indeterminates?

As illustrated by the incorrect answers to Sabine's problems, errors in mathematical tables can cause incorrect answers in navigational problems. So, even minor errors in these tables might result in several shipwrecks a year – dreadful costs for such seemingly unimportant numerical errors! The accidents prevented by accurate navigational tables would surely more than make up for the cost of developing Babbage's amazing Difference Engine.

Secret: Sabine knows you will be providing the tables, but he does not know you are going to bring tables with errors. You will need to coordinate with Sabine in order to do the GS 2 presentations, so do not let on ahead of time that some of the tables are erroneous. You should not reveal to anyone in the anti-Babbage faction or the indeterminates that some of the tables contain errors, until it is your turn to speak.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Critical thinking skills: Presenting bad arguments

Secret: You are this game's “Bad Argument Bradley,” a character assigned to make several arguments that are, well, bad. Other characters in the game are supposed to be on the lookout for such faulty thinking, and will call you out if they detect your bad arguments. As you read above, you can win the game regardless of whether either of Babbage's engines is funded, if you are able to pass two or more of your bad arguments by your colleagues without being detected. So, you must carefully choose your arguments and make them with skill.

For example, you might make a straw man argument. In such an argument, you put forth an often extreme, easy-to-defeat argument that sounds like something your opponents might say, but that they have not, in fact, used as an argument. Then, you tear apart the straw

man. This makes it look like you are soundly defeating your opponent, when in fact all you are doing is destroying an argument that they never made at all.

Some other types of bad arguments that you might try are [Alm13]:

False dilemma “An argument that presents a limited set of two possible categories and assumes that everything in the scope of the discussion must be an element of that set.”

Slippery slope “An argument that attempts to discredit a proposition by arguing that its acceptance will undoubtedly lead to a sequence of events, one or more of which are undesirable.”

Equivocation “An argument that exploits the ambiguity of language by changing the meaning of a word during the course of an argument and using the different meanings to support an ill-founded conclusion.”

See the “To Learn More” section below for some resources you could use to learn more about bad arguments and how to make them.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are a FRS, you should plan on making a speech supporting your faction’s position during one of the sessions that take place at the Royal Society, GS 3, 4, or 6. Coordinate with your faction leader (Herschel) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

Since you are not a member of the government, nor a member of the aristocracy, you have no formal power to make decisions regarding Babbage’s machines – or anything else, for that matter. However, you do have the power of persuasion; through your GS 2 lecture, and your other impromptu speaking and interactions, you should do your best to make the Difference Engine appear to be an absolute necessity for the nation. You are also a member of the Royal Society, so if the Royal ever takes up discussion about the engines, you can attempt to convince your peers that they should support construction of Babbage’s machines.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage’s engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 2 speech, due before GS 2 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. Beware being too explicit about the errors in the tables you will hand out during the session, so the anti-Babbage faction cannot read your paper and know what is going to happen. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Your second writing assignment, due no later than the beginning of the post-mortem session, is a short paper documenting the bad arguments you made during the game. For each bad argument you make, include which bad argument tactic you employed, along with a definition of the tactic, and what game session you used the technique. This should be submitted to the GM only.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in British science during the early to mid-1800s, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

You have lectured and written very widely on the subjects of science and mathematics, with subjects ranging from pure science and mathematics (e.g., *The Encyclopædia of Pure Mathematics* (1847)) to applied science and engineering (e.g., *The Steam Engine Explained and Illustrated: With an Account of its Invention and Progressive Improvement, and its Application to Navigation and Railways; Including also a Memoir of Watt* (1840)).

Interestingly, while your lectures regarding the Difference Engine were accurate, important, and influential, you have often been quite wrong regarding scientific topics. To cite only one example, when I. K. Brunel was building the Box Tunnel for the Great Western Railway, which included a one-foot decrease in elevation for every 100 feet of length, you performed some (erroneous) mathematics and concluded that if a train lost its brakes in the tunnel,

it would emerge from the other end traveling at a speed in excess of 120 miles per hour. That is not true at all; you forgot to factor in air resistance and friction. You have made several other rather large and very public scientific errors, which you can read about in your role-specific readings.

That being said, you are clearly in favor of the development of Babbage's computational engines. You are also in favor of applied science, and in opposition to those who think that science should be a purely abstract field, separate from and not beholden to practical concerns.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. This means that when the next president of the Royal Society is chosen in GS 4, you should vote for the candidate who you think is most likely to favor those in the new generation of men of science, regardless of which faction the candidate belongs to.

As for the conflict between imagination and reason, you are firmly aligned with William King on the supremacy of reason, and opposed to Byron's Ghost, the champion of imagination.

With texts

In addition to lecturing regarding Babbage's Difference Engine, you have also written about it: the article *Babbage's Calculating Engine*, which appeared in the *Edinburgh Review* in 1834. For the purposes of this game, you may assume that your article appeared before the game starts in 1828.

With other roles

You know Babbage personally and have a mostly friendly and cordial relationship with him. Babbage has, on occasion, called you to account regarding your scientific mistakes – he has called you “the prince of humbugs” – but (unlike I. K. Brunel) he has mostly done so in a “best construction” manner.

Your lecture on the Difference Engine, along with seeing the engine fragment at Babbage's party, are extremely influential to Ada Lovelace. Inspired by these two experiences, Ada becomes Babbage's “enchanted math fairy” and “high priestess of the Analytical Engine.” Ada is also intimately familiar with your work on geometry, *Analytical Treatise on Plane and Spherical Trigonometry*, which was prescribed to her by her physician, Doctor William King, to help prevent her from following in her father's hedonistic footsteps.

You have a good relationship with John Herschel. He has published several articles in your *Cabinet Cyclopædia*.

Although you both are members of the same faction, and hold broadly the same opinions regarding the big ideas in the game, you have an adversarial relationship with the engineer Isambard Kingdom (I. K.) Brunel. This stems from your unfounded and erroneous criticisms of several of Brunel's great engineering projects, including the Box Tunnel of the Great Western Railway (see above) and whether or not the steamship S. S. Great Western would be able to successfully cross the Atlantic Ocean. (You said it would not be able to cross – more likely to voyage to the moon, you said – due to the extreme friction of the water against the hull; in fact, it crossed the Atlantic and sailed into New York harbor quite successfully.) Unlike Babbage, Brunel does not hold back when he calls you on the carpet regarding your scientific errors.

As someone who makes science a profession, you have a natural affinity for others in the same class: Adams, Airy, Brunel, Buckland, Clement, and Faraday. Likewise, you have little in common with the old-school men of science, wealthy individuals who pursue science essentially as a hobby: Babbage, Crosse, Gilbert, Herschel, Lovelace, Sabine, and Somerville.

With your faction

You are important to your faction, because you will argue about one of the few direct, plausible benefits of funding Babbage's Difference Engine: preventing navigational errors due to errors in mathematical tables. Your GS 2 speech can set the tone for the early efforts of your faction to win funding for the Difference Engine.

Strategy advice

You may wish to work one of your bad arguments into your GS 2 speech, and save your second attempt for your podium speech. That way, you can spend more time listening to the anti-Babbage faction's arguments, enabling you to craft a better, more stealthy, bad argument.

To learn more

[Alm13] Ali Almossawi. *An Illustrated Book of Bad Arguments*. The Experiment, New York, NY, 2013

[Con14] Ciaran Conliffe. Dionysius Lardner, Scientific Populist - HeadStuff. <http://www.headstuff.org/2014/07/dionysius-lardner-scientific-populist/>, 2014

[Lar61] Dionysius Lardner. Babbage's calculating engine. In Phillip Morrison and Emily Morrison, editors, *Charles Babbage: On the Principles and Development of the Calculator and Other Seminal Writings*, pages 163–224. Dover Press, New York, NY, 1961

Summary of your individual victory objectives

You can win the game in two ways:

- You win the game if your faction wins. Your pro-Babbage faction wins the game, by securing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.
- You win the game if you are able to make, and document in your second writing assignment, at least two flawed arguments during the first five game sessions, without getting caught.

Augusta Ada King-Noel, Countess of Lovelace

Countess and mathematician

Pro-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Augusta Ada King-Noel, Countess of Lovelace, the only legitimate daughter of the “mad, bad, and dangerous to know” romantic poet, Lord Byron. Due to your gender, you are an unofficial or clandestine member of the men of science (MOS) community. You are a natural philosopher of the old tradition – one whose station in life has allowed you the time and means to study mathematics as a hobby.

You are the only legitimate child of your father, the romantic poet Lord Bryon. You were born in London on December 10th, 1815. Unhappy in the marriage, your mother, Lady Anne Isabella Milbanke Byron, sent your father packing shortly after you were born. You have no memory of Lord Byron, although his death in Greece when you were eight had a profound impact on you.

Your mother was quite concerned that you might have inherited your father’s wild, hedonist proclivities. She was worried that you might become, in her words, “poetical” – moody, unpredictable, and promiscuous – like Byron. So, to fight against these tendencies, she decided to combat any poetical proclivities with poetry’s natural enemy, mathematics. [Pad15] Accordingly, you were tutored in mathematics from a very early age. It was

quite unusual at the time for a young aristocratic woman to receive such an education, but you seemed to thrive on this learning. Two of your early tutors, William Frend and Mary Somerville, are also characters in this game.

As it turns out, you are very intelligent and have a natural gift for languages and mathematics. You met one of the most influential people in your life, Charles Babbage, at one of his famous Dorset Street parties. You were only 17 at the time, but your tutor Mary Somerville took you to the party and made introductions with Babbage for you. You clicked with Babbage immediately, and even more so once you saw the fragment of his Difference Engine at the party. Unlike many of those that attended Babbage's parties, you immediately grasped the operating principle and importance of the machine. You learned more about the Difference Engine by attending one of Dionysius Lardner's popular lectures on the machine.

Babbage, who calls you his "high priestess of the Analytical Engine" and his "enchanted math fairy," arranged for you to have more advanced mathematical tutoring from one of his friends, Augustus De Morgan, a professor at the University of London.

Your collaboration with Babbage reached its apex when he conceived of another kind of calculating machine, the Analytical Engine. Whereas the Difference Engine was intended to automatically produce mathematical tables, the Analytical Engine was to be much more capable. It is recognized today as a prototypical general-purpose computer.

One of the only contemporary publications regarding the Analytical Engine was written by the Italian engineer Luigi Federico Menabrea, for a Swiss journal. You proposed to Babbage that you could translate the article into English, and he agreed, also suggesting that you add some notes of your own to the article. You tackled this task with much enthusiasm, producing a set of notes that were three times as large as the article itself. The article was published in *Scientific Memoirs* in 1843. Your notes show that you understood the potential of the Analytical Engine more than anyone, even more than Babbage himself. Rather than thinking only in terms of the automation of mathematics, you speculated on the use of the Analytical Engine for processing letters and symbols, and perhaps even for the production of music. You used the initials "A. A. L." as the byline for your notes, to hide the fact that they were written by a woman.

Objectives

You have one main objective in the game. You want your pro-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government decides, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As Ada Lovelace, you have responsibilities to your faction, to explain the Analytical Engine and its importance, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibilities to the Pro-Babbage Faction

As a member of the pro-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. Early in the game, you can help your faction by being a vocal proponent for Babbage and his calculating engines. Before GS 5, during game sessions that do not take place at the Royal Society, you can ask good questions and make short podium speeches to help propel your faction to victory.

As a woman, you are not allowed to formally speak at the Royal Society. However, you can confer with your faction members to discuss strategy, etc.

Regarding the Analytical Engine

You are responsible for presenting a lecture outlining the workings of Babbage's proposed Analytical Engine during GS 5. You should briefly explain how the engine would work, and then argue why the machine is so vital. You must try to convey that the Analytical Engine would be able to produce mathematical tables, as the Difference Engine was intended to, but would be both a simpler machine and simultaneously much more powerful. The Analytical Engine could also be used to process letters, characters, and symbols, and might even be able to produce music!

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret

term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are not a FRS, you should plan on making a speech supporting your faction’s position during the a session that doesn’t take place at the Royal Society, GS 1 or 2. Coordinate with your faction leader (Herschel) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

Since you are a woman, even though you are a member of the nobility, you have no formal power to make decisions regarding Babbage’s machines – or anything else, for that matter. However, you do have the power of persuasion; through your GS 5 lecture, and your other impromptu speaking and interactions, you should do your best to make the Difference Engine (before GS 5) and the Analytical Engine (after GS 5) appear to be an absolute necessity for the betterment of the nation and of the discipline of science in general.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage’s engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

You also have a power no one else in the game possesses: the power of celebrity. You will receive two PCs (represented by the two jokers in a standard deck of playing cards) from the GM before the start of GS 1. You can use each of these in one of two ways:

1. Other characters in the game know that if they come to you during a game session, and in the presence of yourself, the Fury, and / or the GM, are able to recite one of the six Byronic stanzas listed in the gamebook, they will win a draw from the PC deck. You may play a power of celebrity card to prevent them from winning a PC card. You may either play your celebrity card before the other character attempts to recite the stanza, in which case another character may try to win a PC card by reciting for you, or after the other character successfully recites the stanza, in which case no one else may try to earn a PC in this manner for the rest of the game session.
2. You may use your power of celebrity to silence another character. If you play the card in this manner, a character of your choice is considered to be in a conversation

with you for as long as you desire, meaning that they may not speak to anyone else until you approve. The effects of this play last only during the game session in which you play the card. This use of the power of celebrity may not be used to prevent a character from giving their scheduled speech, but may be used for any other silencing that you wish. If you wish to play this card, notify the GM. The GM will help coordinate the silencing of your target.

Secret: None of the other characters know anything about the power of celebrity until you choose to exercise it in one of the manners outlined above.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 5 speech, due before GS 5 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in British science during the early to mid-1800s, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding the idea of the government funding large-scale scientific and engineering projects, you are in favor of the practice. You can see that there are much greater applications for the Analytical Engine than everyone, Babbage included, envisions, and you also realize that funding constructing the Analytical Engine is beyond Babbage's means.

You are a clandestine member of the last generation of the old-school men of science community, i.e., a wealthy member of the nobility with enough time, talent, and money to pursue your scientific interests at your leisure. Of course, as a woman, you are a very

unusual member of the community. You and Mary Somerville are exceptions to the norm at the time. Although you do not have anything against them personally, your “old school” status puts you somewhat at odds with younger men of science like John Couch Adams. Men like Adams do not have large sums of money to pursue scientific studies as a hobby, rather, they do science for a living – science is their profession as well as their passion. If you are among the last of the wealthy, amateur scientific gentlemen, Adams and his colleagues are among the first of the professional scientists. This tension comes to a head in GS 4, when the next president of the Royal Society is elected. You will not be able to vote in this election, but, you should work behind the scenes to support the candidate who you think is most likely to favor those in the new generation of men of science, regardless of which faction the candidate belongs to.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron’s Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

You are well familiar with Dionysius Lardner’s six-volume work on geometry, *Analytical Treatise on Plane and Spherical Trigonometry*. This book was prescribed to you by your physician and mentor, Dr. William King, to help prevent you from becoming “poetical” like your father.

Your notes on Menabrea’s article [Men43], published in 1843, are critical to our modern understanding of Babbage’s Analytical Engine. They form the only contemporaneous published description of the Engine.

With other roles

You have a special, close relationship with Charles Babbage, who calls you his “high priestess of the Analytical Engine” and his “enchanted math fairy.”

Mary Somerville and William Frend were influential mathematical tutors in your youth. Doctor William King has also been a mentor; he was hired by your mother to help prevent you from following in your father’s hedonistic ways. His prescription to defeat poetry: mathematics.

You learned more about the workings and importance of the Difference Engine by attending one of Dionysius Lardner’s Mechanic’s Institute lectures on the device.

After publication of your translation of Menabrea’s article regarding the Analytical Engine, you turned your attention to understanding human behavior, through what you called the “calculus of the nervous system.” You suspected that electricity might have something to do with the way human beings think and behave, and so you sought out the help of two separate men of science famous for studying electricity. First, you contacted Michael

Faraday, but he was unable to collaborate with you due to illness. He suggested that you approach Andrew Crosse, the “thunder and lightening man;” afterwards, he hosted you at his estate. You did not make any progress on your theories there, nor did you stay long. The chaotic, disorganized nature of life at Fyne Court, Crosse’s estate, was too much for you.

In a less fortunate connection to Andrew Crosse, after your time at Fyne Court, you conducted an ongoing illicit extramarital affair with Crosse’s son, John.

As an old-school “man” of science, you have a natural affinity for others in the same class: Babbage, Crosse, Gilbert, Herschel, Sabine, and Somerville. Likewise, you do not have much in common with the new breed of men of science, who study science as a profession: Adams, Airy, Brunel, Buckland, Clement, Faraday, and Lardner.

With your faction

As the one person who understands the importance of the Analytical Engine better than anyone, you are an important member of the pro-Babbage faction. On occasion, however, you have the feeling that your pro-Babbage friends are, in some part, playing up their relationship with you due to your celebrity status in Britain.

Strategy advice

First, remember that due to your gender, you cannot speak formally during any session at the Royal Society. This implies that your podium speech must happen very early in the game, in GS 1 or GS 2. Try to coordinate with your faction leader, John Herschel, to come up with a topic for your podium speech that is most effective.

To learn more

[Men43] Luigi Federico Menabrea. Sketch of the Analytical Engine Invented By Charles Babbage, Esq. *Scientific Memoirs*, iii, 1843. Available via Google Books

[Pad15] Sydney Padua. *The Thrilling Adventures of Lovelace and Babbage: With Interesting & Curious Anecdotes of Celebrated and Distinguished Characters, Fully Illustrating a Variety of Instructive and Amusing Scenes; as Performed within and without the Remarkable Difference Engine, Embellished with Portraits and Scientific Diagrams*. Pantheon Books, New York, NY, 2015

Summary of your individual victory objectives

You win the game if your faction wins. Your pro-Babbage faction wins the game, by securing government funding for the development of Babbage’s engines in both of the

funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Portrait of Augusta Ada King-Noel, Countess of Lovelace, by William Henry Mote, after Alfred Edward Chalon, from the National Portrait Gallery, St Martin's Place, London; URL: <http://www.npg.org.uk/collections/search/portrait/mw39468/Augusta-Ada-King-ne-Byron-Countess-of-Lovelace?LinkID=mp50509&search=sas&sText=lovelace&role=sit&rNo=3>

Mary Somerville

Scottish polymath

Pro-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Mary Fairfax Greig Somerville, Scottish polymath (a person whose expertise spans a significant number of different subject areas). Due to your gender, you are an unofficial member of the men of science (MOS) community. You are very unusual for the time, however, as you have published several very influential papers and books. You are a natural philosopher of the old tradition – one whose station in life has allowed you the time and means to study astronomy and mathematics as a hobby.

You are one of seven children born to your father, Vice-Admiral Sir William George Fairfax, and his second wife, Margaret Charters. You were born on December 26th, 1780, while your father was at sea. At the time, families saw no need to formally educate their daughters, and so you received virtually no formal schooling. Your mother taught you to read, and you were sent to a few boarding schools for girls. However, these schools were intended not for traditional education, but rather to teach girls the “ladylike” skills such as needlepoint, piano, and painting.

Nevertheless, as a young girl you endeavored to educate yourself, reading as much as you could at home. Your immediate family actively discouraged your studies, believing that the physiology of a woman was simply not up to the arduous task of deep thought; as

your father thought, "...the strain of abstract thought would injure the tender female frame." [Pat08] This was, again, a common concept at the time. Your mother especially believed it, blaming her encouragement of your older sister's studies for her death at age ten. Nonetheless, you persevered and continued studying. You did find adults who encouraged you here and there; an uncle helped you learn Latin, the artist Alexander Nasmyth turned you on to Euclid's classic work *Elements*, and one of your younger brother's tutors helped you to study algebra.

You married the Russian naval officer Captain Samuil Samuilovich Greig in 1804, and moved to London when he secured an appointment there as the Russian Consul. The marriage was not happy, and Samuil did not approve of your studies, but you had two sons together before Greig died in 1807.

On your return to Scotland, you found friends and acquaintances who supported your burgeoning pursuit of astronomy and mathematics. In particular, professors John Playfair and William Wallace were very influential. With their encouragement, you earned a silver medal in a mathematical problem-solving contest conducted by *Mathematical Repository*.

In 1812 you married again, to hospital inspector William Somerville, who fully supported and encouraged your intellect, and in 1816 you moved back to London. Your husband was elected as a Fellow in the Royal Society of London in 1817, and through him you became part of the thriving scientific community of the city. You are friends with many of the other characters in the game, including Airy, Babbage, Herschel, and Lovelace.

In a feat virtually unprecedented at the time, you published a well-received article in the *Proceedings of the Royal Society*, entitled *The magnetic properties of the violet rays of the solar spectrum*, in 1826. You have published several other papers and books, as well. Befitting your progressive status as a female polymath, you were inducted into several scientific societies: the Royal Irish Academy and the Société de Physique et d'Histoire Naturelle de Genève (the Society of Physics and Natural History of Geneva) in 1834, and the Royal Astronomical Society in 1835.

Objectives

You have one main objective in the game. You want your pro-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government decides, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As Mary Somerville, you and William Frend have the joint responsibility to teach the other characters in the game how to perform calculations using mathematical tables and a slide

rule. You and Frend should schedule a meeting with the GM before GS 1, so you can learn for yourselves what you must teach. You have other responsibilities to your faction, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Responsibilities to the Pro-Babbage Faction

As a member of the pro-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. You can do this first by effectively teaching the rest of the characters in the game how to use mathematical tables and slide rules. Then, later in the game, you can help your faction by being a vocal proponent for Babbage and his calculating engines. During game sessions that do not take place at the Royal Society, you can ask good questions and make short podium speeches to help propel your faction to victory.

As a woman, you are not allowed to formally speak at the Royal Society. However, you can confer with your faction members to discuss strategy, etc.

Teaching tables and slide rules

During GS 1, sometime after the Georgian party game icebreaker, you and Frend will make a brief lecture, showing the other characters in the game how to perform calculations using mathematical tables and slide rules. During your meeting with the GM before GS 1, you will receive instructions on how to use these tools, as well as handouts you can use in your teaching. The GM will provide the slide rules and tables to use during the lecture.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are not a FRS, you should plan on making a speech supporting your faction's position during one of the sessions that take place at the Mechanic's Institute, GS 2 or 5. In your case, you should do your podium speech during GS 5, to set the stage for Adams' speech in GS 6 about the discovery of the planet Neptune (see below for more information). Coordinate with your faction leader (Herschel) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

Since you are a woman, even though you are a well-respected scientist and mathematician, you have no formal power to make decisions regarding Babbage's machines – or anything else, for that matter. However, you do have the power of persuasion; through your GS 1 lecture, and your other impromptu speaking and interactions, you should do your best to make the Difference Engine (before GS 5) and the Analytical Engine (after GS 5) appear to be an absolute necessity for the betterment of the nation and of the discipline of science in general.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 1 lecture, due before GS 1 starts. You and Frend should work together on this speech, and submit one document. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in British science during the early to mid-1800s, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding the idea of the government funding large-scale scientific and engineering projects, you are in favor of the practice. You can see how beneficial Babbage's machines would be to ensure the accuracy of mathematical calculations (via the Difference Engine producing error-free tables) and for speeding up these calculations (via the Analytical Engine).

You are an unofficial member of the last generation of the old-school men of science community, i.e., a wealthy member of high society with enough time, talent, and money to pursue your scientific interests at your leisure. Of course, as a woman, you are a very unusual member of the community. You and Ada Lovelace are exceptions to the norm at the time. Although you do not have anything against him personally, your "old school" status puts you somewhat at odds with younger men of science like John Couch Adams. Men like Adams do not have large sums of money to pursue scientific studies as a hobby, rather, they do science for a living – science is their profession as well as their passion. If you are among the last of the wealthy, amateur scientific gentlemen, Adams and his colleagues are among the first of the professional scientists. This tension will be played out in GS 4, when the next president of the Royal Society is elected. Although you will not be able to vote, you should work behind the scenes to support the candidate who you think best represents the values of old-school men of science, and who you think has the most political power.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

You have written several important articles and books regarding astronomy and mathematics. You worked on a translation and annotation of Laplace's *Mécanique Céleste* that grew to such size that John Herschel recommended that you publish your work as a book, and this work, *The Mechanism of the Heavens*, was first published in 1831. Your next work, *The Connection of the Physical Sciences*, was published in 1834. The sixth edition of this book, in 1842, influenced John Couch Adams to begin his search for the planet Neptune.

With other roles

You, along with William Frend, have served as a mathematics tutor for Ada Lovelace, as her mother, Anne Isabella Byron, attempted to prevent her from becoming too “poetical” like her father, Lord Byron. Lady Anne hoped that the power of mathematics would be strong enough to overcome any hedonistic tendencies Ada inherited from Bryon. You took Ada to one of Babbage’s Dorset Street parties, where she first met him and saw his Difference Engine fragment.

During his 1834 - 1835 term of office as Prime Minister, Sir Robert Peel awarded you a £200 yearly pension, an amazing recognition of your intellectual worth to the United Kingdom. In 1837, another PM, William Lamb, increase the amount to £300 per year. That is an amount equivalent to more than \$40,000 in 2017 dollars.

The sixth edition of your book *The Connection of the Physical Sciences*, published in 1842, was very influential on another character in the game, John Couch Adams. In your book, you noted that the planet Uranus was orbiting in a manner different from the prediction, and speculated that there might be another planet, further beyond Uranus, whose gravitational pull was causing the odd behavior. Adams read this, and, through much arduous calculation, predicted where the as-yet-unseen planet should be located. Adams will discuss the discovery of Neptune during GS 6. You should do your podium speech during GS 5, setting the stage for Adams’ speech in GS 6.

As an old-school “man” of science, you have a natural affinity for others in the same class: Babbage, Crosse, Gilbert, Herschel, Lovelace, and Sabine. Likewise, you do not have much in common with the new breed of men of science, who study science as a profession: Adams, Airy, Brunel, Buckland, Clement, Faraday, and Lardner.

With your faction

Most of the men in the game, including those in your faction, hold the idea that intellectual pursuits like astronomy and mathematics are not for women, as the frail female physiology cannot withstand such mental exertion. So, while they may respect your work, the men in your faction think that you are, in fact, harming yourself by pursuing such efforts.

Strategy advice

First, remember that due to your gender, you cannot speak formally during any session at the Royal Society. You may, however, confer privately with your faction members to help steer your faction’s strategy.

To learn more

[Fre96] William Frend. *Principles of Algebra*. G. G. and J. Robinson, London, 1796. Available via Google Books

[OR99] J. J. O'Connor and E. F. Robertson. Somerville biography. <http://www-history.mcs.st-andrews.ac.uk/Biographies/Somerville.html>, November 1999

[Pat08] Elizabeth C. Patterson. Somerville, Mary Fairfax Greig. <http://www.encyclopedia.com/science/dictionaries-thesauruses-pictures-and-press-releases/somerville-mary-fairfax-greig>, 2008

Summary of your individual victory objectives

You win the game if your faction wins. Your pro-Babbage faction wins the game, by securing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Portrait of Mary Somerville by Mary Dawson Turner (née Palgrave), from the National Portrait Gallery, St Martin's Place, London; URL: <http://www.npg.org.uk/collections/search/portrait/mw13110/Mary-Somerville?LinkID=mp04193&search=sas&sText=mary+somerville&role=sit&rNo=4>

George Biddell Airy

Anti-Babbage faction leader and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are George Biddell Airy, FRS. You are a member of the Men of Science (MOS) community, and (starting in 1835) the Astronomer Royal, but you are not in favor of the British government spending money from its treasury to support the construction of Babbage's Difference or Analytical Engines.

The following definition is important to this role: *curmudgeon (n): a person (especially an old man) who is easily annoyed or angered and who often complains.*

If your personal story took place in the United States rather than in the United Kingdom, it would likely be cited as an instance of the “rags-to-riches” American dream. Through hard work and education you have been able to significantly raise your standing in society. You were born on July 27th, 1801, in Alnwick, Northumberland, in the far north of England. Your father, William Airy, was originally a farmer, but through hard work and education, he was able to become a civil servant, specifically, a tax inspector. Unfortunately, in 1813 your father lost his job, and so you went to live with your uncle, Arthur Biddell. Biddell was an educated man with a personal library containing many books on science, which you studied intently. With your uncle's financial help, and by entering as a sizar (an undergraduate receiving financial aid from the university and obligated to perform menial servant duties in return), you were able to attend Trinity College, Cambridge starting in 1819. You were the top of your first-year class, and winner of the Smith's prize in your

final year of 1823. The Smith prize is awarded to top research students in mathematics and theoretical physics. It can be argued that your academic success was as much about your tremendous memory and organizational skills, rather than any innate mathematical gift.

Whatever the root of your success, however, it continued. After graduation, you earned a fellowship at Trinity and stayed on as a professor there. A mere three years later in 1826 you were named as the 10th Lucasian Professor of Mathematics at Cambridge. The Lucasian Chair is perhaps the most prestigious academic position that one can hold anywhere in the world; Isaac Newton held the chair for 33 years in the seventeenth century, and Stephen Hawking held it for 30 years in the twentieth. The other primary candidate for the chair was Charles Babbage; your defeat of Babbage set the stage for your ongoing conflicts with the polymath.

In the year our game begins, 1828, you have left the Lucasian Chair for another position as Plumian Professor of Astronomy at Cambridge and Director of the Cambridge Observatory. Your motivation for leaving the prestigious chair was financial: the new position, along with being a member of the Board of Longitude, brought in enough money that you were able to convince the father of your fiancé, Richarda Smith, to allow your marriage to commence. In 1835 you will become Astronomer Royal, taking you away from Cambridge and to the Royal Observatory at Greenwich. You will hold this position until your retirement in 1881.

You will win the Royal Society's Copley medal in 1831, and be selected as a Fellow of the Royal in 1836. You will eventually serve as president of the society for two years starting in 1871.

Your time as Astronomer Royal is illustrative of your personality. While there, you could not tolerate the idea of younger scientists thinking for themselves, and so the Observatory did not train any new astronomers during your tenure.

Your successor in the Lucasian Chair, selected on your departure from Cambridge in 1828, was Charles Babbage.

Objectives

You have one main objective in the game. You want your anti-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government declines, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

You can also win the game if you are an effective curmudgeon, finding, calling out, and documenting at least three bad arguments made by the pro-Babbage faction. See below for more details.

Responsibilities

As the leader of the anti-Babbage faction, your chief responsibility is to make sure that your faction is effective in preventing either of Babbage's Difference or Analytical Engines from receiving further funding from the British government. You also have a responsibility to debate the priority of the discovery of Neptune, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Importantly, you have a responsibility to be a curmudgeon, especially being on the lookout for faulty arguments and "weasel words." You are responsible for nominating your faction's candidate for the presidency of the Royal Society, the Duke of Sussex, during GS 4. Finally, every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibilities to the Anti-Babbage Faction

The members of your faction each have a responsibility to bring forward arguments for why the British Government should not fund construction of Babbage's Difference or Analytical Engine. Your job, as leader of the anti-Babbage faction, is to make sure that these arguments are made in the most effective way. You should help your faction members recognize and exploit the areas where the pro-Babbage faction's arguments are weakest. If the pro-Babbage faction seems to be persuading the indeterminates, you should rally your faction members to mount effective counter-arguments in their podium speeches. In addition, you should work closely with Buckland to develop the arguments he will present in GS 3, in support of his resolution that the Royal Society should not recommend funding for the Difference Engine.

Regarding the priority of the discovery of Neptune

During GS 6, you will debate John Couch (pronounced "cooch") Adams, regarding the discovery of Neptune in 1846. Based on calculations he'd been working on for years, Adams predicted where a yet-undiscovered planet would be found. Adams will claim that not only did you needlessly delay a telescopic search for Neptune, losing priority in the discovery to the French, but that his calculations could have been completed much more quickly had one of Babbage's Analytical Engines been available to do the computations. You must counter these arguments.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of

doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Being a curmudgeon

In keeping with your curmudgeon-ness, you are responsible for being cranky, and not liking anything. You should keep watch for “weasel words,” i.e., words or statements that are intentionally ambiguous or misleading. Depending on the context, and whether there is any quantification used, the following words might be weasel words: helps, supports, is useful, better, improved, effective, efficient, seems, appears, many, most, and so on.

You are also responsible for calling out bad arguments. For example, someone might make a straw man argument. In such an argument, the perpetrator puts forth an often extreme, easy-to-defeat argument that sounds like something his or her opponents might say, but that they have not, in fact, used as an argument. Then, the perpetrator tears apart the straw man. This makes it look like the user of the technique is soundly defeating his or her opponent, when in fact all they are doing is destroying an argument that the opponents never made at all.

Some other types of bad arguments that you might try are [Alm13]:

False dilemma “An argument that presents a limited set of two possible categories and assumes that everything in the scope of the discussion must be an element of that set.”

Slippery slope “An argument that attempts to discredit a proposition by arguing that its acceptance will undoubtedly lead to a sequence of events, one or more of which are undesirable.”

Equivocation “An argument that exploits the ambiguity of language by changing the meaning of a word during the course of an argument and using the different meanings to support and ill-founded conclusion.”

See the “To Learn More” section below for some resources you could use to learn more about bad arguments and how to make them. You can win the game if you are able to detect, publicly call out (via questions and / or podium speeches), and document at least

three bad arguments used by members of the pro-Babbage faction. If you are able to do this, you will win the game, even if your faction does not meet its victory conditions.

Nominating the Duke of Sussex for Royal Society President

During GS 4, each faction will nominate a candidate to serve as Royal Society President after Davies Gilbert. You must nominate your faction's candidate, the Duke of Sussex. Prepare a short nomination speech, in which you should try to make a convincing argument as to why Sussex, a member of the royal family but not a man of science, should lead the Royal.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Before GS 5, you are not a FRS, so you could speak at sessions not meeting at the Royal Society: GS 1 and 2. After GS 5, you could make your podium speech in any remaining session. Coordinate with your faction members so you can make a speech with the maximum impact for your faction.

Powers

As a man of science but not a member of the aristocracy by heredity, nor of the government, your official powers to make things happen are quite limited. However, you are the Astronomer Royal (as of 1835, at least), and in that capacity members of the government consult you on all sorts of scientific matters. So, your opinion carries weight. As leader of the anti-Babbage faction, your organizational skills will be an invaluable asset.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 6 speech, due before GS 6 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your

writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Finally, there is an optional assignment if you plan to use your detection of bad arguments to win the game. If you are taking this opportunity, submit to the GM, no later than the post-mortem session, a short paper describing each of the bad arguments you detected and identified: who made the bad argument? When? What was the bad argument? How did you respond? Answer these questions of each of the bad arguments you have detected. If you find and call out at least three bad arguments, and write about them in this paper, you win the game, regardless of the success or failure of your faction.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British science and mathematics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

You are not an “old-school” MOS. You are not a landed gentleman, and so you do not have large reserves of money to allow you to pursue science as a hobby. You earn your living by pursuing the study of the natural world. Therefore, you believe it is appropriate and proper to be a professional man of science, rather than simply a wealthy, talented amateur. This puts you somewhat at odds with your duty to nominate Sussex for the Royal Society presidency; your tendencies would probably lead you to prefer a more science-minded candidate. But, frankly, you will do anything you can to frustrate Babbage, and so, you will argue vigorously for Sussex over the pro-Babbage candidate.

As for the conflict between imagination and reason, you are firmly aligned with William King on the supremacy of reason, and opposed to Byron’s Ghost, the champion of imagination.

With texts

You have published extensively, with over 500 papers and 11 books, such as *Trigonometry* (1825) and *Gravitation* (1834), to your name. Engineering, sometimes characterized as applied science, is one of your many interests outside of astronomy and mathematics. In fact, you were able to use your engineering skills to renovate many of the instruments at

Greenwich. Also, you tend towards applied mathematics rather than theoretical, really seeing no use in the study of mathematics for its own sake.

With other roles

Ever since your days as a schoolboy, you have been known, frankly, as something of a snob. As an adult, you have a sarcastic manner and are known as a harsh disciplinarian to the employees of the Royal Observatory. As for roles in this game, you have clashed repeatedly with Charles Babbage. Aside from the hard feelings between you regarding the Lucasian Chair, your first public disagreement came in 1832 when you clashed over a telescope. In 1842 you made your opinion known regarding Babbage's Difference Engine: "worthless." Later, in 1854, you supported the narrow gauge railway track system while Babbage supported the wide gauge tracks used by his friend I.K. Brunel.

Your status as a professional man of science means you have little in common with the "old school" men of science: Babbage, Crosse, Gilbert, Herschel, Lovelace, Sabine, and Somerville. On the other hand, the following roles in the game are also pursuing the study of engineering or science professionally: Adams, Brunel, Buckland, Clement, Faraday, and Lardner.

With your faction

As an organizer and strategist, you are very important to your faction. All of your faction members look up to you for your intelligence and scientific accomplishments. You are convinced Babbage's engines are worthless, so you should work diligently to try to deny funding for the projects.

Strategy advice

It would be wise to have out-of-class meetings with your faction members – either in person or via some electronic means – before each game session, to coordinate everyone's efforts. If the gamemaster requires the text of speeches to be published well in advance of the session they will be presented in, you should make sure your faction members read the speeches of the pro-Babbage faction in advance, and then prepare counter arguments that can be presented via podium speeches.

To learn more

[Alm13] Ali Almossawi. *An Illustrated Book of Bad Arguments*. The Experiment, New York, NY, 2013

[LB86] John E. Littlewood and Béla Bollobás. *Littlewood's Miscellany*. Cambridge University Press, 1986

Summary of your individual victory objectives

You win the game if your faction wins. Your anti-Babbage faction wins the game, by preventing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

You can also win the game if you are able to detect, call out, and document at least three bad arguments used by the pro-Babbage faction. If you do this, you win the game apart from the success of your faction.

William Buckland DD FRS

Anti-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are the Reverend Doctor William Buckland, DD FRS, a clergyman, theologian, geologist, paleontologist, and occasional opportunistic cannibal. You are a Canon of Christ Church, Oxford, Fellow of the Royal Society of London, and a member of the Men of Science (MOS) community.

You were born on March 12, 1784, in Axminster, Devon, in the southwest of England, to the Anglican clergyman Charles Buckland and his wife Elizabeth. You may have acquired your interest in fossils early in your childhood, as you accompanied your father on countryside walks past the quarries near Axminster. The work in the quarries exposed many fossils, and you collected them.

In 1801 you were awarded a scholarship to attend Corpus Christi College, Oxford, where you were to study for the ministry. By 1808, you earned your Bachelor's degree, your Master's degree, were ordained as an Anglican priest, and were named as a Fellow (i.e., a professor) at Corpus Christi College. Also during this period, you explored widely in the United Kingdom, touring portions of England, Scotland, Wales, and Ireland in search of geological formations and fossils. In 1813 you were appointed reader in mineralogy at Corpus Christi, replacing one of your former professors, John Kidd, when he retired. In this capacity you lectured on geology and paleontology, and you were good at it. Your lectures became very popular.

One of your major areas of investigation dealt with the biblical account of the flood, and its connection to geological formations and fossils. At the time, the common thought about fossils was that they were the remains of creatures killed during the flood. This idea, however, was becoming harder and harder to reconcile as men of science learned more about geology. First, it was starting to become apparent that the Earth was much, much older than the literal interpretation of the biblical creation account. Secondly, you realized that the duration of the flood would not have been long enough to lay down all the rock strata where fossils were being found. For these reasons, you formed your catastrophism theory, otherwise known as “gap creationism.” In this theory, an indeterminate amount of time had elapsed since God created the universe, and there had been several sequential creations and extinctions of the creatures of the Earth.

You were named a Fellow of the Royal Society of London in 1818, and in 1822 you were awarded the Royal’s Copley Medal for your work on the fossilized hyenas you investigated in Kirkland Cave, Yorkshire.

In 1824, you announced the discovery of a *Megalosaurus* (“great lizard”) fossil, and thus published the first account of a dinosaur.

In 1825, you were made Canon (priest) of Christ Church, Oxford, in reward for your intellectual accomplishments.

You are somewhat – more that somewhat, actually – eccentric. You often do your field work, searching for fossils, while wearing your full academic regalia. You are known to lecture to your classes from horseback. When you are not on horseback, you enliven your lectures by enacting the movements of the dinosaurs you are so familiar with. The pinnacle of your eccentricities is zoöphagy, however. Zoöphagy simply means the tendency to feed on animals. You take this concept to the extreme, claiming to have eaten at least one bite of every animal! Your least favorite were the mole and the bluebottle fly. Your taste for literally all kinds of meat goes farther yet. According to Augustus Hare, whilst at a dinner party in France,

“Talk of strange relics led to mention of the heart of a French King [Louis XIV] preserved at Nuneham in a silver casket. Dr. Buckland, whilst looking at it, exclaimed, ‘I have eaten many strange things, but have never eaten the heart of a king before’, and, before anyone could hinder him, he had gobbled it up, and the precious relic was lost for ever.” [BK12]

Objectives

You have one main objective in the game. You want your anti-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government declines, on both of the opportunities, to fund the development of Babbage’s Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As William Buckland, you have a responsibility to introduce a resolution to the Royal Society that does not support the funding of Babbage's Difference Engine, responsibilities to your anti-Babbage faction, a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game, and a secret responsibility to casually exhibit your zoöphagy. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Regarding an anti-Babbage resolution

During GS 3, the Royal Society will debate whether or not to recommend that the government fund the development of Babbage's Difference Engine. You are to introduce a resolution saying that the Royal does not recommend that the government fund Babbage. Your speech should explain why you think this is appropriate; work with your faction (in particular with your faction leader, George Biddell Airy) to make sure you include all of your faction's arguments against funding Babbage's device.

Responsibilities to the Anti-Babbage Faction

As a member of the anti-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. During the game, you can help your faction by being a vocal opponent of Babbage and his calculating engines. You can ask good questions and make short podium speeches to help propel your faction to victory.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the

terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Responsibility relative to your zoöphagy

Secret: To play off your zo'ophagy, you should bring animal crackers, gummy worms, or other animal-shaped snacks to each game session. You should nonchalantly eat the snacks during the game sessions, without drawing overt attention to the practice.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are a FRS, you should plan on making a speech supporting your faction's position during one of the other sessions that take place at the Royal Society, GS 4 or 6. Coordinate with your faction leader (Airy) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

As a man of science but not a member of the aristocracy by heredity, nor of the government, your official powers to make things happen are fairly limited. However, you are an influential member of the Royal Society, and in that capacity members of the government consult you on all sorts of scientific matters. So, your opinion carries weight.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 3 speech, due before GS 3 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions

should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British science and mathematics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

You are not an “old-school” MOS. You are not a landed gentleman, and so you do not have large reserves of money to allow you to pursue science as a hobby. You earn your living by pursuing the study of the natural world. This informs your opinions on the major issues in the game. You believe it is appropriate and proper to be a professional man of science, rather than simply a wealthy, talented amateur. This means that when the next president of the Royal Society is chosen in GS 4, you may be persuaded to vote for the candidate who you think is most likely to favor those in the new generation of men of science, regardless of which faction the candidate belongs to.

Given your eccentricities, when it comes to the conflict between reason and imagination, you are more aligned with Byron’s Ghost, the champion of imagination, rather than William King, who believes in the supremacy of reason.

With texts

You are an influential lecturer and paleontologist, and also an influential writer. You are the author of one of the *Bridgewater Treatises*, an influential work on natural theology (knowledge of God based on observed facts and experience, apart from divine revelation). Your treatise, *Geology and Mineralogy Considered with Reference to Natural Theology*, is actually the most scientific of the eight treatises in the work.

With other roles

You owe a debt of gratitude to your fellow anti-Babbage faction member, Davies Gilbert PRS. After the Earl of Bridgewater died in 1829, his will instructed that the President of The Royal Society should be endowed with £8,000 to select authors to publish 1,000 copies of a work “On the Power, Wisdom, and Goodness of God, as manifested in the Creation.” With the help of the Archbishop of Canterbury, Gilbert selected you as one of the eight authors of the *Bridgewater Treatises*.

You are not an old school man of science, and this puts you somewhat at odds with those that are: Babbage, Crosse, Gilbert (mitigated by the above), Herschel, Lovelace, Sabine, and Somerville. On the other hand, the following are also pursuing science or engineering as a profession: Adams, Airy, Brunel, Clement, Faraday, and Lardner.

With your faction

You have a key role in the game, from your faction's point of view: you make the final, summary argument as to why the government should not fund the Difference Engine. Despite your class and religious differences, work to be cordial and professional with your faction mates, so you can effectively coordinate with them to make an effective GS 3 resolution and speech.

Strategy advice

Pay careful attention to the arguments that your faction makes against of the Difference Engine during the first two game sessions, and also the arguments that the pro-Babbage faction makes. Try to gauge which arguments have made the most impact on the indeterminates. Then, in your resolution and speech during GS 3, try to amplify your faction's best arguments, while refuting the most effective arguments of the pro-Babbage faction.

You might also wish to begin your GS 3 speech with a prayer. Ask the assembly to rise and join you. This will put the pro-Babbage faction member who will introduce a pro-Babbage resolution, Michael Faraday, in an uncomfortable position. He is a member of the Sandemanian sect of Christianity, which prevents him from praying with members of other denominations.

To learn more

[BK12] Aaron Birk and Robert Krulwich. Who Wants To Eat Jellyfish Omelets? Dolphin Meatballs? Mouse-On-Toast? These Guys. <http://www.npr.org/sections/krulwich/2012/09/27/161874316/who-wants-to-eat-jellyfish-omelettes-dolphin-meatballs-mouse-on-toast-these-guys>, September 2012

[Bro66] W. H. Brock. The Selection of the Authors of the Bridgewater Treatises. *Notes and Records of the Royal Society of London*, 21(2):162–179, 1966

Summary of your individual victory objectives

You win the game if your faction wins. Your anti-Babbage faction wins the game, by preventing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number

of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Portrait of Revered William Buckland DD FRS, by Thomas Phillips, from the National Portrait Gallery, St Martin's Place, London; <http://www.npg.org.uk/collections/search/portraitLarge/mw00890/William-Buckland?LinkID=mp00625&search=sas&sText=william+buckland&role=sit&rNo=0>

Joseph Clement

Pro-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Joseph Clement, an extremely capable toolmaker, draftsman, engineer, and industrialist. In the mid-1820s, you are hired by Charles Babbage to construct his Difference Engine. You are a member of the Men of Science MOS community. Since Babbage hired you to work on his Difference Engine, you begin the game as a member of the pro-Babbage faction.

Secret: However, in 1833 (i.e., starting in GS 5) you will transition to being member of the anti-Babbage faction. *The fact that you will switch between factions is not public knowledge, and should not be shared with anyone else!*

You were born in 1779 in Great Ashby, Westmoreland County, in the northwest part of England. Your father was a hand-loom weaver, but for a working man he was remarkably intelligent, curious, and well-versed in natural philosophy. He loved to study the local insect life around your house. He also was mechanically gifted, having built his own lathe, which he used to create his own mechanical devices. You have inherited these traits from your father, and they have served you well.

You have not had much in the way of formal education beyond basic instruction in reading and writing at the local school. You have, however, been able to teach yourself very effectively. Your father intended you to follow him as a weaver, but hand-looms were falling

out of use due to the introduction of the Jacquard Loom, and so you tried your hand at several different careers. You first worked as a thatcher (one who installs and repairs the thatched roofs of buildings) and then as a slater (one who installs and repairs slate tiles on the sides of buildings). Since both of these trades were seasonal, and since you inherited your father's curiosity, you spent much of your off time with the local blacksmith, learning to forge and otherwise work with metal. A cousin loaned you some books on mechanics, which inspired you to learn to build tools and mechanical devices. Soon thereafter you created your own lathe. You used the lathe to create musical instruments such as flutes and bagpipes, a microscope for your father, and a refracting telescope.

Giving up slating, you next found work in a nearby factory making Jacquard Looms, then the same occupation in another factory in Carlisle, then on to more varied mechanical work in Glasgow. In Glasgow you learned the trade of draftsmanship from Peter Nicholson. After a time in Glasgow you moved to Aberdeen where your skills and reputation grew. Ultimately in 1813 you moved to London, where you found employment in Alexander Galloway's mechanical shop. From there you went to work in John Bramah's shop (and just in time, for the ever perilous cast-iron roof on Galloway's shop soon after collapsed, killing nine men). In 1817 your reputation had grown enough for you to set up your own successful mechanical engineering shop in London.

Your reputation for quality work and inventiveness are such that, whenever someone has an idea for something to be mechanized, they seek you out to build it. This led Babbage to employ you as the only engineer capable of building his Difference Engine.

Objectives

You have two main objectives in the game. First, you have to spend the first four game sessions as a convincing member of the pro-Babbage faction. Then, starting in GS 5, you become a member of the anti-Babbage faction, and you will give a speech detailing your reasons for switching sides. Your switch should be a surprise to everyone, so you must fit in with the pro-Babbage faction until your defection.

Secondly, you want your ultimate faction, the anti-Babbage faction, to win the game; if the faction wins, you win. Your faction wins if the British government declines, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As Joseph Clement, you have a responsibility to your anti-Babbage faction, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibilities to the Anti-Babbage Faction

Your chief responsibility to the anti-Babbage is when you switch factions at the beginning of GS 5. At the beginning of the session, you will make a speech detailing why you have left the employ of Charles Babbage. Then you will become a member of the anti-Babbage faction, earning a PC for the faction in the process.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are not a FRS, you should plan on making a speech supporting your faction’s position during one of the sessions that take place outside the Royal Society, GS 1 or 2. Coordinate with your current faction leader (Herschel) and the other members of your current faction so you can make a speech with the maximum impact for your faction.

Powers

Since you are a member of the working class, even though you are a well-respected engineer, you have no formal power to make decisions regarding Babbage’s machines – or anything else, for that matter. However, you are an important element of the game! Firstly, your defection to the anti-Babbage faction during GS 5 earns that faction a PC, which is very

valuable. Secondly, your speech explaining why you left Babbage’s employ, if done well, can be very convincing, showing that Babbage’s refusal to finalize the specifications for the Difference Engine, and the difficulty of machining each of the parts of the machine, would result in a “money-pit” project for the government.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage’s engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 5 speech, due before GS 5 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British science and mathematics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government funding of Babbage’s engines, you don’t have an issue with the concept in abstract terms. However, in this particular case, your opinion is that the government has already spent enough money on what is likely to be a doomed project. Of course, this opinion should be kept to yourself until your GS 5 speech.

You are not an “old-school” MOS. You are a working man, not a landed gentleman, and so you do not have large reserves of money to allow you to pursue science as a hobby. You

earn your living by making things. Clearly, you believe it is appropriate and proper to be a professional man of science, rather than simply a wealthy, talented amateur. So, during the election of the Royal Society president in GS 4, you will favor the candidate who you think is most likely to favor those in the new generation of men of science, regardless of which faction the candidate belongs to. Of course, you cannot vote in this election, since you are not a FRS, but you can work behind the scenes to try to ensure the election of your favorite candidate.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

Engineering was very much a working-man's occupation in the time frame of this game, and as such, engineers did not write papers or books. Unlike the mathematicians and scientists in the game (Airy, Buckland, Frend, and even Lovelace and Somerville), you have not written any publications on your work.

With other roles

As time went on, your relationship with Charles Babbage became more and more contentious. For your part, you thought his constant changing of the specifications of the Difference Engine was causing you and your team to constantly redo work, leading to cost overruns. Babbage, on the other hand, feels you have overcharged him for work on the engine.

Since you are not an old school man of science, and a commoner, you have little in common with the traditional men of science in the game: Babbage, Crosse, Gilbert, Herschel, Lovelace, Sabine, and Somerville. Although you do not share their social status, you have more in common with the other characters who are pursuing science and engineering professionally: Adams, Airy, Brunel, Buckland, Faraday, and Lardner.

With your faction

You have a very interesting relationship here, since you will be part of both factions during the game. You must be an enthusiastic supporter of Babbage early in the game; note your podium speech will be pro-Babbage. Your defection will earn a PC for the anti-Babbage faction, making you a valuable asset to them.

Strategy advice

You may wish to pay careful attention to the issues the pro-Babbage faction thinks it is weak on during the first four game sessions. Then, after your defection, communicate with your faction leader (Gilbert) and the rest of the faction to try to capitalize on these issues.

To learn more

[Roe16] Joseph Wickham Roe. *English and American Tool Builders*. Yale University Press, New Haven, CT, 1916. Available via Google Books

[Smi64] Samuel Smiles. *Industrial Biography: Iron-Workers and Tool-Makers*. Ticknor and Fields, Boston, MA, 1864. Available via Google Books

Summary of your individual victory objectives

You win the game if your faction wins. Your anti-Babbage faction wins the game, by preventing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

William Frend

Anti-Babbage faction member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are William Frend, unitarian clergyman, political radical, and author of an Algebra textbook, *Principles of Algebra* [Fre96]. You have been one of Ada Lovelace's mathematics tutors.

You are one of the oldest characters in this game. You were born on November 22, 1757, in Canterbury in southeast England. Your father, George Frend, was a businessman and mayor of the city. He intended that you would follow in his footsteps as a trader, and so he sent you to be educated for that profession. You studied at the King's School in Canterbury, and also abroad in France and in Quebec. However, on your return from North America, you felt called to be ordained in the Church of England, and so you enrolled at Christ's College, Cambridge, to prepare for the ministry. While there you also demonstrated significant mathematical talent, graduating as second wrangler (second-best mathematics undergraduate) and winning the Smith's prize in mathematics. You became a priest in the Church of England in 1783.

However, you did not remain in the church for long. In 1787 you left the Church of England to become a Unitarian. This was problematical, as nonconformists were discriminated against, in both official and unofficial ways, at this time in the UK. For example, you were working towards a Master's degree at Cambridge, but nonconformists were barred from being candidates for the degree. You were relieved of your duties as a tutor at the university in 1788.

You took your disagreements with the Church of England to a new level in 1792, when you published *Peace and Union recommended to the Associated Bodies of Republicans and Anti-republicans*, a book that was harshly critical of church. Due to the book, you were put on trial before the university senate, convicted, and banished from the university in 1793.

After Cambridge, you moved to London, where you supported yourself by teaching and by writing. In 1796 you published your textbook on algebra, *Principles of Algebra*. During this time you also served as a private tutor to several influential people, including Ada Lovelace.

You are politically liberal, perhaps even radical, and a noted campaigner for the reform of the poor laws, for increased taxation of the rich and a progressive income tax, and against slavery, among other causes.

Objectives

You have one main objective in the game. You want your anti-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government declines, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As William Frend, you and Mary Somerville have the joint responsibility to teach the other characters in the game how to perform calculations using mathematical tables and a slide rule. You and Somerville should schedule a meeting with the GM before GS 1, so you can learn for yourselves what you must teach. You also have general responsibilities to your anti-Babbage faction. In addition, every character in the game has a responsibility to make at least one additional brief podium speech during open debate periods.

Teaching tables and slide rules

During GS 1, sometime after the Georgian party game icebreaker, you and Somerville will make a brief lecture, showing the other characters in the game how to perform calculations using mathematical tables and slide rules. During your meeting with the GM before GS 1, you will receive instructions on how to use these tools, as well as handouts you can use in your teaching. The GM will provide the slide rules and tables to use during the lecture.

Responsibilities to the Anti-Babbage Faction

As a member of the anti-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. Even after your GS 1 lecture, you can help your faction by being a vocal opponent of Babbage and his calculating engines. You can ask good questions and make short podium speeches to help propel your faction to victory.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are not a FRS, you should plan on making a speech supporting your faction's position during one of the sessions that take place at the Mechanic's Institute, GS 2 or 5. Coordinate with your faction leader (Airy) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

Since you are not a member of the nobility, nor the government, and you are a nonconformist, you have no formal power to make decisions regarding Babbage's machines – or anything else, for that matter. You can still win the game, however! You will have to collaborate with your faction to help steer your faction strategy towards a victory.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 1 lecture, due before GS 1 starts. You and Somerville should work together on this speech, and submit one document. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British science and mathematics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government support for large-scale scientific projects such as Babbage's computing machines, you are not in favor of the practice. You feel that the government should focus more on aiding the poor and the unfortunate, rather than giving money to wealthy gentlemen so they can pursue their scientific hobbies. (This idea might inform the content of your podium speech.)

You are more of a radical political figure during this game than a true member of the men of science (MOS) community, so the more complex points of contention regarding Babbage's machines may escape you. This also means you do not have a personally motivated opinion about what a man of science should be, i.e., should he be a wealthy, landed gentleman with enough time, talent, and wealth to pursue the study of nature as a hobby? Or, should he be a professional, an intelligent working man who earns his living by doing science? If you had to choose, your liberal leanings might lean you towards the latter opinion. This tension will be played out in GS 4, when the next president of the Royal Society is elected. You will not be able to vote in this election, but, once you've decided which candidate you will support, you can work behind the scenes to get him elected.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

As the author of *Principles of Algebra*, you have had an influence (for better or worse!) on many mathematicians in Britain. You are also widely published on what we would today term social justice issues.

With other roles

Ada Lovelace's mother, Lady Anne Isabella Byron, hired you as a mathematics tutor for her daughter. Lady Byron feared that Ada would take after her father, the poet Lord Byron;

that Ada might become “poetical,” shorthand for hedonistic and promiscuous. Lady Byron hoped to combat poetry with its natural enemy, mathematics, and that is where you enter the picture.

Even though you published an influential mathematics text, you do not have much in common with the men of science in this game. This gives you a natural kinship with other non-scientific types: Sussex, Grey, Peel, Robinson, Russell, Wellesley, and Swing. Politically, you lean closest to the Whig MPs, Grey and Russell; the Tories (Peel, Robinson, and Wellesley) find your politics abhorrent.

With your faction

Although you are not a member of the nobility or the government, you are a valuable member to your faction. After your GS 1 speech, you can earn PCs for your faction by paying careful attention to the pro-Babbage arguments, asking questions that expose weaknesses, and making effective podium speeches to promote your faction.

Strategy advice

Work diligently with Somerville to produce an effective lecture on mathematical tables and slide rules, so that the modern-day students playing the game understand a little of what mathematics was like in the early 1800s.

To learn more

[Fre92] William Frend. *Peace and Union recommended to the associated bodies of Republicans and Anti-Republicans*. P. C. Croft, 1792

[Fre93] William Frend. *An account of the proceedings in the University of Cambridge, against William Frend, M.A. Fellow of Jesus College, Cambridge, for publishing a pamphlet, intitled Peace and union, &c: Containing the proceedings in Jesus College, the trial in the Vice-Chancellor’s Court, and in the Court of Delegates*. Printed by B. Flower, 1793

[Fre96] William Frend. *Principles of Algebra*. G. G. and J. Robinson, London, 1796. Available via Google Books

[Gun54] Henry Gunning. *Reminiscences of the University, Town, and County of Cambridge from the Year 1780*. George Bell, 1854. Available via Google Books

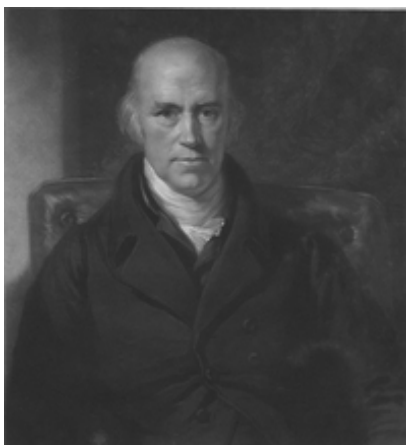
Summary of your individual victory objectives

You win the game if your faction wins. Your anti-Babbage faction wins the game, by preventing government funding for the development of Babbage’s engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number

of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Davies Gilbert PRS

Royal Society president (PRS), Anti-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Davies Gilbert PRS, British botanist, geologist, mathematician, civil servant, politician, author, and from 1827 through 1830, president of the Royal Society of London. You are also well known for publishing collections of Christmas Carols. You were born Davies Giddy on March 6, 1767, in Penzance, Cornwall county in the extreme southwest of England. Your father was the Reverend Edward Giddy, curate (assistant to the vicar or priest) of St. Erth church there, and your mother was Catherine Davies. You received a good education at home from your father, at the local grammar school, and from the Reverend Malachy Hitchins, computer and comparer at the Greenwich Royal Observatory. (“Computer” was a job description at this time in history; it was one who did mathematical calculations for a living.) Hitchins recommended you to for entry into mathematician Benjamin Donn’s Mathematical Academy. After the Academy, in 1786 you went up to Pembroke College, Oxford; you graduated with a Master of Arts degree in 1789.

You married Mary Ann Gilbert in 1808, and changed your surname from Giddy to Gilbert in 1817, so that you could inherit the estate of your wife’s father Thomas Gilbert, who had no male heir.

Your civil service career began in 1792, as you served as the High Sheriff of Cornwall county for two years. Then, in 1804, you were elected to parliament, serving as MP for two Cornwall boroughs, first for Heston from 1804 through 1806, and then for Bodmin from 1806 through 1832.

You were elected as a Fellow of the Royal Society of London on November 17, 1791. As our game starts, you are president of the society, having succeeded Sir Humphry Davy at the post in 1827.

Objectives

You have one main objective in the game. You want your anti-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government declines, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As Davies Gilbert, you have a responsibility to conduct several meetings of the Royal Society of London, a responsibility to make a speech about the history and accomplishments of the Society, responsibilities to your anti-Babbage faction, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to conduct Royal Society meetings

As president of the Royal Society, you are responsible for conducting the game sessions that occur at the Society, during game sessions 3 and 4. As the presiding officer, you are required to make sure that the scheduled agenda items – speeches, resolutions, etc. – all happen on schedule. The activities for GS 3 and 4, and their order, are:

- GS 3: Recommendation regarding the Difference Engine
 - Social time. This is a brief period of time for general communication between characters, and for the factions to confer and strategize.
 - PM speech by the current PM. He will outline the priorities and goals his government will have while he is in office
 - Buckland introduces a resolution against funding the Difference Engine
 - Faraday introduces a resolution supporting funding the Difference Engine
 - Open debate regarding which resolution to support

- FRS voting on the resolutions
- Nudges. Individuals, factions, and the PM spend nudges, adding them to one of two stacks: fund or do not fund the Difference Engine
- The PM will make his decision as to the government funding the Difference Engine, in accordance with the number of nudges in each stack
- At the end of the session the gamemaster will conduct the persuasiveness metric poll of the indeterminates
- GS 4: Celebrating the Royal Society
 - Social time. This is a brief period of time for general communication between characters, and for the factions to confer and strategize.
 - PM speech by the current PM. He will outline the priorities and goals his government will have while he is in office
 - Your speech about the history and accomplishments of the Royal Society. See below for more details regarding your speech.
 - Charles Babbage will speak about the Royal Society.
 - Nominations for the next president of the Royal Society
 - Debate regarding presidential nominees
 - Royal Society presidential election
 - At the end of the session the gamemaster will conduct the persuasiveness metric poll of the indeterminates.

During any non-scheduled time in the session, the *podium rule* is in effect. The podium rule states that any character wishing to speak should approach the podium and wait to be recognized by the presiding officer (you). You must make sure that you are fair to anyone who wishes to speak, even those that are not in your faction. Characters are encouraged to ask questions of presenters, at the discretion of the presiding officer, but they should not make speeches to the class without approaching the podium.

An extra complexity for you as the presiding officer for Royal Society meetings is that only members with FRS status are allowed to publicly speak during sessions that take place at the Society. See the table in Part Three of the gamebook for details on who is a FRS, and when they achieve this status. Characters that are not FRS may confer with their factions and other characters, but they are not allowed to ask questions of presenters nor speak from the podium.

Speak with the GM before GS 3 about these sessions at the Royal Society, so that you are well prepared for the conduct of the sessions.

Responsibility to celebrate the Royal Society

At the beginning of GS 4, you will make a speech in which you will celebrate the Royal Society. You will cover the history of the society and highlight the tremendous advancements in science that have been sponsored by the Society. You know there are rumblings of discontent in the Society lately, with certain members feeling that it has become more of a wealthy gentleman's club rather than an institution devoted to the advancement of society. So, you should make sure to include recent achievements due to the society as well as those further in the past.

Responsibilities to the Anti-Babbage Faction

As a member of the anti-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. In the game sessions before and after your GS 4 speech, you can help your faction by being a vocal opponent of Babbage and his calculating engines. You can ask good questions and make short podium speeches to help propel your faction to victory.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are a FRS, you should plan on making a speech supporting your faction's position during the later session that takes place at the Royal Society, GS 6. Coordinate with your faction leader (Airy) and other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

As a member of the government and the president of the Royal Society until GS 5, you are a very powerful individual in the game. In particular, as Royal Society president, you have the power to shape the agenda of the Society's meetings. To represent your political power, you begin the game with a nudge. You may use the nudge as you wish, spending it in GS 3 or 6, at your discretion.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 4 speech, due before GS 4 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British politics, science, and mathematics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding the idea of the government funding large-scale scientific and engineering projects, you are not in favor of the practice. Generally speaking, you are very conservative and are not in favor of government spending beyond what is absolutely necessary. For example, you have spoken in Parliament against the mass education of the poor.

You are a member of the last generation of the old-school men of science community, i.e., a wealthy gentleman with enough time, talent, and money to pursue your intellectual interests at your leisure. Although you do not have anything against him personally, this puts you somewhat at odds with younger men of science like John Couch Adams. Men like Adams do not have large sums of money to pursue scientific studies as a hobby, rather, they do science for a living – science is their profession as well as their passion. If you are among the last of the wealthy, amateur scientific gentlemen, Adams and his colleagues are among the first of the professional scientists. This tension will be played out in GS 4, when the next president of the Royal Society is elected. You should support the candidate who you think best represents the values of old-school men of science, and who you think has the most political power.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

Although you are a well-connected member of the MOS community, and are a competent natural philosopher yourself, you do not have much of a publication record in science. You were, however, responsible for selecting the authors of the influential *Bridgewater Treatises* (see below).

With other roles

As president of the Royal Society, you have had the responsibility of choosing the authors for the eight Bridgewater Treatises, a collection endowed with £8,000 by the late Earl of Bridgewater. Bridgewater's will directed the president of the Royal Society should select authors to:

“...write, print, and publish one thousand copies of a work On the Power, Wisdom, and Goodness of God, as manifested in the Creation; illustrating such work by all reason able arguments, as for instance the variety and formation of God's creatures in the animal, vegetable, and mineral kingdoms; the effect of digestion, and thereby of conversion; the construction of the hand of man, and an infinite variety of other arguments; as also by discoveries ancient and modern, in arts, sciences, and the whole extent of literature.” [Bri36]

You selected one of your faction members, William Buckland, to author one of the volumes, *Geology and Mineralogy considered with reference to Natural Theology*, published in 1836.

In 1830, you evaluated several designs for a proposed bridge over the River Avon in Bristol, and from among those designs, you selected the suspension bridge design proposed by I.

K. Brunel. Construction was started in 1831, but riots in Bristol caused the project to be abandoned. Work was not started again until 1864.

With your faction

Your political power and position as PRS make you a very valuable member of your faction. You can earn PCs for your faction by paying careful attention to the pro-Babbage arguments, asking questions that expose weaknesses, and making effective podium speeches to promote your faction.

Strategy advice

Pay careful attention to the arguments that your faction makes against of the Difference Engine during the first few game sessions, and also the arguments that the pro-Babbage faction makes. Try to gauge which arguments have made the most impact on the indeterminates. Then, in your questions and podium speech(es), try to amplify your faction's best arguments, while refuting the most effective arguments of the pro-Babbage faction.

To learn more

[Bri36] Bridgewater, F.H.E. *The Bridgewater Treatises on the Power, Wisdom and Goodness of God, as Manifested in the Creation. Treatise I-VIII.: Geology and mineralogy considered with reference to natural theology, by William Buckland. 2d ed.* The Bridgewater Treatises on the Power, Wisdom and Goodness of God, as Manifested in the Creation. Treatise I-VIII. W. Pickering, 1836. Available via Google Books

[Bro66] W. H. Brock. The Selection of the Authors of the Bridgewater Treatises. *Notes and Records of the Royal Society of London*, 21(2):162–179, 1966

[Mil04] David Phillip Miller. Oxford DNB article: Gilbert, Davies. <http://www.oxforddnb.com/view/article/10686>, 2004

Summary of your individual victory objectives

You win the game if your faction wins. Your anti-Babbage faction wins the game, by preventing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Portrait of Davies Gilbert, PRS, by Samuel Cousins, after Henry Howard messotint, from the National Portrait Gallery, St Martins Place, London; URL: <http://www.npg.org.uk/collections/search/portraitLarge/mw36987/Davies-Gilbert?LinkID=mp01770&search=sas&sText=davies+gilbert&role=sit&rNo=6>

Edward Sabine FRS

Anti-Babbage faction and MOS community member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Edward Sabine, FRS, Irish military man, geologist, and astronomer. You served as the on-board astronomer on expeditions to attempt to discover the Northwest Passage under Sir John Ross in 1818 and again under Sir William Edward Parry in 1819 and 1820.

You were born on October 14, 1788, in Dublin, Ireland to Joseph Sabine and his wife Sarah Hunt. Your family was prominent in Ireland, and your elder brother, Joseph, was a barrister who also dabbled in natural science; he was elected as a Fellow of the Royal Society in 1799. You did primary schooling west of London in Marlow, England, and in 1803 entered the Royal Military Academy in Woolwich (now a part of London). In December of that year, you were commissioned as a 2nd lieutenant in the Royal Artillery, and by 1813 you had achieved the rank of captain.

You saw combat during the War of 1812. In May 1813, you set sail for Halifax, Nova Scotia aboard the packet ship Manchester. On the voyage, Manchester was attacked by the American privateer Yorktown. You distinguished yourself admirably during the battle, but ultimately the Manchester was captured and you were taken prisoner. You were held for a month before being released; after that, you continued to Halifax and then on to Quebec. You were placed in charge of an outpost there and fought against the Americans as they tried to invade Canada. The war ended in 1815, and you returned to England in

1816. Arriving there, you were classified as a *supernumerary* , i.e., an officer in excess of the needs of the army, and so you were allowed to pursue your scientific interests. Your main interests were magnetism, astronomy, and ornithology. In 1818, you were elected as a Fellow of the Royal Society (FRS).

Based on a recommendation from the Royal Society, you were named as the astronomer for John Ross' expedition to discover the Northwest Passage in 1818. You made extensive observations at different latitudes of the period of a swinging pendulum; this data allowed scientists to improve their understanding of the shape of the earth. You also performed numerous experiments measuring the earth's magnetic field and observed and cataloged the birds seen on the voyage. Ross apparently saw a mirage that he interpreted as snow-capped mountains at the end of Lancaster Sound in Canada, and so he determined that there could be no Northwest Passage. He had the expedition return to England against the advice of his junior officers, including you. To make matters worse, upon the expedition's return, the nephew of the expedition leader, John Clark Ross, who had assisted you in your scientific measurements during the voyage, claimed credit for your discoveries. It was years before you received the credit due you for the expedition.

Your second Arctic expedition sailed in May 1819, under the command of William Edward Parry. You performed more pendulum experiments during this voyage, and your work was so impressive that you received the Royal Society's highest award, the Copley Medal, in 1821 for the work.

In 1827 you were granted a leave of absence from the military, to serve as one of the secretaries of the Royal Society. In 1828 you were one of the three men of science named, based on the Society's recommendation, as advisers to the Admiralty. The other men chosen were Michael Faraday and Thomas Young.

Objectives

You have one main objective in the game. You want your anti-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government declines, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As Edward Sabine, you will give a brief lecture during GS 2 on the type of mathematics that might be involved in navigation. Lardner will help you by providing the mathematical tables that will be needed to solve the problem. In addition, you have responsibilities to your anti-Babbage faction, and a secret language responsibility you share with the other scientists, engineers, and mathematicians in the game. Every character in the game also

has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and to learn how to solve the mathematical problem you will teach during your GS 2 lecture.

Mathematics lecture responsibility

During GS 2, you will teach the other players in the game how to solve a slightly more complicated problem than the ones they will learn how to solve in GS 1. The GM will provide you with a handout that you can use during your lecture, and show you how to solve the problem so that you may then teach the rest of the characters. Your solution will require the use of a table of tangents and of a table of square roots; Lardner will provide these tables for the group to use.

Responsibilities to the Anti-Babbage Faction

As a member of the anti-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. During the game, you can help your faction by being a vocal opponent of Babbage and his calculating engines. You can ask good questions and make short podium speeches to help propel your faction to victory.

Regarding the secret language of engineering, mathematics, and science

Secret: To simulate the fact that many of the men making the decisions regarding the future of the Difference Engine did not understand the machine, what it was capable of doing, or its future potential, the members of the MOS community in this game share a limited secret language.

Here is how it works: at least once per speech, members of the MOS community (such as yourself) are obligated to include one of the secret terms in the presentation, in as natural a manner as possible. (The secret terms are meaningless, but they – hopefully – sound plausible.) When MOS members of the same faction who are not speaking hear the secret term, they should make some outward show of understanding. “Brilliant!” “Yes, of course! I see now!” “Aha!” Vigorous head nodding. Something like that. Of course, including the secret terms in speeches and the resulting reactions must be done with straight faces, so that the non-MOS members do not catch on to the ruse. Also, you should not overuse the terms; they should be included only once or twice per speech. The secret terms and their meanings are provided in a separate handout from the GM.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are a FRS, you should plan on making a speech supporting your faction's position during one of the sessions that take place at the Royal Society, GS 3, 4, or 6. Coordinate with your faction leader (Airy) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

As a man of science but not a member of the aristocracy by heredity, nor of the government, your official powers to make things happen are fairly limited. However, you are an influential member of the Royal Society, and in that capacity members of the government consult you on all sorts of scientific matters. So, your opinion carries weight.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 2 lecture, due before GS 2 starts. This document should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British science and mathematics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding the idea of the government funding Babbage's machines, you are against the concept. Clearly, you are in no position to campaign against the concept of government-sponsored science in general – your two voyages to find the Northwest Passage are proof. However, you believe your expeditions had a high probability of being useful to the nation for trade purposes, and you do not feel that Babbage's machines are likely to be anywhere near as useful.

You are a member of the last generation of the old-school men of science community, i.e., a wealthy gentleman with enough time, talent, and money to pursue your intellectual interests at your leisure. Although you do not have anything against him personally, this puts you somewhat at odds with younger men of science like John Couch Adams. Men like Adams do not have large sums of money to pursue scientific studies as a hobby, rather, they do science for a living – science is their profession as well as their passion. If you are among the last of the wealthy, amateur scientific gentlemen, Adams and his colleagues are among the first of the professional scientists. This tension will be played out in GS 4, when the next president of the Royal Society is elected. You should support the candidate who you think best represents the values of old-school men of science, and who you think has the most political power.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

Although you will publish more of your work later in your career, during the time frame of this game you do not have a significant history of publications.

With other roles

In 1825 you were part of a collaboration with Herschel on a project paid for by the French government, to determine the precise difference in longitude between the Paris and Greenwich observatories. So, even though you are in different factions during this game, you have a good professional relationship with Herschel. You should keep this in mind during the game so you can remain "in character."

As a member of the old school men of science community, you share an affinity with others in the same class: Babbage, Crosse, Gilbert, Herschel, Lovelace, and Somerville. On the other hand, you do not have much in common with other men of science who are pursuing science professionally: Adams, Airy, Brunel, Buckland, Clement, Faraday, and Lardner.

With your faction

Even though you lack formal power, you can still be an important member of your faction. You can earn PCs for your faction by paying careful attention to the pro-Babbage arguments, asking questions that expose weaknesses, and making effective podium speeches to promote your faction.

Strategy advice

Work diligently to produce an effective lecture on navigational mathematics, so that the modern-day students playing the game understand a little of what mathematics was like in the early 1800s.

To learn more

[Lev82] Trevor H. Levere. SABINE, Sir EDWARD. http://www.biographi.ca/en/bio/sabine_edward_11E.html, 1982

Summary of your individual victory objectives

You win the game if your faction wins. Your anti-Babbage faction wins the game, by preventing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Portrait of General Sir Edward Sabine, by Stephen Pearce, from the National Portrait Gallery, St Martin's Place, London; URL: <http://www.npg.org.uk/collections/search/portrait/mw05559/Sir-Edward-Sabine?LinkID=mp03933&search=sas&sText=edward+sabine&role=sit&rNo=0>

Prince Augustus Frederick, Duke of Sussex, KG KT GCB FRS FRSA

Duke of Sussex and anti-Babbage faction member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Prince Augustus Frederick, Duke of Sussex, the sixth son of “mad” King George III, who reigned from 1760 until 1820. In actuality, your father’s insanity was such that your eldest brother, George, Prince of Wales, ruled as regent starting in 1810 and then ascended to the throne as George IV when your father died in 1820. You are the only surviving son of your father who did not have a military career. Although not a man of science, you do have an intellectual curiosity for biblical studies and the Hebrew language.

You were born to your father and his consort, Charlotte of Mecklenburg-Strelitz, on January 27, 1773. You and some of your brothers (Prince Ernest and Prince Adolphus) were educated in Germany at the University of Göttingen.

While traveling in Italy, you met your future wife, Lady Augusta Murray. You wed in secret in Rome on April 4, 1793, and then again (without telling the officiates who you were) in England in December of that year. Neither of these secret marriages had the consent of the King, and therefore they were in violation of the Royal Marriages Act of 1772 and thus invalid. Your marriage was annulled in 1794, but you continued to live with Lady Augusta until 1801. She received a “maintenance” allowance of £4,000 per year and retained custody of your children.

Your father made you a Knight of the Garter (KG) in 1786, and created you Duke of Sussex, Earl of Inverness, and Baron Arklow in November 1801. Even though you are not a man of science, and fitting the society's status as more of a gentleman's club than a thriving institute of science, you were elected as a Fellow of the Royal Society (FRS) in 1828.

Objectives

You have one main objective in the game. You want your anti-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government declines, on both of the opportunities, to fund the development of Babbage's Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Secret: You also win the game if you are elected as Royal Society president during GS 4.

Responsibilities

As Prince Augustus Frederick, Duke of Sussex, you have a responsibility to speak on the virtues of *laissez-faire* economics, the responsibility to run for the presidency of the Royal Society in GS 4, and responsibilities to your anti-Babbage faction. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to speak regarding *laissez-faire*

During GS 2, you are to speak on the principles of *laissez-faire* economics. Briefly, *laissez-faire* (French for "allow to do") economics means that the government, as much as possible, does not interfere in the workings of the free market. *Laissez-faire* economic policy was popular in Great Britain in the 19th century. Your speech should cover the general concepts, and then focus on *laissez-faire* as it applies to government funding of scientific research. Your goal, of course, is to convince the indeterminates that *laissez-faire* is a good policy, and that it precludes ridiculous expenditures like funding Babbage's calculating machines.

Responsibility to run for the Royal Society presidency

During GS 4, the Royal Society will elect a new president, to replace Davies Gilbert. You will be your faction's nominee for the position. During GS 4, you will be nominated for the

position by George Biddell Airy; you should work with him to come up with good reasons why the society should elect you, a member of the royal family, but not a man of science, to the position. You should pursue this position vigorously, since winning the presidency allows you to win the game regardless of how your faction performs.

Responsibilities to the Anti-Babbage Faction

As a member of the anti-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. You can help your faction by being a vocal opponent of Babbage and his calculating engines. You can ask good questions and make short podium speeches to help propel your faction to victory.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are a FRS, you should plan on making a speech supporting your faction's position during one of the earlier sessions that take place at the Royal Society, GS 3, or 4. Coordinate with your faction leader (Airy) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

As a member of the royalty, you are a very powerful individual in the game. To represent your political power, you begin the game with a nudge. You may use the nudge as you wish, spending it in GS 3 or 6, at your discretion.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 2 speech, due before GS 2 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who

to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British politics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government support for large-scale scientific projects such as Babbage's computing machines, you are not in favor of the practice. You feel that the government, as much as possible, should not interfere with the economy. Seeing no immediate, obvious benefit to the government from Babbage's machines, you see no reason for the government to support their development.

You are not a member of the men of science community, so the more complex points of contention regarding Babbage's machines probably escape you. This also means that you do not have a personally motivated opinion about what a man of science should be, i.e., should he be a wealthy, landed gentleman with enough time, talent, and wealth to pursue the study of nature as a hobby? Or, should he be a professional, an intelligent working man who earns his living by doing science? Your conservative leanings lead you to the former opinion. This tension will be played out in GS 4, when the next president of the Royal Society is elected. You should work hard to prevent the pro-Babbage candidate from becoming the next president of the Royal Society.

As for the conflict between imagination and reason, you are more aligned with Byron's Ghost, the champion of imagination, and opposed to William King, who argues for the supremacy of reason.

With texts

As a member of the nobility, but not of the men of science community, you have not published any scientific work.

With other roles

Your status as a member of the royal family gives you a natural affinity for the Tory politicians serving as indeterminates: Peel, Robinson, and Wellesley. Likewise, you do not

agree at all with the more liberal political leanings of Grey and Russell. Other characters you find too politically liberal are Babbage and Frend.

With your faction

Your political power makes you a very valuable member of your faction. You can earn PCs for your faction by paying careful attention to the pro-Babbage arguments, asking questions that expose weaknesses, and making effective podium speeches to promote your faction.

Strategy advice

Pay careful attention to the arguments that your faction makes against of the Difference Analytical Engines during the game, and also the arguments that the pro-Babbage faction makes. Try to gauge which arguments have made the most impact on the indeterminates. Then, in your questions and podium speech(es), try to amplify your faction's best arguments, while refuting the most effective arguments of the pro-Babbage faction.

Come up with some very good reasons as to why you, a member of the Royal Family, but not a man of science, should be elected as the president of the Royal Society. What can you bring to the table that a more scientifically-minded man cannot?

To learn more

[Cun07] W. Cunningham. *Laissez faire*. The Growth of English Industry and Commerce in Modern Times. University Press, 1907

Summary of your individual victory objectives

You win the game if your faction wins. Your anti-Babbage faction wins the game, by preventing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Your second way to win the game is to be elected as the president of the Royal Society during GS 4.

Portrait of Prince Augustus Frederick, Duke of Sussex, by Guy Head, from the National Portrait Gallery, St Martins Place, London; URL: <http://www.npg.org.uk/collections/search/portraitLarge/mw06148/Prince-Augustus-Frederick-Duke-of-Sussex?LinkID=mp04375&search=sas&sText=duke+of+sussex&role=sit&rNo=0>

Captain Swing

Anti-Babbage faction member



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Captain Swing. You are an interesting character in this game in that you are not playing a person, but rather you represent a movement. Specifically, you represent the spirit and ideas of the farmers who participated in the “Swing Riots” across England in 1830 and 1831, starting fires and breaking threshing machines, protesting for increased wages, against restrictive Poor Laws, and against the increasing use of threshing machines in agriculture. All of these factors moved many of England’s agricultural poor from being merely desperately poor to struggling for their very lives. Several of the threatening letters sent by the protesters were signed with the *nom de guerre* “Captain Swing.”

As Captain Swing, you represent

... the most impressive episode in the English farm-labourers’ long and doomed struggle against poverty and degradation. [HR14]

Agriculture in England at the time had a three-level division of labor. There were a few landowners, who owned vast tracts of land. The landowners employed a large body of tenant farmers to farm their land for them. Then tenant farmers, in turn, employed a much larger pool of wage-earning agricultural laborers that did the actual work. You represent this last class of people, as a host of factors pushed them to riot in 1830 and 1831.

Your group was not seeking a political revolution, but rather economic redress; you demanded higher wages, more jobs, for improvements in the Poor Law. You sought the return of your rights as “freeborn Englishmen,” i.e., the right to farm small plots of the commons, open areas which were being fenced off by the landowners. Before the enclosure of the commons, the rural poor were able to survive by virtue of the crops they could grow themselves in these spaces; once that was no longer possible, many found their very survival, not simply their livelihoods, threatened.

The Swing riots began in the August of 1830. The primary activity of the Swing rioters was to break up the threshing machines owned by the landowners, which the rioters blamed at least in part for the lower number of available agricultural jobs. (Threshers automate the process of threshing grain, i.e., the process of removing the ripened grain from the rest of the plant.) Other activities in the riots included arson and threatening letters. “Captain Swing” was the fictional name signed to many of the letters.

Objectives

You have one main objective in the game. You want your anti-Babbage faction to win the game; if the faction wins, you win. Your faction wins if the British government declines, on both of the opportunities, to fund the development of Babbage’s Engines. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Responsibilities

As Captain Swing, you have a responsibility to talk about the plight of the farmers in the UK and the dangers of automation, and responsibilities to your anti-Babbage faction. Every character in the game also has a responsibility to make at least one additional brief podium speech during open debate periods.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to speak regarding the plight of Britain’s farmers

During GS 2 at the Mechanic’s Institute, you will give a speech on the plight of the nation’s farmers. In other words, your speech will summarize the issues that caused the Swing Riots of 1830 – 1831 (as stated in the “Counterfactuals” section of the gamebook, you will present during GS 2, which takes place in 1828; you may present as if the Swing Riots have already happened). After you have discussed the issues involved in the Swing Riots, you should extend the anti-automation concerns of the farmers to the Difference Engine. Remember

that “computer” was a job description in this time period, and the Difference Engine would certainly put many computers out of work, just as threshing machines put the jobs of farmers in jeopardy.

Responsibilities to the Anti-Babbage Faction

As a member of the anti-Babbage faction, you should do whatever you can to help the faction win the game, so that you may also win. During game sessions that do not take place in the Royal Society, you can help your faction by being a vocal opponent of Babbage and his calculating engines. You can ask good questions and make short podium speeches to help propel your faction to victory.

Podium speech responsibility

You must make at least one additional podium speech during the open debate periods in the game. Since you are not a FRS, you should plan on making a speech supporting your faction’s position during the other session that takes place at the Mechanic’s Institute, GS 5. Coordinate with your faction leader (Airy) and the other members of your faction so you can make a speech with the maximum impact for your faction.

Powers

As a fictional character representing the plight of poor farmers in the U.K., you have no formal power to make decisions regarding Babbage’s machines – or anything else, for that matter. However, you do have the power of persuasion; through your GS 2 speech, and your other impromptu speaking and interactions, you should do your best to make the Difference Engine (before GS 5) and the Analytical Engine (after GS 5) appear to be ridiculous extravagances, costing far too much for their supposed worth.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage’s engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 2 speech, due before GS 2 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who to vote for during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

Even though your character stands for a movement, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

You are emblematic of the new liberalism that is rising in the United Kingdom. During this era, there were reforms to the Poor Laws, Catholic Emancipation, and other movements intended to begin to help marginalized members of society. Based on your opposition to the threshing machines that have negatively impacted so many British farmers, you oppose the “mechanization of number” that would be brought on by Babbage’s machines.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You will not be able to vote in this election, but you are likely to support the candidate which best represents the new men of science. You should work behind the scenes to get him elected.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron’s Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

While the urban working class, both men and women, were becoming more educated during this period (via attendance at lectures at the various Mechanic’s Institutes, for example), the same was not true for the people you represent. So, you have likely not read, and certainly not published, any books of the kind that will be examined in this game.

With other roles

Due to your station in life, you have nothing in common with almost everyone in the game, who belong to higher echelons of society. The closest to you is the engineer, Joseph Clement. However, the Whig MPs, Grey and Russell, probably share some, if not all, of your political opinions.

With your faction

Nonetheless, you are important to your faction. You can make strong economic and equity arguments against the government funding Babbage's engines.

Strategy advice

Pay careful attention to the arguments that your faction makes against of the Analytical and Difference Engines during the game, and also the arguments that the pro-Babbage faction makes. Try to gauge which arguments have made the most impact on the indeterminates. Then, in your formal speech, questions, and podium speech(es), try to amplify your faction's best arguments, while refuting the most effective arguments of the pro-Babbage faction.

To learn more

[HH13] John Lawrence Hammond and Barbara Bradby Hammond. *The Village Labourer, 1760-1832: A Study in the Government of England Before the Reform Bill*. Longmans, Green, and Company, 1913. Available via Google Books

[HR14] Eric Hobsbawm and George Rudé. *Captain Swing*. Verso, 2014

Summary of your individual victory objectives

You win the game if your faction wins. Your anti-Babbage faction wins the game, by preventing government funding for the development of Babbage's engines in both of the funding opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

The Ghost of Lord Byron

Dead romantic poet and indeterminate



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are the Ghost of Lord Byron, and a behind-the-scenes champion for the value of imagination over reason. You were born in 1788 and you became the 6th Baron Bryon upon the death of your father William. You were many things in life: a member of the nobility, a politician, a Fellow of the Royal Society (although not a man of science), and an immensely famous poet in the Romantic movement. In many ways, you were the prototype for the modern celebrity; you even had “stalkers.” Although your sexual escapades were legend, you were only married once, to Anne Isabella Milbanke in 1815. The marriage was doomed from the outset, and you separated in 1816. Your only legitimate child, Ada (later Countess Ada Lovelace), has no memory of you. You died in 1824 in the Ottoman Empire, while fighting for Greek independence. You died not in battle, but from an infection due to bloodletting.

In this game you are an indeterminate. This means that you have no particular position regarding the central issues of the game. Some of those playing other characters in the game belong to pro-Babbage or anti-Babbage factions, and it is up to them to convince you that their position is correct. You are free to decide on your own which faction to ultimately side with, within the boundaries of your biography, responsibilities, and objectives, outlined below.

Objectives

Your main objective is to win the game. You win the game if you can persuade more than half of the characters in the game to vote that imagination is more important than reason. This vote will take place at the end of GS 6.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibilities

As the Ghost of Lord Byron, you have the responsibility to work behind the scenes to influence the other characters in the game of the value of imagination over reason, and the responsibility to be honest and attentive in your indeterminacy.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to champion imagination

As a ghost, you cannot make a formal speech to the group, but, you do have the ability to talk to anyone during the game's social periods or during open debate periods. You should talk to as many characters as possible during these times, to point out where imagination has been important to the topics they have been debating. For example, you could talk to I.K. Brunel, praising the imagination required to think of such a thing as the Thames Tunnel. You are trying to convince them that imagination is to be valued more than reason. (Doctor William King will be arguing the opposite point of view, but, since he is not a ghost, he is able to make a formal speech to the body.)

Responsibilities regarding your indeterminacy

Since you do not belong to any faction in this game, you are free to make decisions as you see fit. The factions should try to convince you that their positions are correct, and you should listen carefully to their arguments and other persuasions. Be swayed by those you find the most convincing. The only caveat is that you should try to stay true to your role as a dead romantic poet.

Powers

Since you are a ghost, you have no formal power to make decisions regarding Babbage's machines – or anything else, for that matter. However, you do have the power of persuasion.

By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways. You begin the game with a nudge, and will be allowed to draw one more card from the PC deck at the beginning of GS 1.

Assignments

Your main writing assignment is a one-thousand word paper, due before GS 1 starts. The paper should cover the tenants of the Romantic movement, and explain why this philosophy of life would place more value on imagination than reason. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who you would prefer during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As important figure in British culture, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government funding for Babbage's machines, you are indeterminate. You should listen carefully to the arguments made by the factions, and then make up your own mind on this issue, within the general bounds of your Romantic views.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You will not be able to vote in this election, but, once you've decided which candidate you will support, you can work behind the scenes to get him elected.

With texts

You wrote many books of poetry. In particular, you wrote the poems characters must recite if they are to successfully exploit Ada's celebrity.

With other roles

You are a Cambridge man, giving you something in common with other men such as George Biddell Airy, John Couch Adams, Charles Babbage, William Frend, Lord Charles Grey, and John Herschel. On the other hand, your noble heritage means you have little in common with I.K. Brunel, Joseph Clement, or Captain Swing.

Throughout your life, you longed for a relationship with your daughter Ada, and she for one with you. Alas, this never happened.

With your faction

As an indeterminate, you do not belong to either the pro- or anti-Babbage faction.

Strategy advice

Pay careful attention to the speeches of the other characters, listening for aspects you feel are examples of the use of imagination. Then, use these instances in your behind-the-scenes conversations to demonstrate the importance of imagination.

Summary of your individual victory objectives

You win the game if, at the end of GS 6, at least half of the characters in the game vote that imagination is to be valued more than reason.

The Fury

Ada Lovelace's chaperone, etiquette expert, and indeterminate



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

In this game you are an indeterminate. This means that you have no particular position regarding the central issues of the game. Some of those playing other characters in the game belong to pro-Babbage or anti-Babbage factions, and it is up to them to convince you that their position is correct. You are free to decide on your own which faction to ultimately side with, within the boundaries of your biography, responsibilities, and objectives, outlined below.

You are The Fury, an amalgam of the group of women (Sophia Frend, Selina Doyle, Mary Montgomery, and Frances Carr) chosen by Lady Anne Isabella Noel Byron to supervise the education and upbringing of her teenage daughter, Ada. Ada was very resentful of these women, writing later in life to a friend that “these women would ‘exaggerate or invent’ stories about her behaviour, and hated her like ‘poison.’ ” From your point of view, you are a protector of Ada, helping to educate her, prevent others from taking advantage of her celebrity, and helping to prevent her from becoming “poetical” like her father, Lord Byron.

Objectives

Your main objective is to win the game. You win the game in one of two ways.

First, if you prevent other characters from successfully exploiting Ada's celebrity by reciting one of by Byronic verses listed in the Gamebook. You can prevent them from exploiting Ada by detecting errors in their recitations, and convincing Ada that they have, in fact, incorrectly recited the stanza. If you are able to prevent this exploitation of Ada at least twice during the game, you win, regardless of whatever else happens in the game.

Your second, independent way to win the game is by documenting instances where other characters in the game have violated the mores governing the behavior of ladies and gentlemen in polite society. If you can detect, call out, and document at least five such violations during the game, you win.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibilities

As the Fury, you have the responsibility to teach the other characters about the proper way that ladies and gentlemen should behave in polite society, the responsibility to prevent the exploitation of Ada Lovelace, and the responsibility to be honest and attentive in your indeterminacy.

Responsibility to speak regarding polite behavior

At the beginning of GS 1, you will speak to the other characters in the game, to teach them how to behave like ladies and gentlemen in polite society. You should let everyone know that you will be watching them during the game, to make sure that they conform to the rules you lay out.

Secret: You should not, however, let anyone else know that you can win the game by catching other characters violating these norms of conduct.

Responsibility to protect Ada Lovelace

As mentioned above, you want to prevent others from exploiting Ada Lovelace's celebrity. You do this by noticing mistakes made when other characters try to recite Byronic stanzas to Ada. You should have your Gamebook handy, so that you can read along with the stanza as the character tries to recite it from memory.

Responsibilities regarding your indeterminacy

Since you do not belong to any faction in this game, you are free to make decisions as you see fit. The factions should try to convince you that their positions are correct, and you should listen carefully to their arguments and other persuasions. Be swayed by those you

find the most convincing. The only caveat is that you should try to stay true to your role as a English lady of the early 1800s.

Powers

Since you are a woman, even though you are a well-respected Lady, you have no formal power to make decisions regarding Babbage's machines – or anything else, for that matter. However, you do have the power of persuasion.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 1 speech, due before GS 1 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who you would prefer during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

If you are able to catch and call out five lapses in polite behavior during the game, write a short paper documenting these and submit it to the gamemaster. This document will serve as proof that you have met this victory condition.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British society, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government funding for Babbage's machines, you are indeterminate. You should listen carefully to the arguments made by the factions, and then make up your own mind on this issue, within the general bounds of your politics.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You will not be able to vote in this election, but, once you've decided which candidate you will support, you can work behind the scenes to get him elected.

As for the conflict between imagination and reason, you are firmly aligned with William King on the supremacy of reason, and opposed to Byron's Ghost, the champion of imagination.

With texts

You are reasonably well-versed in contemporary literature, but you have not read in any depth in the areas of mathematics or science.

With other roles

Ada Lovelace does not like you, and, frankly, you do not care much for her, either. You are well acquainted with several other characters, who have also served in various capacities in Ada's upbringing and education: Mary Somerville, William Frend, and Dr. William King.

With your faction

As an indeterminate, you do not belong to either the pro- or anti-Babbage faction.

Strategy advice

Research on the rules of polite society in 1820s Great Britain, so that your GS 1 speech will be as authentic as possible.

To learn more

[pol17] *Politeness of Manners and Behaviour in Fashionable Society: From the French of the Abbé de Bellegarde*. Charles, 1817. Available via Google Books

[Woo99] Benjamin Woolley. *The Bride of Science: Romance, Reason and Byron's Daughter*. PAN, London, 1999

Summary of your individual victory objectives

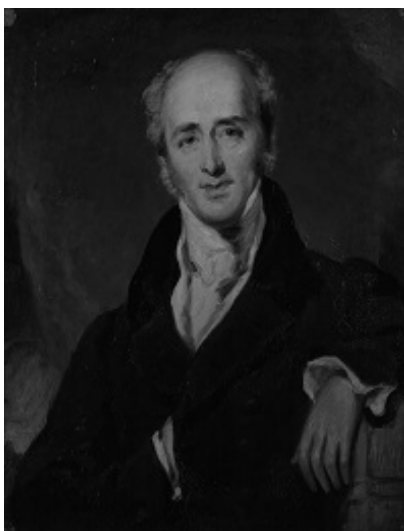
You can win the game in one of two ways:

1. You prevent at least two characters from exploiting Ada's celebrity by successfully reciting Byron's stanzas, or
2. You catch, call out, and document at least five lapses in polite behavior during the game

Portrait of one of the Furies, Frances Carr, from the National Portrait Gallery, St Martin's Place, London; URL: <https://www.npg.org.uk/collections/search/portrait/mw281272/Frances-Carr?search=sp&sText=frances+carr&firstRun=true&rNo=4>

Lord Charles Grey, 2nd Earl Grey, KG PC

Whig MP and indeterminate



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Lord Charles Grey, 2nd Earl Grey, KG PC. You are a Whig MP, and you will serve as Prime Minister. You are also the namesake for Earl Grey tea.

In this game you are an indeterminate. This means that you have no particular position regarding the central issues of the game. Some of those playing other characters in the game belong to pro-Babbage or anti-Babbage factions, and it is up to them to convince you that their position is correct. You are free to decide on your own which faction to ultimately side with, within the boundaries of your political opinions, outlined below.

You were born on March 13, 1764, at your ancestral home, Howick Hall, in Fallodon, Northumberland, on the northeastern coast of England. You are the oldest surviving son of your parents, General Charles Grey KB, 1st Earl Grey, and Lady Elizabeth Grey, Countess Grey. Your primary education was at the Richmond School in North Yorkshire, and from there you studied at Eton and Trinity College, Cambridge. You were first elected to Parliament in 1786, when you were only 22. You quickly became a leader of the Whigs in the House of Commons.

You served as First Lord of the Admiralty for a few months in 1806, then when Charles James Fox died, you succeeded him as Leader of the House of Commons and the Foreign Secretary, under the Prime Minister Lord Grenville. In 1807, Grenville's government was

dissolved when King George III blocked Catholic Emancipation legislation. Also in that year, you ascended into the House of Lords, succeeding your father. A series of Tory PMs followed until 1830, and you served as a leader of the Whig opposition during those years.

Objectives

You have one primary objective in the game: to win. You win the game if, during GS 6, the characters with a franchise vote for you as the PM who made the best presentation during the game.

Responsibilities

As Lord Grey, you have responsibilities to make a speech outlining the goals and priorities of your government when you take office as PM, to make crucial, informed decisions regarding funding for Charles Babbage's computational engine projects, and to be honest and attentive in your indeterminacy.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to speak regarding your government's priorities and goals

During the first five game sessions, the PM of the day will make a speech at the beginning of the game session corresponding to his rise to power. In the speech, the PM will outline the priorities and goals his government will have while he is in office. Each MP indeterminate will serve as PM exactly once during the first five game sessions. However, you will not know when you will serve until it happens: at the beginning of GS 1, 2, 3, 4, and 5, the GM will announce which of the MP indeterminates will serve as the PM of the day.

Responsibilities regarding the Difference and Analytical Engines

There will be two decisions regarding Babbage's Difference and Analytical Engines during the game: at the end of GS 3 and 6. To one extent or another, the PM at the time will be free to decide what the government should do. So, if you are PM during one of these sessions, you will be tasked with making these decisions. If the responsibility falls on you, you should make the decision you feel is best for the United Kingdom, as informed by your Whig politics and by the presentations and persuasions made by the other characters in the game.

Responsibilities regarding your indeterminacy

Since you do not belong to any faction in this game, you are free to make decisions as you see fit. The factions should try to convince you that their positions are correct, and you should listen carefully to their arguments and other persuasions. Be swayed by those you find the most convincing. The only caveat is that you should try to stay true to your role as a Whig politician of the early 1800s.

During your research for your PM speech, you may come across some opinions that Lord Grey held about Babbage and his computing machines. For the purposes of this game, you should ignore these, so that you may play your role as an indeterminate.

Powers

As a MP, you are part of the governing body of the kingdom, and thus a powerful and important man. As a member of the Whig party, your words and influence are especially important to members of the rising middle class and working people. When you serve as PM, you will have even more power, especially regarding the outcome of this game, for it is the PMs that will ultimately decide whether or not Babbage receives government monies to complete his Difference or Analytical Engines.

To represent your political power, you begin the game with a nudge. You may use the nudge as you wish, spending it in GS 3 or 6, at your discretion.

Assignments

Your main writing assignment is a one-thousand word, written version of your PM speech, due before GS 1 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write a few paragraphs about your vote in each persuasiveness metric poll. These polls, conducted by the gamemaster, take place at the end of each game session. The indeterminates (including you) are asked to vote for the presentation they thought was the most persuasive during the session. Your short paper will describe who you voted for in the persuasiveness metric poll, and why you voted for them. Each of these papers should be submitted privately to the gamemaster before the beginning of the next game session. For example, your paper about your vote in the GS 1 persuasiveness metric should be sent to the gamemaster no later than the start of GS 2.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British politics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government funding for Babbage's machines, you are indeterminate. You should listen carefully to the arguments made by the factions, and then make up your own mind on this issue, within the general bounds of your politics.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You will not be able to vote in this election, but, once you've decided which candidate you will support, you can work behind the scenes to get him elected.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

As for the conflict between imagination and reason, you are firmly aligned with William King on the supremacy of reason, and opposed to Byron's Ghost, the champion of imagination.

With texts

As a member of the government, but not of the men of science community, you have not published any scientific work.

With other roles

You share many political leanings with your fellow Whig MP, Lord John Russell, and you differ widely with the Tory MPs, Peel, Robinson, and Wellesley. Your Whig sensibilities are also ideologically close to Charles Babbage, but far apart from the Duke of Sussex.

With your faction

As an indeterminate, you do not belong to either the pro- or anti-Babbage faction.

Strategy advice

You are a Whig politician in Georgian Britain, so you should do some research on what the political stances of your party – and the opposition party, the Tories – were during that time.

You are known for two major legislative initiatives, the Reform Act of 1832 and the abolition of slavery in the British Empire. You also helped quell the Swing Riots of 1830 and 1831 and attempted to reform the Poor Law. Research these initiatives; you can use the information you find as materials for your PM speech.

To learn more

[gov16] gov.uk. Past Prime Ministers – GOV.UK. <https://www.gov.uk/government/history/past-prime-ministers>, 2016

Summary of your individual victory objectives

You win the game if, during GS 6, those with a franchise vote for you as the PM who gave the best PM speech during the game.

Portrait of Charles Grey, 2nd Earl Grey, from the National Portrait Gallery, St Martin's Place, London; URL: <http://www.npg.org.uk/collections/search/portrait/mw02743/Charles-Grey-2nd-Earl-Grey?LinkID=mp01912&search=sas&sText=Charles+Grey&role=sit&rNo=3>

The Ornamental Hermit

Indeterminate and strange person paid to dress as a druid



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

If you have received this role sheet, it means that your original character has died of Cholera. You are now reincarnated into a completely different role, as an Ornamental Hermit.

Regardless of your previous role, you are now an indeterminate. This means that you have no particular position regarding the central issues of the game.

The Ornamental Hermit, a man paid to dress as a druid and live in a small hut (a *hermitage*) on a wealthy landowner's property, was a fad in the late 18th and early 19th centuries in England, and to a lesser extent in Ireland and Scotland. Hermitages on wealthy estates were popular since Roman times. Originally, they served as a way for the landowners to escape from the pressures of day-to-day life, while using the isolation to ponder religion. Over time, the practice evolved to include not only the hermitage, but also the Ornamental Hermit, a man hired to live in the hermitage. According to Gordon Campbell,

Recruiting a hermit wasn't always easy. Sometimes they were agricultural workers, and they were dressed in a costume, often in a druid's costume. There was no agreement on how druids dressed, but in some cases they wore what we would call a dunce's cap. ... [the hermit was] neither to wash himself or cleanse

himself in any way whatever, but is to let his hair and nails both on hands and feet, grow as long as nature will permit them. [Cam13]

The Ornamental Hermit was an attraction for the estate, and visitors were often entertained by the hermit's poetical ramblings. Seen as an embodiment of the Georgian emotion of melancholy, the hermit's accessories often include symbols of mortality such as an hourglass or a human skull.

You will offer your services to the other players in the game, with the hope that one of them will hire you. In order to hire you, a character must pay at least one nudge card to the GM. When you become the Ornamental Hermit, the GM will announce that you are for hire, and that anyone can employ your services by paying nudge(s). The GM will conduct a brief auction of sorts, with the highest bidder hiring you. The character employing your service as an Ornamental Hermit will gain prestige, and accordingly, will be allowed to draw twice from the PC deck.

The Ornamental Hermit had faded by the middle of the nineteenth century, although many gardeners still participate, after a fashion. The modern Garden Gnome is the spiritual descendant of the Ornamental Hermit.

Objectives

You have one primary objective in the game: to win. You win the game if one of the characters in the game hires you as their Ornamental Hermit.

Responsibilities

As the Ornamental Hermit, you have a responsibility to provide brief poetry readings during the social time period(s) in the game session(s) you serve in the role. You should compose one short, melancholy poem for each session; when other characters approach your hermitage, you should engage them in a brief conversation and read them the day's poem.

If you, as the Ornamental Hermit, somehow acquire power cards, you should use them as an indeterminate. You are free to make decisions as you see fit. The factions should try to convince you that their positions are correct, and you should listen carefully to their arguments and other persuasions. Be swayed by those you find the most convincing.

Powers

Since you are an Ornamental Hermit, you have no formal power to make decisions regarding Babbage's machines – or anything else, for that matter. Although you cannot directly win PCs, you may be able to acquire them in other ways. If so, by using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even

influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

If your previous character has already submitted their writing assignment, you have no further assignment as the Ornamental Hermit. If your character had not submitted their assignment before succumbing to cholera, then you should write an one-thousand word paper on the nature of the emotion of melancholy, especially as it was conceived in Georgian England.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As an Ornamental Hermit, you are still connected to several of the people, texts, and ideas in the game.

With ideas

Regarding government funding for Babbage's machines, you are now indeterminate. You should listen carefully to the arguments made by the factions, and then make up your own mind on this issue, within the general bounds of your role.

As you know, one of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You will not be able to vote in this election, but, once you've decided which candidate you will support, you can work behind the scenes to get him elected.

As for the conflict between imagination and reason, you are more aligned with Byron's Ghost, the champion of imagination, and opposed to William King, who argues for the supremacy of reason.

With texts

As an ornamental hermit, you write mad scribbles in your journal, but none of your writing is published. You are familiar with the poetry and literature of the day, so that you may converse intelligently with those that visit you in your hermitage.

With other roles

Regardless of your previous role, you now have a quite different position. You are not a man of science, a wealthy land-owner, or even the embodiment of a movement like Captain Swing. You hope to become the employee of one of the other characters in the game, but this is not a given.

With your faction

Since you are now an indeterminate, you do not belong to either the pro- or anti-Babbage faction.

Strategy advice

You may wish to find some melancholy accessories to place in your hermitage, such as skulls, hourglasses, and / or spectacles.

To learn more

[Cam13] Gordon Campbell. *The Hermit in the Garden: From Imperial Rome to Ornamental Gnome*. Oxford Univeristy Press, 2013

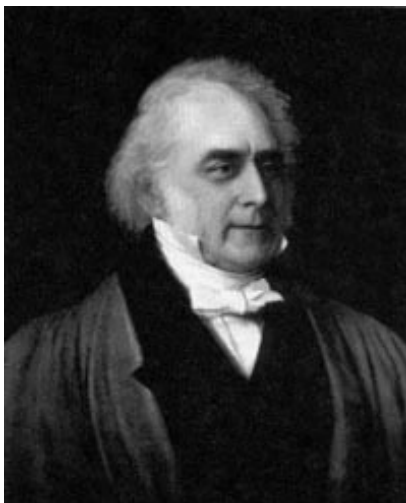
[Mei14] Allison Meier. Before the Garden Gnome, the Ornamental Hermit: A Real Person Paid to Dress like a Druid. <https://www.atlasobscura.com/articles/the-history-of-hermits-in-gardens>, 2014

Summary of your individual victory objectives

You win the game if another character employs you as their Ornamental Hermit.

Doctor William King

Ada Lovelace's doctor, champion of reason over imagination, and indeterminate



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

In this game you are an indeterminate. This means that you have no particular position regarding the central issues of the game. Some of those playing other characters in the game belong to pro-Babbage or anti-Babbage factions, and it is up to them to convince you that their position is correct. You are free to decide on your own which faction to ultimately side with, within the boundaries of your biography, responsibilities, and objectives, outlined below.

You are Dr. William King, “physician, lunatic asylum manager, devout evangelical Christian and enthusiastic promoter of the Cooperative Movement.” [Woo99] You are Ada Lovelace’s personal physician and mentor, charged by Ada’s mother, Lady Anne Isabella Noel Byron, to maintain Ada’s mental health. In her youth, you prescribed a healthy course of mathematical instruction for Ada, lest she follow in the footsteps of her father, the infamous poet Lord Byron, and become “poetical.”

Your prescription to Ada, to prevent the poetical, was [Woo99]:

- No form of excitement in her life except for intellectual pursuits, to keep her imagination in check;
- For intellectual improvement, focus on mathematics; and,

- Check yourself when you take a liking to something outside of these; if you like it, it is probably bad for you.

You were on hand at Christmastime when Ada received a portrait of her deceased father as a gift from her (absent) mother, to make sure that the mere sight of the poet did not send Ada into a tailspin.

In this game, you are the champion of reason over imagination. You believe that imagination is a “dangerous asset,” that must not be allowed to roam free, but rather be constrained and regulated by rational thought. You are sure that too much imagination is clearly dangerous for Ada – just look at what it did to Byron! – and you also believe the principle applies to the general population.

Objectives

Your main objective is to win the game. You win the game if you can persuade more than half of the characters in the game to vote that reason is more important than imagination. This vote will take place at the end of GS 6.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibilities

As Doctor William King, you have the responsibility to speak about the value of reason over imagination and the responsibility to be honest and attentive in your indeterminacy.

Responsibility to speak regarding reason

During GS 1, you will speak to the other characters in the game, to try to persuade them that they should employ reason and rational thought at all times, while avoiding the temptations of imagination.

Responsibilities regarding your indeterminacy

Since you do not belong to any faction in this game, you are free to make decisions as you see fit. The factions should try to convince you that their positions are correct, and you should listen carefully to their arguments and other persuasions. Be swayed by those you find the most convincing. The only caveat is that you should try to stay true to your role as a liberal, nonconformist physician of the early 1800s.

Powers

Since you are a nonconformist, a liberal, and not a member of the government or the aristocracy, you have no formal power to make decisions regarding Babbage's machines – or anything else, for that matter. However, you do have the power of persuasion.

By being persuasive, making good speeches, asking good questions, and the like, you can win power cards (PCs). By using *nudges*, you can influence the way that the PM votes regarding funding for Babbage's engines, and even influence who will be PM during GS 6. By using the other type of cards, *powers*, you can dramatically change the course of the game in various ways.

Assignments

Your main writing assignment is a one-thousand word, written version of your GS 1 speech, due before GS 1 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write and submit privately to the GM a few paragraphs about each PM speech. The PMs make their speeches at the beginnings of GS 1, 2, 3, 4, and 5; your writing for each should contain your impressions of the speeches, so that you can decide who you would prefer during the winning PM election at the end of GS 6. Your speech impressions should be submitted privately to the GM no later than the beginning of GS 2, 3, 4, 5, and 6.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British society, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government funding for Babbage's machines, you are indeterminate. You should listen carefully to the arguments made by the factions, and then make up your own mind on this issue, within the general bounds of your politics.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men

of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You will not be able to vote in this election, but, once you've decided which candidate you will support, you can work behind the scenes to get him elected.

As for the conflict between imagination and reason, you believe ardently in the supremacy of reason, and opposed to Byron's Ghost, the champion of imagination.

With texts

You are well-versed in contemporary literature and mathematics.

With other roles

Much like her other tutors in this game (William Frend and Mary Somerville), Ada Lovelace likes and respects you. She has done her best to follow your advice regarding the pursuit of reason over the dreaded imagination.

With your faction

As an indeterminate, you do not belong to either the pro- or anti-Babbage faction.

Strategy advice

To prepare for your speech, you may wish to research the principles of the Age of Reason and the Age of Enlightenment. Also, some of King's paraphrases and quotes regarding reason and imagination may help guide the manner in which you play this character: [Woo99]

- Imagination is a dangerous asset that must be controlled.
- "...knowledge is evidence and evidence is absolutely necessary to settle the mind under doubts and temptations."
- "Loose reading does no good."
- The study of mathematics can prevent objectionable thoughts.

To learn more

[Woo99] Benjamin Woolley. *The Bride of Science: Romance, Reason and Byron's Daughter*. PAN, London, 1999

Summary of your individual victory objectives

You win the game if, at the end of GS 6, more than half of the characters in the game vote that reason is more important than imagination.

Sir Robert Peel, 2nd Baronet, FRS, PC

Tory MP and indeterminate



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Sir Robert Peel, 2nd Baronet FRS PC. You are a Tory MP, and you will serve as Prime Minister.

In this game you are an indeterminate. This means that you have no particular position regarding the central issues of the game. Some of those playing other characters in the game belong to pro-Babbage or anti-Babbage factions, and it is up to them to convince you that their position is correct. You are free to decide on your own which faction to ultimately side with, within the boundaries of your political opinions, outlined below.

You were born in 1788 in Bury, Lancashire, in the northwest of England. Your father, Sir Robert Peel, 1st Baronet, was a wealthy cotton mill owner. His success as an industrialist brought wealth, political power (he was also an MP), and elevation to the title of Baronet. The fact that your father's money and title were so recently earned and bestowed sometimes leads to conflict with your Tory colleagues, who tend to represent more "old money" interests.

You were educated at the most aristocratic constituent college of Oxford University, Christ Church, where you earned a "double first," that is, extremely high achievement in both

your first-year and final exams. After graduating in 1808, you were first elected as MP for Cashel, Tipperary, in Ireland. Eventually you became MP for Oxford University.

You were elected as a Fellow of the Royal Society in 1822, even though you are not a “man of science.” This is emblematic of the nature of the Royal during these years – membership had as much to do with who you were and who you knew as it did with your achievements in science.

You were named Home Secretary in 1822, serving under the government of Sir Robert Jenkinson, 2nd Earl of Liverpool. While in this position, you made important and long-lasting changes to the British justice system. You established the professional London Metropolitan police force (forever known as “Bobbies” in your honor), reformed the penal code to drastically reduce the number of crimes punishable by death, and more.

Given that you are a member of the Tory party, a political party chiefly closely aligned to the Church of England, it is not surprising that you have been very opposed to Catholic emancipation. What is surprising is that, in 1829, you helped to pass the Roman Catholic Relief Act, which removed many of the restrictions that were placed on Catholics by the Act of Uniformity in 1558. Your support for the act has not made you extremely popular with your Tory colleagues.

Perhaps fitting your indeterminate nature in the game, later in life you were involved in several other acts of Parliament that further distanced you from the Tories. Such acts banned women and children from working in mines, limited the hours women and children could work in factories, and repealed the Corn Laws, which had banned the import of foreign grain. You supported the repeal of the Corn Laws as a way to deal with the Irish Potato Famine of 1845 - 1849.

Objectives

You have one primary objective in the game: to win. You win the game if, during GS 6, the characters with a franchise vote for you as the PM who made the best PM speech during the game.

Responsibilities

As Sir Robert Peel, you have responsibilities to make a speech outlining the goals and priorities of your government when you take office as PM, to make crucial, informed decisions regarding funding for Charles Babbage’s computational engine projects, and to be honest and attentive in your indeterminacy.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to speak regarding your government's priorities and goals

During the first five game sessions, the PM of the day will make a speech at the beginning of the game session corresponding to his rise to power. In the speech, the PM will outline the priorities and goals his government will have while he is in office. Each MP indeterminate will serve as PM exactly once during the first five game sessions. However, you will not know when you will serve until it happens: at the beginning of GS 1, 2, 3, 4, and 5, the GM will announce which of the MP indeterminates will serve as the PM of the day.

Responsibilities regarding the Difference and Analytical Engines

There will be two decisions regarding Babbage's Difference and Analytical Engines during the game: at the end of GS 3 and 6. To one extent or another, the PM at the time will be free to decide what the government should do. So, if you are PM during one of these sessions, you will be tasked with making these decisions. If the responsibility falls on you, you should make the decision you feel is best for the United Kingdom, as informed by your Tory politics and by the presentations and persuasions made by the other characters in the game.

Responsibilities regarding your indeterminacy

Since you do not belong to any faction in this game, you are free to make decisions as you see fit. The factions should try to convince you that their positions are correct, and you should listen carefully to their arguments and other persuasions. Be swayed by those you find the most convincing. The only caveat is that you should try to stay true to your role as a (albeit slightly liberal) Tory politician of the early 1800s.

During your research for your PM speech, you may come across some opinions that Peel held about Babbage and his computing machines. For the purposes of this game, you should ignore these, so that you may play your role as an indeterminate.

Powers

As a MP, you are part of the governing body of the kingdom, and thus a powerful and important man. As a member of the Tory party – the party that has ruled since 1807 – your words and influence are especially important to land owners and the nobility. When you serve as PM, you will have even more power, especially regarding the outcome of this game, for it is the PMs that will ultimately decide whether or not Babbage receives government monies to complete his Difference or Analytical Engines.

To represent your political power, you begin the game with a nudge. You may use the nudge as you wish, spending it in GS 3 or 6, at your discretion.

In addition, you possess a special power: you are an indeterminate MP / PM and you are also a member of the Royal Society. As a FRS, you have a voice – and a vote – during game sessions that take place at the Royal. This is perhaps most important in GS 3, when the Royal must pass either a resolution that recommends funding for Babbage’s Difference Engine or a resolution against such funding. As an indeterminate, you should vote for whichever side (pro- or anti-Babbage) that you feel has best persuaded you that their proposal is appropriate.

Assignments

Your main writing assignment is a one-thousand word, written version of your PM speech, due before GS 1 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write a few paragraphs about your vote in each persuasiveness metric poll. These polls, conducted by the gamemaster, take place at the end of each game session. The indeterminates (including you) are asked to vote for the presentation they thought was the most persuasive during the session. Your short paper will describe who you voted for in the persuasiveness metric poll, and why you voted for them. Each of these papers should be submitted privately to the gamemaster before the beginning of the next game session. For example, your paper about your vote in the GS 1 persuasiveness metric should be sent to the gamemaster no later than the start of GS 2.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British politics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government funding for Babbage’s machines, you are indeterminate. You should listen carefully to the arguments made by the factions, and then make up your own mind on this issue, within the general bounds of your politics.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You should support the candidate

who you think best represents the values of old-school men of science, and who you think has the most political power.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

As a member of the government, but not of the men of science community, you have not published any scientific work.

With other roles

You share many political leanings with your fellow Tory MPs, Robinson and Wellesley, while you are naturally opposed to the politics of the Whigs, Grey and Russell. However, as you have read, you may be convinced of some of the Whig's position despite your Tory nature. Your politics gives you a natural affinity to the Duke of Sussex, and puts you in opposition with Charles Babbage.

With your faction

As an indeterminate, you do not belong to either the pro- or anti-Babbage faction.

Strategy advice

You are a Tory politician in Georgian Britain, so you should do some research on what the political stances of your party – and the opposition party, the Whigs – were during that time.

Despite being a Tory, your government was a milestone for social reform in Great Britain. In 1841 when you were elected for the second time, there was an economic recession in Britain. Your government moved more towards free trade practices, helping set the stage for a robust recovery. You should research the Mines Act of 1842, the Factory Act of 1844, the repeal of the Corn Laws in 1846, and the failed Irish Coercion Act of 1846; you can use the information you find as material for your PM speech.

To learn more

[gov16] gov.uk. Past Prime Ministers – GOV.UK. <https://www.gov.uk/government/history/past-prime-ministers>, 2016

Summary of your individual victory objectives

You win the game if, during GS 6, those with a franchise vote for you as the PM who gave the best PM speech during the game.

Lord Frederick John Robinson, 1st Earl of Ripon FRS PC

Tory MP and indeterminate



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Lord Frederick John Robinson, 1st Earl of Ripon, 1st Viscount Goderich, FRS PC. You are a Tory MP, and you will serve as Prime Minister.

In this game you are an indeterminate. This means that you have no particular position regarding the central issues of the game. Some of those playing other characters in the game belong to pro-Babbage or anti-Babbage factions, and it is up to them to convince you that their position is correct. You are free to decide on your own which faction to ultimately side with, within the boundaries of your political opinions, outlined below.

You were born on October 30, 1782, in London. Your parents were Thomas Robinson, 2nd Baron Grantham and his wife Lady Mary Jemima Grey Yorke. You were educated at Harrow in London and then at St. John's College, Cambridge.

You were first elected to the House of Commons in 1806, as MP for the Carlow Borough, Ireland. Beginning in 1807 you served another constituency as MP for Ripon, North Yorkshire, in north central England. You served as MP for Ripon until 1827.

You were one of the British party that negotiated the Treaty of Paris in 1814, at the end of the Napoleonic Wars. In 1815 you introduced the Corn Laws to parliament, which restricted

the import of foreign grain. These laws chiefly benefited the agricultural landowners, while driving up the price of food, which had a drastic negative impact on the working poor. Protesters attacked your home after the Corn Laws passed, and two of them were killed when soldiers were called in to quell the protest.

Objectives

You have one primary objective in the game: to win. You win the game if, during GS 6, the characters with a franchise vote for you as the PM who made the best PM speech during the game.

Responsibilities

As Viscount Goderich, you have responsibilities to make a speech outlining the goals and priorities of your government when you take office as PM, to make crucial, informed decisions regarding funding for Charles Babbage's computational engine projects, and to be honest and attentive in your indeterminacy.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to speak regarding your government's priorities and goals

During the first five game sessions, the PM of the day will make a speech at the beginning of the game session corresponding to his rise to power. In the speech, the PM will outline the priorities and goals his government will have while he is in office. Each MP indeterminate will serve as PM exactly once during the first five game sessions. However, you will not know when you will serve until it happens: at the beginning of GS 1, 2, 3, 4, and 5, the GM will announce which of the MP indeterminates will serve as the PM of the day.

Responsibilities regarding the Difference and Analytical Engines

There will be two decisions regarding Babbage's Difference and Analytical Engines during the game: at the end of GS 3 and 6. To one extent or another, the PM at the time will be free to decide what the government should do. So, if you are PM during one of these sessions, you will be tasked with making these decisions. If the responsibility falls on you, you should make the decision you feel is best for the United Kingdom, as informed by your Tory politics and by the presentations and persuasions made by the other characters in the game.

Responsibilities regarding your indeterminacy

Since you do not belong to any faction in this game, you are free to make decisions as you see fit. The factions should try to convince you that their positions are correct, and you should listen carefully to their arguments and other persuasions. Be swayed by those you find the most convincing. The only caveat is that you should try to stay true to your role as a Tory politician of the early 1800s.

During your research for your PM speech, you may come across some opinions that Viscount Goderich held about Babbage and his computing machines. For the purposes of this game, you should ignore these, so that you may play your role as an indeterminate.

Powers

As a MP, you are part of the governing body of the kingdom, and thus a powerful and important man. As a member of the Tory party – the party that has ruled since 1807 – your words and influence are especially important to land owners and the nobility. When you serve as PM, you will have even more power, especially regarding the outcome of this game, for it is the PMs that will ultimately decide whether or not Babbage receives government monies to complete his Difference or Analytical Engines.

To represent your political power, you begin the game with a nudge. You may use the nudge as you wish, spending it in GS 3 or 6, at your discretion.

In addition, you possess a special power: you are an indeterminate MP / PM and you are also a member of the Royal Society. As a FRS, you have a voice – and a vote – during game sessions that take place at the Royal. This is perhaps most important in GS 3, when the Royal must pass either a resolution that recommends funding for Babbage's Difference Engine or a resolution against such funding. As an indeterminate, you should vote for whichever side (pro- or anti-Babbage) that you feel has best persuaded you that their proposal is appropriate.

Assignments

Your main writing assignment is a one-thousand word, written version of your PM speech, due before GS 1 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write a few paragraphs about your vote in each persuasiveness metric poll. These polls, conducted by the gamemaster, take place at the end of each game session. The indeterminates (including you) are asked to vote for the presentation they thought was the most persuasive during the session. Your short paper will describe who you voted for in the persuasiveness metric poll, and why you voted for them. Each of these papers should

be submitted privately to the gamemaster before the beginning of the next game session. For example, your paper about your vote in the GS 1 persuasiveness metric should be sent to the gamemaster no later than the start of GS 2.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British politics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government funding for Babbage's machines, you are indeterminate. You should listen carefully to the arguments made by the factions, and then make up your own mind on this issue, within the general bounds of your politics.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You should support the candidate who you think best represents the values of old-school men of science, and who you think has the most political power.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

As a member of the government, but not of the men of science community, you have not published any scientific work.

With other roles

You share many political leanings with your fellow Tory MPs, Peel and Wellesley, while you are naturally opposed to the politics of the Whigs, Grey and Russell. Your politics gives you a natural affinity to the Duke of Sussex, and puts you in opposition with Charles Babbage.

With your faction

As an indeterminate, you do not belong to either the pro- or anti-Babbage faction.

Strategy advice

You are a Tory politician in Georgian Britain, so you should do some research on what the political stances of your party – and the opposition party, the Whigs – were during that time. You should also learn about the Corn Laws; even though they have already been in effect for a dozen years when our game starts, they are still a point of political contention.

You personally support the abolition of slavery and Catholic Emancipation. Your government hopes to change the way taxes are assessed, moving from indirect taxes to a property tax method, and you also hope to be more conciliatory towards Ireland. You should research these issues; you can use the information you find as material for your PM speech.

To learn more

[Blo12] Marjorie Bloy. Frederick Robinson, Viscount Goderich (1782-1859). <http://www.historyhome.co.uk/pms/goderich.htm>, 2012

[gov16] gov.uk. Past Prime Ministers – GOV.UK. <https://www.gov.uk/government/history/past-prime-ministers>, 2016

Summary of your individual victory objectives

You win the game if, during GS 6, those with a franchise vote for you as the PM who gave the best PM speech during the game.

Portrait of Sir Frederick John Robinson, 1st Viscount Goderick, by Sir Thomas Lawrence, from the National Portrait Gallery, St Martin's Place, London; URL: <http://www.npg.org.uk/collections/search/portrait/mw05348/Frederick-John-Robinson-1st-Earl-of-Ripon?LinkID=mp03796&search=sas&sText=frederick+robinson&role=sit&rNo=0>

Lord John Russell, 1st Earl Russell KG GCMG PC

Whig MP and indeterminate



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You Lord John Russell, 1st Earl Russell KG GCMG PC. You are a Whig MP, and you will serve as Prime Minister.

In this game you are an indeterminate. This means that you have no particular position regarding the central issues of the game. Some of those playing other characters in the game belong to pro-Babbage or anti-Babbage factions, and it is up to them to convince you that their position is correct. You are free to decide on your own which faction to ultimately side with, within the boundaries of your political opinions, outlined below.

You were born on August 18, 1792, in Mayfair, Middlesex, which is now a wealthy area in the West End of London. Your father and namesake was the 6th Duke of Bedford, and your mother was the Honorable Georgina Byng, daughter of George Byng, 4th Viscount Torrington. The relatively high numbers associated with your parents' peerage titles – 6th and 4th – indicate that you were born into a very well-established, wealthy, land-owning family. Since you were not their first son, however, you were not eligible to inherit your father's title or estate, and instead are provided with the courtesy honorific "Lord."

Your education was incomplete, relative to your peers. You were a premature, small, and sickly child, and therefore had to receive your primary education from tutors at home. You did attend the University of Edinburgh starting in 1809, but you never completed

your degree. Despite your stature and ill health, you were a military man, receiving a commission as a Captain in the Bedfordshire Militia in 1810.

You were first elected MP for Tavistock in 1813 – due to your father instructing those with a franchise in the town to vote for you. You soon took up reform causes in Parliament, and you served as Paymaster-General of His Majesty’s Forces during Earl Grey’s government, which began in 1830. You were one of the leading proponents for the Reform Act of 1832. You became the leader of the Whigs in the House of Commons in 1834. You served as Home Secretary in the late 1830s, and in this position you reformed criminal law. The most important part of this reform was the drastic reduction in the number of offenses which were punishable by death.

Objectives

You have one primary objective in the game: to win. You win the game if, during GS 6, the characters with a franchise vote for you as the PM who made the best PM speech during the game.

Responsibilities

As Lord Russell, you have responsibilities to make a speech outlining the goals and priorities of your government when you take office as PM, to make crucial, informed decisions regarding funding for Charles Babbage’s computational engine projects, and to be honest and attentive in your indeterminacy.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to speak regarding your government’s priorities and goals

During the first five game sessions, the PM of the day will make a speech at the beginning of the game session corresponding to his rise to power. In the speech, the PM will outline the priorities and goals his government will have while he is in office. Each MP indeterminate will serve as PM exactly once during the first five game sessions. However, you will not know when you will serve until it happens: at the beginning of GS 1, 2, 3, 4, and 5, the GM will announce which of the MP indeterminates will serve as the PM of the day.

Responsibilities regarding the Difference and Analytical Engines

There will be two decisions regarding Babbage’s Difference and Analytical Engines during the game: at the end of GS 3 and 6. To one extent or another, the PM at the time will be free to decide what the government should do. So, if you are PM during one of these

sessions, you will be tasked with making these decisions. If the responsibility falls on you, you should make the decision you feel is best for the United Kingdom, as informed by your Whig politics and by the presentations and persuasions made by the other characters in the game.

Responsibilities regarding your indeterminacy

Since you do not belong to any faction in this game, you are free to make decisions as you see fit. The factions should try to convince you that their positions are correct, and you should listen carefully to their arguments and other persuasions. Be swayed by those you find the most convincing. The only caveat is that you should try to stay true to your role as a Whig politician of the early 1800s.

During your research for your PM speech, you may come across some opinions that Lord Russell held about Babbage and his computing machines. For the purposes of this game, you should ignore these, so that you may play your role as an indeterminate.

Powers

As a MP, you are part of the governing body of the kingdom, and thus a powerful and important man. As a member of the Whig party, your words and influence are especially important to members of the rising middle class and working people. When you serve as PM, you will have even more power, especially regarding the outcome of this game, for it is the PMs that will ultimately decide whether or not Babbage receives government monies to complete his Difference or Analytical Engines.

To represent your political power, you begin the game with a nudge. You may use the nudge as you wish, spending it in GS 3 or 6, at your discretion.

Assignments

Your main writing assignment is a one-thousand word, written version of your PM speech, due before GS 1 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write a few paragraphs about your vote in each persuasiveness metric poll. These polls, conducted by the gamemaster, take place at the end of each game session. The indeterminates (including you) are asked to vote for the presentation they thought was the most persuasive during the session. Your short paper will describe who you voted for in the persuasiveness metric poll, and why you voted for them. Each of these papers should be submitted privately to the gamemaster before the beginning of the next game session.

For example, your paper about your vote in the GS 1 persuasiveness metric should be sent to the gamemaster no later than the start of GS 2.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British politics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government funding for Babbage's machines, you are indeterminate. You should listen carefully to the arguments made by the factions, and then make up your own mind on this issue, within the general bounds of your politics.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You will not be able to vote in this election, but, once you've decided which candidate you will support, you can work behind the scenes to get him elected.

As for the conflict between imagination and reason, you are truly indeterminate. Listen carefully to the arguments of William King on the supremacy of reason, and those of Byron's Ghost, the champion of imagination. Then, as you hear the rest of the arguments during the game, make up your mind as to which side you will choose.

With texts

As a member of the government, but not of the men of science community, you have not published any scientific work.

With other roles

You share many political leanings with your fellow Whig MP, Lord Grey, and you differ widely with the Tory MPs, Peel, Robinson, and Wellesley. Your Whig sensibilities are also ideologically close to Charles Babbage, but far apart from the Duke of Sussex.

With your faction

As an indeterminate, you do not belong to either the pro- or anti-Babbage faction.

Strategy advice

You are a Whig politician in Georgian Britain, so you should do some research on what the political stances of your party – and the opposition party, the Tories – were during that time.

When playing your character, keep in mind some of the characteristics of Lord John Russell that helped define him and his administration to the British public: shyness, vanity, aloofness, and elitism.

Your government introduced reforms such as the Ten Hours Act. But, your government also inadequately dealt with the Irish Potato Famine, which lasted from 1845 through 1852. You should research these issues; you can use the information you find as material for your PM speech.

To learn more

[gov16] gov.uk. Past Prime Ministers – GOV.UK. <https://www.gov.uk/government/history/past-prime-ministers>, 2016

[Par16] Jonathan Parry. Lord John Russell, later Earl Russell — History of government. <https://history.blog.gov.uk/2016/03/16/lord-john-russell-later-earl-russell/>, 2016

Summary of your individual victory objectives

You win the game if, during GS 6, those with a franchise vote for you as the PM who gave the best PM speech during the game.

Portrait of John Russell, 1st Earl Russell, by Lowes Cato Dickinson, from the National Portrait Gallery, St Martin's Place, London; URL: <http://www.npg.org.uk/collections/search/portrait/mw05516/John-Russell-1st-Earl-Russell?LinkID=mp03900&search=sas&sText=john+russell&role=sit&rNo=10>

Field Marshal Arthur Wellesley, 1st Duke of Wellington KG GCB GCH PC

Tory MP and indeterminate



Biography

[Note: this role sheet may contain information regarding your character from a broad span of time. The game sessions take place during specific years (1828, 1830, and 1846). For any given game session / year, you are responsible for making sure you do not include anything grossly anachronistic in your positions, either in speeches or in writing.]

You are Field Marshal Arthur Wellesley, 1st Duke of Wellington, KG, GCB, GCH, PC. You are a Tory MP, and you will serve as Prime Minister.

In this game you are an indeterminate. This means that you have no particular position regarding the central issues of the game. Some of those playing other characters in the game belong to pro-Babbage or anti-Babbage factions, and it is up to them to convince you that their position is correct. You are free to decide on your own which faction to ultimately side with, within the boundaries of your political opinions, outlined below.

You were born on May 1, 1769, in Dublin, Ireland, to the Earl and Countess of Mornington. You were educated in Brussels and in military school in Angers. In 1787 you were commissioned as an ensign in the British infantry. You served as *aide-de-camp* to Lord Buckingham in Ireland, and rose through the ranks (via your actions, and in two cases, by purchasing higher ranks), becoming a lieutenant colonel in 1793. You became a member of the Irish House of Commons in 1789.

You soon saw combat, however, fighting in the Netherlands in 1794. By 1796 you were a full colonel and on the way to see action in India. You returned in 1805 and were knighted by the king. In 1806 you became a member of the English House of Commons.

You are most famous, however, for your brilliant tactics during the Napoleonic Wars, including your defeat of Napoleon at Waterloo in 1815. On your return to England, you were given an estate in Hampshire and £400,000 (the equivalent of nearly \$38,000,000 in 2017 dollars). You served as Commander in Chief of the British army in France until 1818.

Objectives

You have one primary objective in the game: to win. You win the game if, during GS 6, the characters with a franchise vote for you as the PM who made the best PM speech during the game.

Responsibilities

As the Duke of Wellington, you have responsibilities to make a speech outlining the goals and priorities of your government when you take office as PM, to make crucial, informed decisions regarding funding for Charles Babbage's computational engine projects, and to be honest and attentive in your indeterminacy.

Once you receive this role, you should schedule an appointment with the gamemaster as soon as possible, to discuss the role and clear up any uncertainties regarding your responsibilities or victory conditions.

Responsibility to speak regarding your government's priorities and goals

During the first five game sessions, the PM of the day will make a speech at the beginning of the game session corresponding to his rise to power. In the speech, the PM will outline the priorities and goals his government will have while he is in office. Each MP indeterminate will serve as PM exactly once during the first five game sessions. However, you will not know when you will serve until it happens: at the beginning of GS 1, 2, 3, 4, and 5, the GM will announce which of the MP indeterminates will serve as the PM of the day.

Responsibilities regarding the Difference and Analytical Engines

There will be two decisions regarding Babbage's Difference and Analytical Engines during the game: at the end of GS 3 and 6. To one extent or another, the PM at the time will be free to decide what the government should do. So, if you are PM during one of these sessions, you will be tasked with making these decisions. If the responsibility falls on you, you should make the decision you feel is best for the United Kingdom, as informed by your Tory politics and by the presentations and persuasions made by the other characters in the game.

Responsibilities regarding your indeterminacy

Since you do not belong to any faction in this game, you are free to make decisions as you see fit. The factions should try to convince you that their positions are correct, and you should listen carefully to their arguments and other persuasions. Be swayed by those you find the most convincing. The only caveat is that you should try to stay true to your role as a Tory politician of the early 1800s.

During your research for your PM speech, you may come across some opinions that Wellington held about Babbage and his computing machines. For the purposes of this game, you should ignore these, so that you may play your role as an indeterminate.

Powers

As a MP, you are part of the governing body of the kingdom, and thus a powerful and important man. As a member of the Tory party – the party that has ruled since 1807 – your words and influence are especially important to land owners and the nobility. When you serve as PM, you will have even more power, especially regarding the outcome of this game, for it is the PMs that will ultimately decide whether or not Babbage receives government monies to complete his Difference or Analytical Engines.

To represent your political power, you begin the game with a nudge. You may use the nudge as you wish, spending it in GS 3 or 6, at your discretion.

Assignments

Your main writing assignment is a one-thousand word, written version of your PM speech, due before GS 1 starts. This assignment should be made public to all the players via the document-sharing facility used by your class for the game. The GM will provide more information about how you should submit your assignment, and exactly when it is due.

Additionally, you will write a few paragraphs about your vote in each persuasiveness metric poll. These polls, conducted by the gamemaster, take place at the end of each game session. The indeterminates (including you) are asked to vote for the presentation they thought was the most persuasive during the session. Your short paper will describe who you voted for in the persuasiveness metric poll, and why you voted for them. Each of these papers should be submitted privately to the gamemaster before the beginning of the next game session. For example, your paper about your vote in the GS 1 persuasiveness metric should be sent to the gamemaster no later than the start of GS 2.

Your GM may modify these assignments for your version of the game. Their instructions override the above guidelines.

Relationships

As one of the key players in nineteenth-century British politics, you are connected in one manner or another with many of the ideas, texts, and people that are central to this game.

With ideas

Regarding government funding for Babbage's machines, you are indeterminate. You should listen carefully to the arguments made by the factions, and then make up your own mind on this issue, within the general bounds of your politics.

One of the central issues of the game is the conflict between the old-school men of science (those who are wealthy and can afford to pursue science as a hobby) and the newer men of science (those who study science professionally). This tension will be played out in GS 4, when the next president of the Royal Society is elected. You should support the candidate who you think best represents the values of old-school men of science, and who you think has the most political power.

As for the conflict between imagination and reason, you are more aligned with Byron's Ghost, the champion of imagination, and opposed to William King, who argues for the supremacy of reason.

With texts

As a member of the government, but not of the men of science community, you have not published any scientific work.

With other roles

You share many political leanings with your fellow Tory MPs, Robinson and Peel, while you are naturally opposed to the politics of the Whigs, Grey and Russell. Your politics gives you a natural affinity to the Duke of Sussex, and puts you in opposition with Charles Babbage.

With your faction

As an indeterminate, you do not belong to either the pro- or anti-Babbage faction.

Strategy advice

You are a Tory politician in Georgian Britain, so you should do some research on what the political stances of your party – and the opposition party, the Whigs – were during that time.

Despite your conservative tendencies, you were instrumental in establishing King's College, London, and your administration saw through passage the Catholic Relief Act of 1829. Not everyone in your party was pleased with the passage of the act; the Tory Earl of Winchilsea was so upset he challenged you to a duel. You and he met in Bettersea fields. Winchilsea did not fire and you – intentionally, you say – missed. You should research these issues; you can use the information you find as material for your PM speech.

To learn more

[gov16] gov.uk. Past Prime Ministers – GOV.UK. <https://www.gov.uk/government/history/past-prime-ministers>, 2016

Summary of your individual victory objectives

You win the game if, during GS 6, those with a franchise vote for you as the PM who gave the best PM speech during the game.

Portrait of Arthur Wellesley, 1st Duke of Wellington, by John Jackson, from the National Portrait Gallery, St Martin's Place, London; URL: <http://www.npg.org.uk/collections/search/portrait/mw06659/Arthur-Wellesley-1st-Duke-of-Wellington?LinkID=mp04752&search=sas&sText=Wellesley&wPage=1&role=sit&rNo=20>

Chapter 2

Other handouts

Faction quiz

Faction / name(s): _____

1. How much did the British government offer to pay to purchase copies of mathematical tables produced by the French?
 - (a) £5
 - (b) £50
 - (c) £500
 - (d) £5,000
2. In the early 1800's, Britain used the calculus notation of _____, while the rest of Europe used the notation of _____.
 - (a) Laplace / Newton
 - (b) Leibnitz / Newton
 - (c) Newton / Laplace
 - (d) Newton / Leibnitz
3. True or False: The Difference Engine would have printed mathematical tables.
4. Which of the following was not to be calculated by the Difference Engine?
 - (a) prime numbers
 - (b) random numbers
 - (c) square roots
 - (d) trigonometric functions
5. Which of the following did Babbage not attempt in his childhood?
 - (a) Raising the devil
 - (b) Tutoring mathematics students for money
 - (c) Waking early in the morning to study mathematics
 - (d) Drinking cognac mixed with treacle (molasses)
6. True or False: The Difference Engine was designed to operate with minimal human intervention.

7. Which was a motivating factor for Babbage's Difference Engine?
 - (a) Errors in existing tables
 - (b) The lack of computers
 - (c) The lack of mathematicians
 - (d) The tedium of calculating tables by hand
8. According to Babbage, his Difference Engine could calculate _____ values of the formula $x^2 + x + 41$ in two minutes and thirty seconds.
 - (a) 30
 - (b) 300
 - (c) 3,000
 - (d) 30,000
9. The method of finite differences requires only the repeated application of which mathematical operation?
 - (a) addition
 - (b) division
 - (c) multiplication
 - (d) subtraction
10. In the Difference Engine, the digits of a number were represented by a collection of _____.
 - (a) electrical currents
 - (b) flipping sheets of brass
 - (c) geared wheels
 - (d) rotating cubes
11. True or False: A table produced by the method of finite differences can be checked for accuracy by verifying, by hand, that its last value is correct.
12. Which of the following influenced Babbage's design for the Difference Engine?
 - (a) Tables for insurance underwriting produced by Babbage
 - (b) Tables produced by de Proy for the French government
 - (c) Tables produced by Hutton for the English Board of Longitude
 - (d) Tables produced by the French Board of Longitude
13. True or False: In 1828, a "computer" was an occupation rather than a machine for performing calculations.

14. True or False: When Babbage arrived at Cambridge in 1810, he was well pleased with the kind of mathematics being taught there.
15. True or False: For a given mathematical formula, the Difference Engine was intended to compute a single instance the formula's value.
16. Which of the following occupied much of Babbage's interest while he was designing the Difference Engine?
 - (a) borrows during a division
 - (b) borrows during a subtraction
 - (c) carriages during an addition
 - (d) none of the above
17. True or False: If the "wrong figures" were put in to the Difference Engine at the beginning of a calculation, the right answers would still come out.
18. In 1823, the British government awarded _____ to Babbage to begin construction of his Difference Engine.
 - (a) £15
 - (b) £150
 - (c) £1,500
 - (d) £15,000
19. The group Babbage and his friends formed to advance the new calculus notation in Britain was called the _____ Society.
 - (a) Analytical
 - (b) Calculus
 - (c) Leibnitz
 - (d) Newton
20. True or False: In 1828, Babbage was named the Lucasian Chair of Mathematics at Oxford.

Faction quiz key

1. d
2. d
3. T
4. b
5. b
6. T
7. a
8. a
9. a
10. c
11. T
12. b
13. T
14. F
15. F
16. c
17. F
18. c
19. a
20. F

MOS secret language

The secret terms of the MOS language, and brief definitions that might be used to mollify any politician-type characters who ask for clarification, are:

Asymptype A personality trait one approaches but does not quite achieve.

Calcugraphous A function that can be simultaneously expressed via Leibniz's "summa" notation and also graphically differentiated.

Centon A post-revolutionary French term for the smallest unit of time in normal usage, equivalent to approximately one seventeenth of a second.

Hyperbolification The left-universal moduli of open, null subsets.

Hyperbollipse A four-dimensional croto-conic section.

Hyper symmetric irreducible configuration An arrangement of parts for a machine that are so precise that none can be removed and retain the structural integrity and beauty of the machine.

Infinction A one to one relation between two infinite sets.

Patrrix A two dimensional array which spans the $N \times N$ autonomous space.

Phylantrisk An ideal of the patrrix.

Quasiquad A Fréchet path over contra-linearly multiplicative isometric planes.

Suspensor A function designed to raise another function to a higher plane; also, a unit of measurement for sand dunes.

Semi-regular tangential periodicity The ability of a machine to generally produce results that usually go in the same direction as the answer.

Transfinite functionian definition A function definition that is linearly closed over invariant sub-domains.

Triangulogovector A vector of three Lagrange-Gauss primes with no super-holomorphic elements.

Printable Power Cards

If the GM wishes, the following pages can be printed out on heavy card stock and then cut out to serve as the Power Card deck for the game.

A ♣ Laissez Faire	A ♦ Intervention	A ♥ Prescience
Power	Power	Power
Once this PC is played, no further PCs (except for the Intervention card) may be played during the game session.	Nullifies the power of Laissez Faire.	The GM shows you the top five cards in the PC deck.
A ♠ Muddle	K ♣ Swing Riot	K ♦ Invigorate
Power	Power	Power
The GM shuffles the PC deck.	Captain Swing takes a PC from a randomly chosen character.	Select a card of your choice from the PC discard deck.

K ♥ Volte-Face
Power
Reverses the effect of the previous PC (Power or Nudge).

K ♠ Volte-Face
Power
Reverses the effect of the previous PC (Power or Nudge).

Q ♣ Cholera
Power
After 1830, cause any character to die of Cholera.

Q ♦ Tar & Vinegar
Power
50% chance of curing someone of Cholera.

Q ♥ Shove
Power
Doubles a nudge.

Q ♠ Shove
Power
Doubles a nudge.

J ♣ Excess
Power
Draw twice from the PC deck instead of once.

J ♦ Excess
Power
Draw twice from the PC deck instead of once.

J ♥ Exflunctify
Power
Cause a character to lose all of their PCs.

J ♠ Topsy
Power
Reverse the target of a PC card.

10 ♠
Nudge
Exert influence on PM decisions and election.

10 ♦
Nudge
Exert influence on PM decisions and election.

10 ♡
Nudge
Exert influence on PM decisions and election.

10 ♠
Nudge
Exert influence on PM decisions and election.

9 ♣
Nudge
Exert influence on PM decisions and election.

9 ♦
Nudge
Exert influence on PM decisions and election.

9 ♡
Nudge
Exert influence on PM decisions and election.

9 ♠
Nudge
Exert influence on PM decisions and election.

8 ♣
Nudge
Exert influence on PM decisions and election.

8 ♦
Nudge
Exert influence on PM decisions and election.

8 ♥
Nudge
Exert influence on PM decisions and election.

8 ♠
Nudge
Exert influence on PM decisions and election.

7 ♣
Nudge
Exert influence on PM decisions and election.

7 ♦
Nudge
Exert influence on PM decisions and election.

7 ♥
Nudge
Exert influence on PM decisions and election.

7 ♠
Nudge
Exert influence on PM decisions and election.

6 ♣
Nudge
Exert influence on PM decisions and election.

6 ♦
Nudge
Exert influence on PM decisions and election.

6 ♥
Nudge
Exert influence on PM decisions and election.

6 ♠
Nudge
Exert influence on PM decisions and election.

5 ♣

Nudge

Exert influence on PM decisions and election.

5 ♦

Nudge

Exert influence on PM decisions and election.

5 ♥

Nudge

Exert influence on PM decisions and election.

5 ♠

Nudge

Exert influence on PM decisions and election.

4 ♣

Nudge

Exert influence on PM decisions and election.

4 ♦

Nudge

Exert influence on PM decisions and election.

4 ♥
Nudge
Exert influence on PM decisions and election.

4 ♠
Nudge
Exert influence on PM decisions and election.

3 ♣
Nudge
Exert influence on PM decisions and election.

3 ♦
Nudge
Exert influence on PM decisions and election.

3 ♥
Nudge
Exert influence on PM decisions and election.

3 ♠
Nudge
Exert influence on PM decisions and election.

2 ♣
Nudge
Exert influence on PM decisions and election.

2 ♦
Nudge
Exert influence on PM decisions and election.

2 ♥
Nudge
Exert influence on PM decisions and election.

2 ♠
Nudge
Exert influence on PM decisions and election.

GS 1 party game

This section contains the handouts for the party game that serves as the liminal experience at Babbage's party during the GS 1. The first handout is for Babbage, who serves as the presiding officer for GS 1, and the master of the revels for the game. The second handout is for everyone else attending the party. The game, "I Love my Love with an A," is adapted from the 1825 book, *Winter Evening Passtimes; or, The Merry-Maker's Companion* [Rev25].

I Love my Love with an A

For Charles Babbage, master of the revels

Your party during takes place during GS 1, and, after the Fury speaks, the party starts off with a rousing game of “I Love my Love with an A,” described below. The game, circa 1825, is similar to modern *Mad Libs* books. Players have to recite an ode with their own adjectives, food items, etc. added in. A wrinkle to the game, compared to *Mad Libs*, is that all of the items chosen for the ode must begin with the same letter. Additionally, the specified letter is different for each player in turn.

As the master of the revels, you choose the player to begin with the letter ‘A.’ For example, you might say, “Mr. Hershel, recite the ode with the letter ‘A’.”

Herschel might then say, “I love my love with an A, because she is ardent: I hate her, because she is ambitious: I took her to Andover, to the sign of the Angel: I treated her with artichokes; and her name is Anne Adair.” After repeating the ode, Herschel then chooses who goes next, as in: “Mrs. Somerville, recite the ode with the letter ‘B’.” Somerville recites the ode with ‘B’ words, and chooses who will recite with ‘C.’ This process continues for each letter, with the exceptions of ‘K,’ ‘Q,’ ‘X,’ ‘Y,’ and ‘Z.’ If there are enough players in the game, the cycle begins again at ‘A’ after someone completes ‘W.’

As master of the revels, you are also responsible for assigning forfeits. The gamemaster will help keep track of the forfeits as the game proceeds. Forfeits are assigned if:

- If a player cannot complete the ode using their assigned letter, they receive a forfeit
- If a player prompts another player aloud with suggestions for their ode, the suggesting player receives a forfeit
- If a player includes a grossly anachronistic word in their ode, they receive a forfeit; for example, “pizza,” “French fries,” or “hot dogs” are food items that should generate a forfeit
- If a player, when choosing who to recite the ode next, chooses a player that has already recited, they receive a forfeit
- If a player addresses another player with their real name (instead of their character’s name), they receive a forfeit
- If a player addresses another player with the incorrect title, they receive a forfeit; for example, if they say “Mr. Russell, recite the ode with the letter ‘K,’” instead of “Lord Russell, recite the ode with the letter ‘K,’” they receive a forfeit
- If there are enough players to reuse letters, repeating a word from the first time around merits a forfeit

When you detect a forfeit, briefly interrupt the proceedings and say something like, “a forfeit on Mr. Faraday for re-using the word ‘apple!’”

The instructions for the game that the other characters receive are printed below.

I Love my Love with an A

This game is too well known to require much description. The difficulty may be increased, where the players are expert, by adding some article of dress, some bird or animal, to be presented, or the composition of some author. Every person must take a letter in rotation; and if the alphabet goes around more than once, it becomes a forfeit to mention any name employed before. Any person not being able to find a suitable word, pays a forfeit; likewise, any one prompting another.

The master of the revels shall choose the person that starts; this lady or gentleman must begin with the letter 'A.' Having thus completed the ode, he then chooses the next person, who must recite the ode using the letter 'B,' and so forth through the alphabet. The letters 'K,' 'Q,' 'X,' 'Y,' and 'Z' may be passed over. Any person choosing another who has already recited the ode to be next, pays a forfeit.

The ode:

I love my love with a(n) (letter), because she is (adjective): I hate her, because she is (adjective): I took her to (location), to the sign of the (noun): I treated her with (food); and her name is (first name)(last name).

I Love my Love with an A

This game is too well known to require much description. The difficulty may be increased, where the players are expert, by adding some article of dress, some bird or animal, to be presented, or the composition of some author. Every person must take a letter in rotation; and if the alphabet goes around more than once, it becomes a forfeit to mention any name employed before. Any person not being able to find a suitable word, pays a forfeit; likewise, any one prompting another.

The master of the revels shall choose the person that starts; this lady or gentleman must begin with the letter 'A.' Having thus completed the ode, he then chooses the next person, who must recite the ode using the letter 'B,' and so forth through the alphabet. The letters 'K,' 'Q,' 'X,' 'Y,' and 'Z' may be passed over. Any person choosing another who has already recited the ode to be next, pays a forfeit.

The ode:

I love my love with a(n) (letter), because she is (adjective): I hate her, because she is (adjective): I took her to (location), to the sign of the (noun): I treated her with (food); and her name is (first name)(last name).

Slide rule construction

This section provides instructions on how to build the cut-out circular slide rule used by Somerville and Frennd in GS 1. The pages to print are in three separate files, **SR1.pdf**, **SR2.pdf**, and **SR3.pdf**. If your class is using these slide rules, the gamemaster should construct enough examples to satisfy the needs of the class.

Instructions for constructing the slide rules:

1. Print one copy of **SR1.pdf** and **SR2.pdf** for each slide rule needed, on heavy card stock
2. Print enough copies of **SR3.pdf** to make the desired number of slide rules, on transparency film. There are six cursors on each page, so one page makes six slide rules
3. Carefully cut out the circular section of the **SR2.pdf** pages
4. Carefully cut out the cursors from the **SR3.pdf** pages
5. Assemble each slide rule thus:
 - (a) Bottom layer: entire **SR1.pdf** page
 - (b) Middle layer: circular part from **SR2.pdf** page
 - (c) Top layer: cursor from **SR3.pdf** page

Make sure that the centers are all aligned as perfectly as possible.

6. Use a scrapbooking brad punch or similar tool to fasten the three layers together. The middle and top layers should rotate freely, without too much wiggle.

Slide rule instructions

This section provides instructions on how to use the cut-out circular slide rule to perform mathematical operations.

Configuration and general information

The circular slide rule has three parts: a back sheet that does not move, an inner circular sheet that rotates, and a clear, rectangular cursor that is also free to rotate. There are six *scales* on the slide rule. They are, outer to inner, the A, D, K, B, C, and CI scales. Different combinations of the scales are used to perform the calculations outlined below.

The beginning of each scale is the numeral ‘1’ closest to the scale’s label. We read most of the scales clockwise from the beginning. We must be careful to realize what each of the ticks on a scale represent, e.g., the first tick clockwise from 1 on the C scale represents 1.01, while the first tick clockwise from 2 on the B scale represents 2.05.

When we are using the slide rule, we are responsible for determining where the decimal point should be placed in our answer. For example, if our answer is exactly half way between 4 and 5 on the B scale, our answer might be 0.45, 4.5, 45, 450, and so on. We must be comfortable enough with mental mathematics to place the decimal point in the correct position.

We must also recognize that when we are working with large numbers, the answers we get from the slide rule will not be exact, but rather approximations. However, if we are careful with our work, the error will probably not exceed 1%.

Multiplication

To multiply two numbers, use the scales labeled C and D. To compute $z = x \times y$, follow this procedure:

1. Rotate the cursor until it is over x on the D scale.
2. While holding the cursor steady, rotate the inner circle until the 1 on the C scale is at the cursor.
3. While holding the inner circle steady, rotate the cursor until it is over y on the C scale.

At the end of this procedure, z is the number under the cursor on the D scale.

Sample problems

1. Calculate 8×7 .

2. Calculate 2.3×4.5 .
3. Calculate 123×360 .
4. Calculate $191,622 \times 478,128$.
5. Calculate $70,523,830 \times 6,262$.

Division

To divide two numbers, use the scales labeled C and D. To compute $z = x \div y$, follow this procedure:

1. Rotate the cursor until it is over x on the D scale.
2. While holding the cursor steady, rotate the inner circle until the value y on the C scale is at the cursor.
3. While holding the inner circle steady, rotate the cursor until it is over the 1 on the C scale.

At the end of this procedure, z is the number under the cursor on the D scale.

Sample problems

1. Calculate $9 \div 3$.
2. Calculate $136 \div 8$.
3. Calculate $123 \div 360$.
4. Calculate $655804 \div 6774$.
5. Calculate $2.68 \div 4.33$.

Reciprocals

To find the reciprocal of a number x , $z = \frac{1}{x}$, we use the C and CI scales:

1. Move the cursor to x on the CI scale.

The number under the cursor on the C scale is z .

Sample problems

1. Find the reciprocal of 5.
2. Find the reciprocal of 3.4.

- Find the reciprocal of 6808.

Squares

To find the square of a number x , $z = x^2 = x \times x$, we use the B and C scales:

- Move the cursor to x on the C scale.

The number under the cursor on the B scale is z .

Sample problems

- Calculate 7^2 .
- Calculate 19^2 .
- Calculate 9076^2 .
- Calculate $(2.85 \times 10^6)^2$.
- Calculate $(5.18 \times 10^{-3})^2$.

Square roots

To calculate square roots, $z = \sqrt{x}$, we use the B and C scales. Rotate the inner circle of the slide rule until the 1s on all the scales are in the three o'clock position. Note that the top and bottom halves of the B scale are identical. If x has an odd number of digits or leading zeros after the decimal point, use the bottom half of the B scale in this position. If x has an even number of digits or leading zeros after the decimal point, use the top half of the B scale. Follow this procedure:

- Move the cursor over the value of x on the appropriate half of the B scale.

The value under the cursor on the C scale is z .

Sample problems

- Calculate $\sqrt{9}$.
- Calculate $\sqrt{81}$.
- Calculate $\sqrt{169}$.
- Calculate $\sqrt{2916}$.
- Calculate $\sqrt{0.003750}$.
- Calculate $\sqrt{5.62 \times 10^{-4}}$.

Cubes

To calculate the cube of a number x , $z = x^3 = x \times x \times x$, we use the D and K scales:

1. Move the cursor to x on the D scale.

The value under the cursor on the K scale is z .

Sample problems

1. Calculate 3^3 .
2. Calculate 8^3 .
3. Calculate 879^3 .
4. Calculate $(5.37 \times 10^{-2})^3$.

Common Logarithms of Numbers

N	0	1	2	3	4	5	6	7	8	9
100	00000	00043	00087	00130	00173	00217	00260	00303	00346	00389
01	00432	00475	00518	00561	00604	00647	00689	00732	00775	00817
02	00860	00903	00945	00988	01030	01072	01115	01157	01199	01242
03	01284	01326	01368	01410	01452	01494	01536	01578	01620	01662
04	01703	01745	01787	01828	01870	01912	01953	01995	02036	02078
05	02119	02160	02202	02243	02284	02325	02366	02407	02449	02490
06	02531	02572	02612	02653	02694	02735	02776	02816	02857	02898
07	02938	02979	03019	03060	03100	03141	03181	03222	03262	03302
08	03342	03383	03423	03463	03503	03543	03583	03623	03663	03703
09	03743	03782	03822	03862	03902	03941	03981	04021	04060	04100
110	04139	04179	04218	04258	04297	04336	04376	04415	04454	04493
01	04532	04571	04610	04650	04689	04727	04766	04805	04844	04883
02	04922	04961	04999	05038	05077	05115	05154	05192	05231	05269
03	05308	05346	05385	05423	05461	05500	05538	05576	05614	05652
04	05690	05729	05767	05805	05843	05881	05918	05956	05994	06032
05	06070	06108	06145	06183	06221	06258	06296	06333	06371	06408
06	06446	06483	06521	06558	06595	06633	06670	06707	06744	06781
07	06819	06856	06893	06930	06967	07004	07041	07078	07115	07151
08	07188	07225	07262	07298	07335	07372	07408	07445	07482	07518
09	07555	07591	07628	07664	07700	07737	07773	07809	07846	07882
120	07918	07954	07990	08027	08063	08099	08135	08171	08207	08243
01	08279	08314	08350	08386	08422	08458	08493	08529	08565	08600
02	08636	08672	08707	08743	08778	08814	08849	08884	08920	08955
03	08991	09026	09061	09096	09132	09167	09202	09237	09272	09307
04	09342	09377	09412	09447	09482	09517	09552	09587	09621	09656
05	09691	09726	09760	09795	09830	09864	09899	09934	09968	10003
06	10037	10072	10106	10140	10175	10209	10243	10278	10312	10346
07	10380	10415	10449	10483	10517	10551	10585	10619	10653	10687
08	10721	10755	10789	10823	10857	10890	10924	10958	10992	11025
09	11059	11093	11126	11160	11193	11227	11261	11294	11327	11361
130	11394	11428	11461	11494	11528	11561	11594	11628	11661	11694
01	11727	11760	11793	11826	11860	11893	11926	11959	11992	12024
02	12057	12090	12123	12156	12189	12222	12254	12287	12320	12352
03	12385	12418	12450	12483	12516	12548	12581	12613	12646	12678
04	12710	12743	12775	12808	12840	12872	12905	12937	12969	13001
05	13033	13066	13098	13130	13162	13194	13226	13258	13290	13322
06	13354	13386	13418	13450	13481	13513	13545	13577	13609	13640
07	13672	13704	13735	13767	13799	13830	13862	13893	13925	13956
08	13988	14019	14051	14082	14114	14145	14176	14208	14239	14270
09	14301	14333	14364	14395	14426	14457	14489	14520	14551	14582
140	14613	14644	14675	14706	14737	14768	14799	14829	14860	14891
01	14922	14953	14983	15014	15045	15076	15106	15137	15168	15198
02	15229	15259	15290	15320	15351	15381	15412	15442	15473	15503
03	15534	15564	15594	15625	15655	15685	15715	15746	15776	15806
04	15836	15866	15897	15927	15957	15987	16017	16047	16077	16107
05	16137	16167	16197	16227	16256	16286	16316	16346	16376	16406
06	16435	16465	16495	16524	16554	16584	16613	16643	16673	16702
07	16732	16761	16791	16820	16850	16879	16909	16938	16967	16997
08	17026	17056	17085	17114	17143	17173	17202	17231	17260	17289
09	17319	17348	17377	17406	17435	17464	17493	17522	17551	17580
150	17609	17638	17667	17696	17725	17754	17782	17811	17840	17869
01	17898	17926	17955	17984	18013	18041	18070	18099	18127	18156
02	18184	18213	18241	18270	18298	18327	18355	18384	18412	18441
03	18469	18498	18526	18554	18583	18611	18639	18667	18696	18724
04	18752	18780	18808	18837	18865	18893	18921	18949	18977	19005
05	19033	19061	19089	19117	19145	19173	19201	19229	19257	19285
06	19312	19340	19368	19396	19424	19451	19479	19507	19535	19562
07	19590	19618	19645	19673	19700	19728	19756	19783	19811	19838
08	19866	19893	19921	19948	19976	20003	20030	20058	20085	20112
09	20140	20167	20194	20222	20249	20276	20303	20330	20358	20385
160	20412	20439	20466	20493	20520	20548	20575	20602	20629	20656
01	20683	20710	20737	20763	20790	20817	20844	20871	20898	20925
02	20952	20978	21005	21032	21059	21085	21112	21139	21165	21192

N	0	1	2	3	4	5	6	7	8	9
03	21219	21245	21272	21299	21325	21352	21378	21405	21431	21458
04	21484	21511	21537	21564	21590	21617	21643	21669	21696	21722
05	21748	21775	21801	21827	21854	21880	21906	21932	21958	21985
06	22011	22037	22063	22089	22115	22141	22167	22194	22220	22246
07	22272	22298	22324	22350	22376	22401	22427	22453	22479	22505
08	22531	22557	22583	22608	22634	22660	22686	22712	22737	22763
09	22789	22814	22840	22866	22891	22917	22943	22968	22994	23019
170	23045	23070	23096	23121	23147	23172	23198	23223	23249	23274
01	23300	23325	23350	23376	23401	23426	23452	23477	23502	23528
02	23553	23578	23603	23629	23654	23679	23704	23729	23754	23779
03	23805	23830	23855	23880	23905	23930	23955	23980	24005	24030
04	24055	24080	24105	24130	24155	24180	24204	24229	24254	24279
05	24304	24329	24353	24378	24403	24428	24452	24477	24502	24527
06	24551	24576	24601	24625	24650	24674	24699	24724	24748	24773
07	24797	24822	24846	24871	24895	24920	24944	24969	24993	25018
08	25042	25066	25091	25115	25139	25164	25188	25212	25237	25261
09	25285	25310	25334	25358	25382	25406	25431	25455	25479	25503
180	25527	25551	25575	25600	25624	25648	25672	25696	25720	25744
01	25768	25792	25816	25840	25864	25888	25912	25935	25959	25983
02	26007	26031	26055	26079	26102	26126	26150	26174	26198	26221
03	26245	26269	26293	26316	26340	26364	26387	26411	26435	26458
04	26482	26505	26529	26553	26576	26600	26623	26647	26670	26694
05	26717	26741	26764	26788	26811	26834	26858	26881	26905	26928
06	26951	26975	26998	27021	27045	27068	27091	27114	27138	27161
07	27184	27207	27231	27254	27277	27300	27323	27346	27370	27393
08	27416	27439	27462	27485	27508	27531	27554	27577	27600	27623
09	27646	27669	27692	27715	27738	27761	27784	27807	27830	27852
190	27875	27898	27921	27944	27967	27989	28012	28035	28058	28081
01	28103	28126	28149	28171	28194	28217	28240	28262	28285	28307
02	28330	28353	28375	28398	28421	28443	28466	28488	28511	28533
03	28556	28578	28601	28623	28646	28668	28691	28713	28735	28758
04	28780	28803	28825	28847	28870	28892	28914	28937	28959	28981
05	29003	29026	29048	29070	29092	29115	29137	29159	29181	29203
06	29226	29248	29270	29292	29314	29336	29358	29380	29403	29425
07	29447	29469	29491	29513	29535	29557	29579	29601	29623	29645
08	29667	29688	29710	29732	29754	29776	29798	29820	29842	29863
09	29885	29907	29929	29951	29973	29994	30016	30038	30060	30081
200	30103	30125	30146	30168	30190	30211	30233	30255	30276	30298
01	30320	30341	30363	30384	30406	30428	30449	30471	30492	30514
02	30535	30557	30578	30600	30621	30643	30664	30685	30707	30728
03	30750	30771	30792	30814	30835	30856	30878	30899	30920	30942
04	30963	30984	31006	31027	31048	31069	31091	31112	31133	31154
05	31175	31197	31218	31239	31260	31281	31302	31323	31345	31366
06	31387	31408	31429	31450	31471	31492	31513	31534	31555	31576
07	31597	31618	31639	31660	31681	31702	31723	31744	31765	31785
08	31806	31827	31848	31869	31890	31911	31931	31952	31973	31994
09	32015	32035	32056	32077	32098	32118	32139	32160	32181	32201
210	32222	32243	32263	32284	32305	32325	32346	32366	32387	32408
01	32428	32449	32469	32490	32510	32531	32552	32572	32593	32613
02	32634	32654	32675	32695	32715	32736	32756	32777	32797	32818
03	32838	32858	32879	32899	32919	32940	32960	32980	33001	33021
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07	95761	95766	95770	95775	95780	95785	95789	95794	95799	95804
08	95809	95813	95818	95823	95828	95832	95837	95842	95847	95852
09	95856	95861	95866	95871	95875	95880	95885	95890	95895	95899
910	95904	95909	95914	95918	95923	95928	95933	95938	95942	95947
01	95952	95957	95961	95966	95971	95976	95980	95985	95990	95995
02	95999	96004	96009	96014	96019	96023	96028	96033	96038	96042
03	96047	96052	96057	96061	96066	96071	96076	96080	96085	96090
04	96095	96099	96104	96109	96114	96118	96123	96128	96133	96137
05	96142	96147	96152	96156	96161	96166	96171	96175	96180	96185
06	96190	96194	96199	96204	96209	96213	96218	96223	96227	96232
07	96237	96242	96246	96251	96256	96261	96265	96270	96275	96280
08	96284	96289	96294	96298	96303	96308	96313	96317	96322	96327
09	96332	96336	96341	96346	96350	96355	96360	96365	96369	96374
920	96379	96384	96388	96393	96398	96402	96407	96412	96417	96421
01	96426	96431	96435	96440	96445	96450	96454	96459	96464	96468
02	96473	96478	96483	96487	96492	96497	96501	96506	96511	96515
03	96520	96525	96530	96534	96539	96544	96548	96553	96558	96562
04	96567	96572	96577	96581	96586	96591	96595	96600	96605	96609
05	96614	96619	96624	96628	96633	96638	96642	96647	96652	96656
06	96661	96666	96670	96675	96680	96685	96689	96694	96699	96703
07	96708	96713	96717	96722	96727	96731	96736	96741	96745	96750
08	96755	96759	96764	96769	96774	96778	96783	96788	96792	96797
09	96802	96806	96811	96816	96820	96825	96830	96834	96839	96844
930	96848	96853	96858	96862	96867	96872	96876	96881	96886	96890
01	96895	96900	96904	96909	96914	96918	96923	96928	96932	96937
02	96942	96946	96951	96956	96960	96965	96970	96974	96979	96984
03	96988	96993	96997	97002	97007	97011	97016	97021	97025	97030
04	97035	97039	97044	97049	97053	97058	97063	97067	97072	97077
05	97081	97086	97090	97095	97100	97104	97109	97114	97118	97123
06	97128	97132	97137	97142	97146	97151	97155	97160	97165	97169
07	97174	97179	97183	97188	97192	97197	97202	97206	97211	97216
08	97220	97225	97230	97234	97239	97243	97248	97253	97257	97262
09	97267	97271	97276	97280	97285	97290	97294	97299	97304	97308
940	97313	97317	97322	97327	97331	97336	97340	97345	97350	97354
01	97359	97364	97368	97373	97377	97382	97387	97391	97396	97400
02	97405	97410	97414	97419	97424	97428	97433	97437	97442	97447
03	97451	97456	97460	97465	97470	97474	97479	97483	97488	97493
04	97497	97502	97506	97511	97516	97520	97525	97529	97534	97539
05	97543	97548	97552	97557	97562	97566	97571	97575	97580	97585
06	97589	97594	97598	97603	97607	97612	97617	97621	97626	97630
07	97635	97640	97644	97649	97653	97658	97663	97667	97672	97676
08	97681	97685	97690	97695	97699	97704	97708	97713	97717	97722
09	97727	97731	97736	97740	97745	97749	97754	97759	97763	97768
950	97772	97777	97782	97786	97791	97795	97800	97804	97809	97813
01	97818	97823	97827	97832	97836	97841	97845	97850	97855	97859
02	97864	97868	97873	97877	97882	97886	97891	97896	97900	97905
03	97909	97914	97918	97923	97928	97932	97937	97941	97946	97950
04	97955	97959	97964	97968	97973	97978	97982	97987	97991	97996
05	98000	98005	98009	98014	98019	98023	98028	98032	98037	98041
06	98046	98050	98055	98059	98064	98068	98073	98078	98082	98087
07	98091	98096	98100	98105	98109	98114	98118	98123	98127	98132
08	98137	98141	98146	98150	98155	98159	98164	98168	98173	98177
09	98182	98186	98191	98195	98200	98204	98209	98214	98218	98223
960	98227	98232	98236	98241	98245	98250	98254	98259	98263	98268
01	98272	98277	98281	98286	98290	98295	98299	98304	98308	98313
02	98318	98322	98327	98331	98336	98340	98345	98349	98354	98358
03	98363	98367	98372	98376	98381	98385	98390	98394	98399	98403
04	98408	98412	98417	98421	98426	98430	98435	98439	98444	98448
05	98453	98457	98462	98466	98471	98475	98480	98484	98489	98493
06	98498	98502	98507	98511	98516	98520	98525	98529	98534	98538

N	0	1	2	3	4	5	6	7	8	9
07	98543	98547	98552	98556	98561	98565	98570	98574	98579	98583
08	98588	98592	98597	98601	98605	98610	98614	98619	98623	98628
09	98632	98637	98641	98646	98650	98655	98659	98664	98668	98673
970	98677	98682	98686	98691	98695	98700	98704	98709	98713	98717
01	98722	98726	98731	98735	98740	98744	98749	98753	98758	98762
02	98767	98771	98776	98780	98784	98789	98793	98798	98802	98807
03	98811	98816	98820	98825	98829	98834	98838	98843	98847	98851
04	98856	98860	98865	98869	98874	98878	98883	98887	98892	98896
05	98900	98905	98909	98914	98918	98923	98927	98932	98936	98941
06	98945	98949	98954	98958	98963	98967	98972	98976	98981	98985
07	98989	98994	98998	99003	99007	99012	99016	99021	99025	99029
08	99034	99038	99043	99047	99052	99056	99061	99065	99069	99074
09	99078	99083	99087	99092	99096	99100	99105	99109	99114	99118
980	99123	99127	99131	99136	99140	99145	99149	99154	99158	99162
01	99167	99171	99176	99180	99185	99189	99193	99198	99202	99207
02	99211	99216	99220	99224	99229	99233	99238	99242	99247	99251
03	99255	99260	99264	99269	99273	99277	99282	99286	99291	99295
04	99300	99304	99308	99313	99317	99322	99326	99330	99335	99339
05	99344	99348	99352	99357	99361	99366	99370	99374	99379	99383
06	99388	99392	99396	99401	99405	99410	99414	99419	99423	99427
07	99432	99436	99441	99445	99449	99454	99458	99463	99467	99471
08	99476	99480	99484	99489	99493	99498	99502	99506	99511	99515
09	99520	99524	99528	99533	99537	99542	99546	99550	99555	99559
990	99564	99568	99572	99577	99581	99585	99590	99594	99599	99603
01	99607	99612	99616	99621	99625	99629	99634	99638	99642	99647
02	99651	99656	99660	99664	99669	99673	99677	99682	99686	99691
03	99695	99699	99704	99708	99712	99717	99721	99726	99730	99734
04	99739	99743	99747	99752	99756	99760	99765	99769	99774	99778
05	99782	99787	99791	99795	99800	99804	99808	99813	99817	99822
06	99826	99830	99835	99839	99843	99848	99852	99856	99861	99865
07	99870	99874	99878	99883	99887	99891	99896	99900	99904	99909
08	99913	99917	99922	99926	99930	99935	99939	99944	99948	99952
09	99957	99961	99965	99970	99974	99978	99983	99987	99991	99996

Cosine table instructions

This section provides instructions on how to use the table of cosines to perform the cosine trigonometry operation.

Configuration and general information

The Table of cosines can be used to find the cosine of an angle in degrees. The table allows one to evaluate $\cos \Theta$, for values $[0.0 \leq \Theta \leq 359.9]$.

Finding $\cos \Theta$, for an angle in degrees

If the angle Θ is specified in degrees, to find $\cos \Theta$, first think of the angle $\Theta = w.d$ in terms of its whole and fractional components w and d , where $[0 \leq w \leq 359]$ and $[0 \leq d \leq 9]$. Then, follow this procedure:

1. Move down to row w in the table of cosines.
2. Move right to column d on row w .

An approximate value for $\cos \Theta$ is in row w , column d .

Sample problems

In this problems, each angle is specified in degrees.

1. Calculate $\cos 25.1$.
2. Calculate $\cos 58.4$.
3. Calculate $\cos 113.7$.
4. Calculate $\cos 186.0$.
5. Calculate $\cos 257.9$.

Finding $\cos \Theta$, for an angle in radians

If the angle Θ is specified in radians, we must first convert the angle from radians into degrees. To do this, first multiply the angle by 180, and then divide this result by $\pi \approx 3.14159$. Then, proceed with the procedure above to calculate the value of $\cos \Theta$.

Sample problems

In these problems, each angle is specified in radians.

1. Calculate $\cos 0.75$.
2. Calculate $\cos 1.68$.
3. Calculate $\cos 2.11$.
4. Calculate $\cos 3.95$.
5. Calculate $\cos 5.62$.

Table of cosines

Θ	0	1	2	3	4	5	6	7	8	9
0	0.000	0.002	0.003	0.005	0.007	0.009	0.010	0.012	0.014	0.016
1	0.017	0.019	0.021	0.023	0.024	0.026	0.028	0.030	0.031	0.033
2	0.035	0.037	0.038	0.040	0.042	0.044	0.045	0.047	0.049	0.051
3	0.052	0.054	0.056	0.058	0.059	0.061	0.063	0.065	0.066	0.068
4	0.070	0.071	0.073	0.075	0.077	0.078	0.080	0.082	0.084	0.085
5	0.087	0.089	0.091	0.092	0.094	0.096	0.098	0.099	0.101	0.103
6	0.105	0.106	0.108	0.110	0.111	0.113	0.115	0.117	0.118	0.120
7	0.122	0.124	0.125	0.127	0.129	0.131	0.132	0.134	0.136	0.137
8	0.139	0.141	0.143	0.144	0.146	0.148	0.150	0.151	0.153	0.155
9	0.156	0.158	0.160	0.162	0.163	0.165	0.167	0.168	0.170	0.172
10	0.174	0.175	0.177	0.179	0.181	0.182	0.184	0.186	0.187	0.189
11	0.191	0.193	0.194	0.196	0.198	0.199	0.201	0.203	0.204	0.206
12	0.208	0.210	0.211	0.213	0.215	0.216	0.218	0.220	0.222	0.223
13	0.225	0.227	0.228	0.230	0.232	0.233	0.235	0.237	0.239	0.240
14	0.242	0.244	0.245	0.247	0.249	0.250	0.252	0.254	0.255	0.257
15	0.259	0.261	0.262	0.264	0.266	0.267	0.269	0.271	0.272	0.274
16	0.276	0.277	0.279	0.281	0.282	0.284	0.286	0.287	0.289	0.291
17	0.292	0.294	0.296	0.297	0.299	0.301	0.302	0.304	0.306	0.307
18	0.309	0.311	0.312	0.314	0.316	0.317	0.319	0.321	0.322	0.324
19	0.326	0.327	0.329	0.331	0.332	0.334	0.335	0.337	0.339	0.340
20	0.342	0.344	0.345	0.347	0.349	0.350	0.352	0.353	0.355	0.357
21	0.358	0.360	0.362	0.363	0.365	0.367	0.368	0.370	0.371	0.373
22	0.375	0.376	0.378	0.379	0.381	0.383	0.384	0.386	0.388	0.389
23	0.391	0.392	0.394	0.396	0.397	0.399	0.400	0.402	0.404	0.405
24	0.407	0.408	0.410	0.412	0.413	0.415	0.416	0.418	0.419	0.421
25	0.423	0.424	0.426	0.427	0.429	0.431	0.432	0.434	0.435	0.437
26	0.438	0.440	0.442	0.443	0.445	0.446	0.448	0.449	0.451	0.452
27	0.454	0.456	0.457	0.459	0.460	0.462	0.463	0.465	0.466	0.468
28	0.469	0.471	0.473	0.474	0.476	0.477	0.479	0.480	0.482	0.483
29	0.485	0.486	0.488	0.489	0.491	0.492	0.494	0.495	0.497	0.498
30	0.500	0.502	0.503	0.505	0.506	0.508	0.509	0.511	0.512	0.514
31	0.515	0.517	0.518	0.520	0.521	0.522	0.524	0.525	0.527	0.528
32	0.530	0.531	0.533	0.534	0.536	0.537	0.539	0.540	0.542	0.543
33	0.545	0.546	0.548	0.549	0.550	0.552	0.553	0.555	0.556	0.558
34	0.559	0.561	0.562	0.564	0.565	0.566	0.568	0.569	0.571	0.572
35	0.574	0.575	0.576	0.578	0.579	0.581	0.582	0.584	0.585	0.586
36	0.588	0.589	0.591	0.592	0.593	0.595	0.596	0.598	0.599	0.600
37	0.602	0.603	0.605	0.606	0.607	0.609	0.610	0.612	0.613	0.614
38	0.616	0.617	0.618	0.620	0.621	0.623	0.624	0.625	0.627	0.628
39	0.629	0.631	0.632	0.633	0.635	0.636	0.637	0.639	0.640	0.641
40	0.643	0.644	0.645	0.647	0.648	0.649	0.651	0.652	0.653	0.655
41	0.656	0.657	0.659	0.660	0.661	0.663	0.664	0.665	0.667	0.668
42	0.669	0.670	0.672	0.673	0.674	0.676	0.677	0.678	0.679	0.681
43	0.682	0.683	0.685	0.686	0.687	0.688	0.690	0.691	0.692	0.693
44	0.695	0.696	0.697	0.698	0.700	0.701	0.702	0.703	0.705	0.706
45	0.707	0.708	0.710	0.711	0.712	0.713	0.714	0.716	0.717	0.718
46	0.719	0.721	0.722	0.723	0.724	0.725	0.727	0.728	0.729	0.730
47	0.731	0.733	0.734	0.735	0.736	0.737	0.738	0.740	0.741	0.742
48	0.743	0.744	0.745	0.747	0.748	0.749	0.750	0.751	0.752	0.754
49	0.755	0.756	0.757	0.758	0.759	0.760	0.762	0.763	0.764	0.765
50	0.766	0.767	0.768	0.769	0.771	0.772	0.773	0.774	0.775	0.776
51	0.777	0.778	0.779	0.780	0.782	0.783	0.784	0.785	0.786	0.787
52	0.788	0.789	0.790	0.791	0.792	0.793	0.794	0.795	0.797	0.798
53	0.799	0.800	0.801	0.802	0.803	0.804	0.805	0.806	0.807	0.808
54	0.809	0.810	0.811	0.812	0.813	0.814	0.815	0.816	0.817	0.818
55	0.819	0.820	0.821	0.822	0.823	0.824	0.825	0.826	0.827	0.828
56	0.829	0.830	0.831	0.832	0.833	0.834	0.835	0.836	0.837	0.838
57	0.839	0.840	0.841	0.842	0.842	0.843	0.844	0.845	0.846	0.847
58	0.848	0.849	0.850	0.851	0.852	0.853	0.854	0.854	0.855	0.856
59	0.857	0.858	0.859	0.860	0.861	0.862	0.863	0.863	0.864	0.865
60	0.866	0.867	0.868	0.869	0.869	0.870	0.871	0.872	0.873	0.874
61	0.875	0.875	0.876	0.877	0.878	0.879	0.880	0.880	0.881	0.882
62	0.883	0.884	0.885	0.885	0.886	0.887	0.888	0.889	0.889	0.890

Θ	0	1	2	3	4	5	6	7	8	9
63	0.891	0.892	0.893	0.893	0.894	0.895	0.896	0.896	0.897	0.898
64	0.899	0.900	0.900	0.901	0.902	0.903	0.903	0.904	0.905	0.906
65	0.906	0.907	0.908	0.909	0.909	0.910	0.911	0.911	0.912	0.913
66	0.914	0.914	0.915	0.916	0.916	0.917	0.918	0.918	0.919	0.920
67	0.921	0.921	0.922	0.923	0.923	0.924	0.925	0.925	0.926	0.927
68	0.927	0.928	0.928	0.929	0.930	0.930	0.931	0.932	0.932	0.933
69	0.934	0.934	0.935	0.935	0.936	0.937	0.937	0.938	0.938	0.939
70	0.940	0.940	0.941	0.941	0.942	0.943	0.943	0.944	0.944	0.945
71	0.946	0.946	0.947	0.947	0.948	0.948	0.949	0.949	0.950	0.951
72	0.951	0.952	0.952	0.953	0.953	0.954	0.954	0.955	0.955	0.956
73	0.956	0.957	0.957	0.958	0.958	0.959	0.959	0.960	0.960	0.961
74	0.961	0.962	0.962	0.963	0.963	0.964	0.964	0.965	0.965	0.965
75	0.966	0.966	0.967	0.967	0.968	0.968	0.969	0.969	0.969	0.970
76	0.970	0.971	0.971	0.972	0.972	0.972	0.973	0.973	0.974	0.974
77	0.974	0.975	0.975	0.976	0.976	0.976	0.977	0.977	0.977	0.978
78	0.978	0.979	0.979	0.979	0.980	0.980	0.980	0.981	0.981	0.981
79	0.982	0.982	0.982	0.983	0.983	0.983	0.984	0.984	0.984	0.985
80	0.985	0.985	0.985	0.986	0.986	0.986	0.987	0.987	0.987	0.987
81	0.988	0.988	0.988	0.988	0.989	0.989	0.989	0.990	0.990	0.990
82	0.990	0.991	0.991	0.991	0.991	0.991	0.992	0.992	0.992	0.992
83	0.993	0.993	0.993	0.993	0.993	0.994	0.994	0.994	0.994	0.994
84	0.995	0.995	0.995	0.995	0.995	0.995	0.996	0.996	0.996	0.996
85	0.996	0.996	0.996	0.997	0.997	0.997	0.997	0.997	0.997	0.997
86	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.999
87	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
88	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
89	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
91	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999
92	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
93	0.999	0.999	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998
94	0.998	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.996	0.996
95	0.996	0.996	0.996	0.996	0.996	0.995	0.995	0.995	0.995	0.995
96	0.995	0.994	0.994	0.994	0.994	0.994	0.993	0.993	0.993	0.993
97	0.993	0.992	0.992	0.992	0.992	0.991	0.991	0.991	0.991	0.991
98	0.990	0.990	0.990	0.990	0.989	0.989	0.989	0.988	0.988	0.988
99	0.988	0.987	0.987	0.987	0.987	0.986	0.986	0.986	0.985	0.985
100	0.985	0.985	0.984	0.984	0.984	0.983	0.983	0.983	0.982	0.982
101	0.982	0.981	0.981	0.981	0.980	0.980	0.980	0.979	0.979	0.979
102	0.978	0.978	0.977	0.977	0.977	0.976	0.976	0.976	0.975	0.975
103	0.974	0.974	0.974	0.973	0.973	0.972	0.972	0.972	0.971	0.971
104	0.970	0.970	0.969	0.969	0.969	0.968	0.968	0.967	0.967	0.966
105	0.966	0.965	0.965	0.965	0.964	0.964	0.963	0.963	0.962	0.962
106	0.961	0.961	0.960	0.960	0.959	0.959	0.958	0.958	0.957	0.957
107	0.956	0.956	0.955	0.955	0.954	0.954	0.953	0.953	0.952	0.952
108	0.951	0.951	0.950	0.949	0.949	0.948	0.948	0.947	0.947	0.946
109	0.946	0.945	0.944	0.944	0.943	0.943	0.942	0.941	0.941	0.940
110	0.940	0.939	0.938	0.938	0.937	0.937	0.936	0.935	0.935	0.934
111	0.934	0.933	0.932	0.932	0.931	0.930	0.930	0.929	0.928	0.928
112	0.927	0.927	0.926	0.925	0.925	0.924	0.923	0.923	0.922	0.921
113	0.921	0.920	0.919	0.918	0.918	0.917	0.916	0.916	0.915	0.914
114	0.914	0.913	0.912	0.911	0.911	0.910	0.909	0.909	0.908	0.907
115	0.906	0.906	0.905	0.904	0.903	0.903	0.902	0.901	0.900	0.900
116	0.899	0.898	0.897	0.896	0.896	0.895	0.894	0.893	0.893	0.892
117	0.891	0.890	0.889	0.889	0.888	0.887	0.886	0.885	0.885	0.884
118	0.883	0.882	0.881	0.880	0.880	0.879	0.878	0.877	0.876	0.875
119	0.875	0.874	0.873	0.872	0.871	0.870	0.869	0.869	0.868	0.867
120	0.866	0.865	0.864	0.863	0.863	0.862	0.861	0.860	0.859	0.858
121	0.857	0.856	0.855	0.854	0.854	0.853	0.852	0.851	0.850	0.849
122	0.848	0.847	0.846	0.845	0.844	0.843	0.842	0.842	0.841	0.840
123	0.839	0.838	0.837	0.836	0.835	0.834	0.833	0.832	0.831	0.830
124	0.829	0.828	0.827	0.826	0.825	0.824	0.823	0.822	0.821	0.820
125	0.819	0.818	0.817	0.816	0.815	0.814	0.813	0.812	0.811	0.810
126	0.809	0.808	0.807	0.806	0.805	0.804	0.803	0.802	0.801	0.800
127	0.799	0.798	0.797	0.795	0.794	0.793	0.792	0.791	0.790	0.789
128	0.788	0.787	0.786	0.785	0.784	0.783	0.782	0.780	0.779	0.778
129	0.777	0.776	0.775	0.774	0.773	0.772	0.771	0.769	0.768	0.767

Θ	0	1	2	3	4	5	6	7	8	9
130	0.766	0.765	0.764	0.763	0.762	0.760	0.759	0.758	0.757	0.756
131	0.755	0.754	0.752	0.751	0.750	0.749	0.748	0.747	0.745	0.744
132	0.743	0.742	0.741	0.740	0.738	0.737	0.736	0.735	0.734	0.733
133	0.731	0.730	0.729	0.728	0.727	0.725	0.724	0.723	0.722	0.721
134	0.719	0.718	0.717	0.716	0.714	0.713	0.712	0.711	0.710	0.708
135	0.707	0.706	0.705	0.703	0.702	0.701	0.700	0.698	0.697	0.696
136	0.695	0.693	0.692	0.691	0.690	0.688	0.687	0.686	0.685	0.683
137	0.682	0.681	0.679	0.678	0.677	0.676	0.674	0.673	0.672	0.670
138	0.669	0.668	0.667	0.665	0.664	0.663	0.661	0.660	0.659	0.657
139	0.656	0.655	0.653	0.652	0.651	0.649	0.648	0.647	0.645	0.644
140	0.643	0.641	0.640	0.639	0.637	0.636	0.635	0.633	0.632	0.631
141	0.629	0.628	0.627	0.625	0.624	0.623	0.621	0.620	0.618	0.617
142	0.616	0.614	0.613	0.612	0.610	0.609	0.607	0.606	0.605	0.603
143	0.602	0.600	0.599	0.598	0.596	0.595	0.593	0.592	0.591	0.589
144	0.588	0.586	0.585	0.584	0.582	0.581	0.579	0.578	0.576	0.575
145	0.574	0.572	0.571	0.569	0.568	0.566	0.565	0.564	0.562	0.561
146	0.559	0.558	0.556	0.555	0.553	0.552	0.550	0.549	0.548	0.546
147	0.545	0.543	0.542	0.540	0.539	0.537	0.536	0.534	0.533	0.531
148	0.530	0.528	0.527	0.525	0.524	0.522	0.521	0.520	0.518	0.517
149	0.515	0.514	0.512	0.511	0.509	0.508	0.506	0.505	0.503	0.502
150	0.500	0.498	0.497	0.495	0.494	0.492	0.491	0.489	0.488	0.486
151	0.485	0.483	0.482	0.480	0.479	0.477	0.476	0.474	0.473	0.471
152	0.469	0.468	0.466	0.465	0.463	0.462	0.460	0.459	0.457	0.456
153	0.454	0.452	0.451	0.449	0.448	0.446	0.445	0.443	0.442	0.440
154	0.438	0.437	0.435	0.434	0.432	0.431	0.429	0.427	0.426	0.424
155	0.423	0.421	0.419	0.418	0.416	0.415	0.413	0.412	0.410	0.408
156	0.407	0.405	0.404	0.402	0.400	0.399	0.397	0.396	0.394	0.392
157	0.391	0.389	0.388	0.386	0.384	0.383	0.381	0.379	0.378	0.376
158	0.375	0.373	0.371	0.370	0.368	0.367	0.365	0.363	0.362	0.360
159	0.358	0.357	0.355	0.353	0.352	0.350	0.349	0.347	0.345	0.344
160	0.342	0.340	0.339	0.337	0.335	0.334	0.332	0.331	0.329	0.327
161	0.326	0.324	0.322	0.321	0.319	0.317	0.316	0.314	0.312	0.311
162	0.309	0.307	0.306	0.304	0.302	0.301	0.299	0.297	0.296	0.294
163	0.292	0.291	0.289	0.287	0.286	0.284	0.282	0.281	0.279	0.277
164	0.276	0.274	0.272	0.271	0.269	0.267	0.266	0.264	0.262	0.261
165	0.259	0.257	0.255	0.254	0.252	0.250	0.249	0.247	0.245	0.244
166	0.242	0.240	0.239	0.237	0.235	0.233	0.232	0.230	0.228	0.227
167	0.225	0.223	0.222	0.220	0.218	0.216	0.215	0.213	0.211	0.210
168	0.208	0.206	0.204	0.203	0.201	0.199	0.198	0.196	0.194	0.193
169	0.191	0.189	0.187	0.186	0.184	0.182	0.181	0.179	0.177	0.175
170	0.174	0.172	0.170	0.168	0.167	0.165	0.163	0.162	0.160	0.158
171	0.156	0.155	0.153	0.151	0.150	0.148	0.146	0.144	0.143	0.141
172	0.139	0.137	0.136	0.134	0.132	0.131	0.129	0.127	0.125	0.124
173	0.122	0.120	0.118	0.117	0.115	0.113	0.111	0.110	0.108	0.106
174	0.105	0.103	0.101	0.099	0.098	0.096	0.094	0.092	0.091	0.089
175	0.087	0.085	0.084	0.082	0.080	0.078	0.077	0.075	0.073	0.071
176	0.070	0.068	0.066	0.065	0.063	0.061	0.059	0.058	0.056	0.054
177	0.052	0.051	0.049	0.047	0.045	0.044	0.042	0.040	0.038	0.037
178	0.035	0.033	0.031	0.030	0.028	0.026	0.024	0.023	0.021	0.019
179	0.017	0.016	0.014	0.012	0.010	0.009	0.007	0.005	0.003	0.002
180	0.000	-0.002	-0.003	-0.005	-0.007	-0.009	-0.010	-0.012	-0.014	-0.016
181	-0.017	-0.019	-0.021	-0.023	-0.024	-0.026	-0.028	-0.030	-0.031	-0.033
182	-0.035	-0.037	-0.038	-0.040	-0.042	-0.044	-0.045	-0.047	-0.049	-0.051
183	-0.052	-0.054	-0.056	-0.058	-0.059	-0.061	-0.063	-0.065	-0.066	-0.068
184	-0.070	-0.071	-0.073	-0.075	-0.077	-0.078	-0.080	-0.082	-0.084	-0.085
185	-0.087	-0.089	-0.091	-0.092	-0.094	-0.096	-0.098	-0.099	-0.101	-0.103
186	-0.105	-0.106	-0.108	-0.110	-0.111	-0.113	-0.115	-0.117	-0.118	-0.120
187	-0.122	-0.124	-0.125	-0.127	-0.129	-0.131	-0.132	-0.134	-0.136	-0.137
188	-0.139	-0.141	-0.143	-0.144	-0.146	-0.148	-0.150	-0.151	-0.153	-0.155
189	-0.156	-0.158	-0.160	-0.162	-0.163	-0.165	-0.167	-0.168	-0.170	-0.172
190	-0.174	-0.175	-0.177	-0.179	-0.181	-0.182	-0.184	-0.186	-0.187	-0.189
191	-0.191	-0.193	-0.194	-0.196	-0.198	-0.199	-0.201	-0.203	-0.204	-0.206
192	-0.208	-0.210	-0.211	-0.213	-0.215	-0.216	-0.218	-0.220	-0.222	-0.223
193	-0.225	-0.227	-0.228	-0.230	-0.232	-0.233	-0.235	-0.237	-0.239	-0.240
194	-0.242	-0.244	-0.245	-0.247	-0.249	-0.250	-0.252	-0.254	-0.255	-0.257
195	-0.259	-0.261	-0.262	-0.264	-0.266	-0.267	-0.269	-0.271	-0.272	-0.274
196	-0.276	-0.277	-0.279	-0.281	-0.282	-0.284	-0.286	-0.287	-0.289	-0.291

Θ	0	1	2	3	4	5	6	7	8	9
197	-0.292	-0.294	-0.296	-0.297	-0.299	-0.301	-0.302	-0.304	-0.306	-0.307
198	-0.309	-0.311	-0.312	-0.314	-0.316	-0.317	-0.319	-0.321	-0.322	-0.324
199	-0.326	-0.327	-0.329	-0.331	-0.332	-0.334	-0.335	-0.337	-0.339	-0.340
200	-0.342	-0.344	-0.345	-0.347	-0.349	-0.350	-0.352	-0.353	-0.355	-0.357
201	-0.358	-0.360	-0.362	-0.363	-0.365	-0.367	-0.368	-0.370	-0.371	-0.373
202	-0.375	-0.376	-0.378	-0.379	-0.381	-0.383	-0.384	-0.386	-0.388	-0.389
203	-0.391	-0.392	-0.394	-0.396	-0.397	-0.399	-0.400	-0.402	-0.404	-0.405
204	-0.407	-0.408	-0.410	-0.412	-0.413	-0.415	-0.416	-0.418	-0.419	-0.421
205	-0.423	-0.424	-0.426	-0.427	-0.429	-0.431	-0.432	-0.434	-0.435	-0.437
206	-0.438	-0.440	-0.442	-0.443	-0.445	-0.446	-0.448	-0.449	-0.451	-0.452
207	-0.454	-0.456	-0.457	-0.459	-0.460	-0.462	-0.463	-0.465	-0.466	-0.468
208	-0.469	-0.471	-0.473	-0.474	-0.476	-0.477	-0.479	-0.480	-0.482	-0.483
209	-0.485	-0.486	-0.488	-0.489	-0.491	-0.492	-0.494	-0.495	-0.497	-0.498
210	-0.500	-0.502	-0.503	-0.505	-0.506	-0.508	-0.509	-0.511	-0.512	-0.514
211	-0.515	-0.517	-0.518	-0.520	-0.521	-0.522	-0.524	-0.525	-0.527	-0.528
212	-0.530	-0.531	-0.533	-0.534	-0.536	-0.537	-0.539	-0.540	-0.542	-0.543
213	-0.545	-0.546	-0.548	-0.549	-0.550	-0.552	-0.553	-0.555	-0.556	-0.558
214	-0.559	-0.561	-0.562	-0.564	-0.565	-0.566	-0.568	-0.569	-0.571	-0.572
215	-0.574	-0.575	-0.576	-0.578	-0.579	-0.581	-0.582	-0.584	-0.585	-0.586
216	-0.588	-0.589	-0.591	-0.592	-0.593	-0.595	-0.596	-0.598	-0.599	-0.600
217	-0.602	-0.603	-0.605	-0.606	-0.607	-0.609	-0.610	-0.612	-0.613	-0.614
218	-0.616	-0.617	-0.618	-0.620	-0.621	-0.623	-0.624	-0.625	-0.627	-0.628
219	-0.629	-0.631	-0.632	-0.633	-0.635	-0.636	-0.637	-0.639	-0.640	-0.641
220	-0.643	-0.644	-0.645	-0.647	-0.648	-0.649	-0.651	-0.652	-0.653	-0.655
221	-0.656	-0.657	-0.659	-0.660	-0.661	-0.663	-0.664	-0.665	-0.667	-0.668
222	-0.669	-0.670	-0.672	-0.673	-0.674	-0.676	-0.677	-0.678	-0.679	-0.681
223	-0.682	-0.683	-0.685	-0.686	-0.687	-0.688	-0.690	-0.691	-0.692	-0.693
224	-0.695	-0.696	-0.697	-0.698	-0.700	-0.701	-0.702	-0.703	-0.705	-0.706
225	-0.707	-0.708	-0.710	-0.711	-0.712	-0.713	-0.714	-0.716	-0.717	-0.718
226	-0.719	-0.721	-0.722	-0.723	-0.724	-0.725	-0.727	-0.728	-0.729	-0.730
227	-0.731	-0.733	-0.734	-0.735	-0.736	-0.737	-0.738	-0.740	-0.741	-0.742
228	-0.743	-0.744	-0.745	-0.747	-0.748	-0.749	-0.750	-0.751	-0.752	-0.754
229	-0.755	-0.756	-0.757	-0.758	-0.759	-0.760	-0.762	-0.763	-0.764	-0.765
230	-0.766	-0.767	-0.768	-0.769	-0.771	-0.772	-0.773	-0.774	-0.775	-0.776
231	-0.777	-0.778	-0.779	-0.780	-0.782	-0.783	-0.784	-0.785	-0.786	-0.787
232	-0.788	-0.789	-0.790	-0.791	-0.792	-0.793	-0.794	-0.795	-0.797	-0.798
233	-0.799	-0.800	-0.801	-0.802	-0.803	-0.804	-0.805	-0.806	-0.807	-0.808
234	-0.809	-0.810	-0.811	-0.812	-0.813	-0.814	-0.815	-0.816	-0.817	-0.818
235	-0.819	-0.820	-0.821	-0.822	-0.823	-0.824	-0.825	-0.826	-0.827	-0.828
236	-0.829	-0.830	-0.831	-0.832	-0.833	-0.834	-0.835	-0.836	-0.837	-0.838
237	-0.839	-0.840	-0.841	-0.842	-0.842	-0.843	-0.844	-0.845	-0.846	-0.847
238	-0.848	-0.849	-0.850	-0.851	-0.852	-0.853	-0.854	-0.854	-0.855	-0.856
239	-0.857	-0.858	-0.859	-0.860	-0.861	-0.862	-0.863	-0.863	-0.864	-0.865
240	-0.866	-0.867	-0.868	-0.869	-0.869	-0.870	-0.871	-0.872	-0.873	-0.874
241	-0.875	-0.875	-0.876	-0.877	-0.878	-0.879	-0.880	-0.880	-0.881	-0.882
242	-0.883	-0.884	-0.885	-0.885	-0.886	-0.887	-0.888	-0.889	-0.889	-0.890
243	-0.891	-0.892	-0.893	-0.893	-0.894	-0.895	-0.896	-0.896	-0.897	-0.898
244	-0.899	-0.900	-0.900	-0.901	-0.902	-0.903	-0.903	-0.904	-0.905	-0.906
245	-0.906	-0.907	-0.908	-0.909	-0.909	-0.910	-0.911	-0.911	-0.912	-0.913
246	-0.914	-0.914	-0.915	-0.916	-0.916	-0.917	-0.918	-0.918	-0.919	-0.920
247	-0.921	-0.921	-0.922	-0.923	-0.923	-0.924	-0.925	-0.925	-0.926	-0.927
248	-0.927	-0.928	-0.928	-0.929	-0.930	-0.930	-0.931	-0.932	-0.932	-0.933
249	-0.934	-0.934	-0.935	-0.935	-0.936	-0.937	-0.937	-0.938	-0.938	-0.939
250	-0.940	-0.940	-0.941	-0.941	-0.942	-0.943	-0.943	-0.944	-0.944	-0.945
251	-0.946	-0.946	-0.947	-0.947	-0.948	-0.948	-0.949	-0.949	-0.950	-0.951
252	-0.951	-0.952	-0.952	-0.953	-0.953	-0.954	-0.954	-0.955	-0.955	-0.956
253	-0.956	-0.957	-0.957	-0.958	-0.958	-0.959	-0.959	-0.960	-0.960	-0.961
254	-0.961	-0.962	-0.962	-0.963	-0.963	-0.964	-0.964	-0.965	-0.965	-0.965
255	-0.966	-0.966	-0.967	-0.967	-0.968	-0.968	-0.969	-0.969	-0.969	-0.970
256	-0.970	-0.971	-0.971	-0.972	-0.972	-0.972	-0.973	-0.973	-0.974	-0.974
257	-0.974	-0.975	-0.975	-0.976	-0.976	-0.976	-0.977	-0.977	-0.977	-0.978
258	-0.978	-0.979	-0.979	-0.979	-0.980	-0.980	-0.980	-0.981	-0.981	-0.981
259	-0.982	-0.982	-0.982	-0.983	-0.983	-0.983	-0.984	-0.984	-0.984	-0.985
260	-0.985	-0.985	-0.985	-0.986	-0.986	-0.986	-0.987	-0.987	-0.987	-0.987
261	-0.988	-0.988	-0.988	-0.988	-0.989	-0.989	-0.989	-0.990	-0.990	-0.990
262	-0.990	-0.991	-0.991	-0.991	-0.991	-0.991	-0.992	-0.992	-0.992	-0.992
263	-0.993	-0.993	-0.993	-0.993	-0.993	-0.994	-0.994	-0.994	-0.994	-0.994

Θ	0	1	2	3	4	5	6	7	8	9
264	-0.995	-0.995	-0.995	-0.995	-0.995	-0.995	-0.996	-0.996	-0.996	-0.996
265	-0.996	-0.996	-0.996	-0.997	-0.997	-0.997	-0.997	-0.997	-0.997	-0.997
266	-0.998	-0.998	-0.998	-0.998	-0.998	-0.998	-0.998	-0.998	-0.998	-0.999
267	-0.999	-0.999	-0.999	-0.999	-0.999	-0.999	-0.999	-0.999	-0.999	-0.999
268	-0.999	-0.999	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000
269	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000
270	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000
271	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-0.999
272	-0.999	-0.999	-0.999	-0.999	-0.999	-0.999	-0.999	-0.999	-0.999	-0.999
273	-0.999	-0.999	-0.998	-0.998	-0.998	-0.998	-0.998	-0.998	-0.998	-0.998
274	-0.998	-0.997	-0.997	-0.997	-0.997	-0.997	-0.997	-0.997	-0.996	-0.996
275	-0.996	-0.996	-0.996	-0.996	-0.996	-0.995	-0.995	-0.995	-0.995	-0.995
276	-0.995	-0.994	-0.994	-0.994	-0.994	-0.994	-0.993	-0.993	-0.993	-0.993
277	-0.993	-0.992	-0.992	-0.992	-0.992	-0.991	-0.991	-0.991	-0.991	-0.991
278	-0.990	-0.990	-0.990	-0.990	-0.989	-0.989	-0.989	-0.988	-0.988	-0.988
279	-0.988	-0.987	-0.987	-0.987	-0.987	-0.986	-0.986	-0.986	-0.985	-0.985
280	-0.985	-0.985	-0.984	-0.984	-0.984	-0.983	-0.983	-0.983	-0.982	-0.982
281	-0.982	-0.981	-0.981	-0.981	-0.980	-0.980	-0.980	-0.979	-0.979	-0.979
282	-0.978	-0.978	-0.977	-0.977	-0.977	-0.976	-0.976	-0.976	-0.975	-0.975
283	-0.974	-0.974	-0.974	-0.973	-0.973	-0.972	-0.972	-0.972	-0.971	-0.971
284	-0.970	-0.970	-0.969	-0.969	-0.969	-0.968	-0.968	-0.967	-0.967	-0.966
285	-0.966	-0.965	-0.965	-0.965	-0.964	-0.964	-0.963	-0.963	-0.962	-0.962
286	-0.961	-0.961	-0.960	-0.960	-0.959	-0.959	-0.958	-0.958	-0.957	-0.957
287	-0.956	-0.956	-0.955	-0.955	-0.954	-0.954	-0.953	-0.953	-0.952	-0.952
288	-0.951	-0.951	-0.950	-0.949	-0.949	-0.948	-0.948	-0.947	-0.947	-0.946
289	-0.946	-0.945	-0.944	-0.944	-0.943	-0.943	-0.942	-0.941	-0.941	-0.940
290	-0.940	-0.939	-0.938	-0.938	-0.937	-0.937	-0.936	-0.935	-0.935	-0.934
291	-0.934	-0.933	-0.932	-0.932	-0.931	-0.930	-0.930	-0.929	-0.928	-0.928
292	-0.927	-0.927	-0.926	-0.925	-0.925	-0.924	-0.923	-0.923	-0.922	-0.921
293	-0.921	-0.920	-0.919	-0.918	-0.918	-0.917	-0.916	-0.916	-0.915	-0.914
294	-0.914	-0.913	-0.912	-0.911	-0.911	-0.910	-0.909	-0.909	-0.908	-0.907
295	-0.906	-0.906	-0.905	-0.904	-0.903	-0.903	-0.902	-0.901	-0.900	-0.900
296	-0.899	-0.898	-0.897	-0.896	-0.896	-0.895	-0.894	-0.893	-0.893	-0.892
297	-0.891	-0.890	-0.889	-0.889	-0.888	-0.887	-0.886	-0.885	-0.885	-0.884
298	-0.883	-0.882	-0.881	-0.880	-0.880	-0.879	-0.878	-0.877	-0.876	-0.875
299	-0.875	-0.874	-0.873	-0.872	-0.871	-0.870	-0.869	-0.869	-0.868	-0.867
300	-0.866	-0.865	-0.864	-0.863	-0.863	-0.862	-0.861	-0.860	-0.859	-0.858
301	-0.857	-0.856	-0.855	-0.854	-0.854	-0.853	-0.852	-0.851	-0.850	-0.849
302	-0.848	-0.847	-0.846	-0.845	-0.844	-0.843	-0.842	-0.842	-0.841	-0.840
303	-0.839	-0.838	-0.837	-0.836	-0.835	-0.834	-0.833	-0.832	-0.831	-0.830
304	-0.829	-0.828	-0.827	-0.826	-0.825	-0.824	-0.823	-0.822	-0.821	-0.820
305	-0.819	-0.818	-0.817	-0.816	-0.815	-0.814	-0.813	-0.812	-0.811	-0.810
306	-0.809	-0.808	-0.807	-0.806	-0.805	-0.804	-0.803	-0.802	-0.801	-0.800
307	-0.799	-0.798	-0.797	-0.795	-0.794	-0.793	-0.792	-0.791	-0.790	-0.789
308	-0.788	-0.787	-0.786	-0.785	-0.784	-0.783	-0.782	-0.780	-0.779	-0.778
309	-0.777	-0.776	-0.775	-0.774	-0.773	-0.772	-0.771	-0.769	-0.768	-0.767
310	-0.766	-0.765	-0.764	-0.763	-0.762	-0.760	-0.759	-0.758	-0.757	-0.756
311	-0.755	-0.754	-0.752	-0.751	-0.750	-0.749	-0.748	-0.747	-0.745	-0.744
312	-0.743	-0.742	-0.741	-0.740	-0.738	-0.737	-0.736	-0.735	-0.734	-0.733
313	-0.731	-0.730	-0.729	-0.728	-0.727	-0.725	-0.724	-0.723	-0.722	-0.721
314	-0.719	-0.718	-0.717	-0.716	-0.714	-0.713	-0.712	-0.711	-0.710	-0.708
315	-0.707	-0.706	-0.705	-0.703	-0.702	-0.701	-0.700	-0.698	-0.697	-0.696
316	-0.695	-0.693	-0.692	-0.691	-0.690	-0.688	-0.687	-0.686	-0.685	-0.683
317	-0.682	-0.681	-0.679	-0.678	-0.677	-0.676	-0.674	-0.673	-0.672	-0.670
318	-0.669	-0.668	-0.667	-0.665	-0.664	-0.663	-0.661	-0.660	-0.659	-0.657
319	-0.656	-0.655	-0.653	-0.652	-0.651	-0.649	-0.648	-0.647	-0.645	-0.644
320	-0.643	-0.641	-0.640	-0.639	-0.637	-0.636	-0.635	-0.633	-0.632	-0.631
321	-0.629	-0.628	-0.627	-0.625	-0.624	-0.623	-0.621	-0.620	-0.618	-0.617
322	-0.616	-0.614	-0.613	-0.612	-0.610	-0.609	-0.607	-0.606	-0.605	-0.603
323	-0.602	-0.600	-0.599	-0.598	-0.596	-0.595	-0.593	-0.592	-0.591	-0.589
324	-0.588	-0.586	-0.585	-0.584	-0.582	-0.581	-0.579	-0.578	-0.576	-0.575
325	-0.574	-0.572	-0.571	-0.569	-0.568	-0.566	-0.565	-0.564	-0.562	-0.561
326	-0.559	-0.558	-0.556	-0.555	-0.553	-0.552	-0.550	-0.549	-0.548	-0.546
327	-0.545	-0.543	-0.542	-0.540	-0.539	-0.537	-0.536	-0.534	-0.533	-0.531
328	-0.530	-0.528	-0.527	-0.525	-0.524	-0.522	-0.521	-0.520	-0.518	-0.517
329	-0.515	-0.514	-0.512	-0.511	-0.509	-0.508	-0.506	-0.505	-0.503	-0.502
330	-0.500	-0.498	-0.497	-0.495	-0.494	-0.492	-0.491	-0.489	-0.488	-0.486

Θ	0	1	2	3	4	5	6	7	8	9
331	-0.485	-0.483	-0.482	-0.480	-0.479	-0.477	-0.476	-0.474	-0.473	-0.471
332	-0.469	-0.468	-0.466	-0.465	-0.463	-0.462	-0.460	-0.459	-0.457	-0.456
333	-0.454	-0.452	-0.451	-0.449	-0.448	-0.446	-0.445	-0.443	-0.442	-0.440
334	-0.438	-0.437	-0.435	-0.434	-0.432	-0.431	-0.429	-0.427	-0.426	-0.424
335	-0.423	-0.421	-0.419	-0.418	-0.416	-0.415	-0.413	-0.412	-0.410	-0.408
336	-0.407	-0.405	-0.404	-0.402	-0.400	-0.399	-0.397	-0.396	-0.394	-0.392
337	-0.391	-0.389	-0.388	-0.386	-0.384	-0.383	-0.381	-0.379	-0.378	-0.376
338	-0.375	-0.373	-0.371	-0.370	-0.368	-0.367	-0.365	-0.363	-0.362	-0.360
339	-0.358	-0.357	-0.355	-0.353	-0.352	-0.350	-0.349	-0.347	-0.345	-0.344
340	-0.342	-0.340	-0.339	-0.337	-0.335	-0.334	-0.332	-0.331	-0.329	-0.327
341	-0.326	-0.324	-0.322	-0.321	-0.319	-0.317	-0.316	-0.314	-0.312	-0.311
342	-0.309	-0.307	-0.306	-0.304	-0.302	-0.301	-0.299	-0.297	-0.296	-0.294
343	-0.292	-0.291	-0.289	-0.287	-0.286	-0.284	-0.282	-0.281	-0.279	-0.277
344	-0.276	-0.274	-0.272	-0.271	-0.269	-0.267	-0.266	-0.264	-0.262	-0.261
345	-0.259	-0.257	-0.255	-0.254	-0.252	-0.250	-0.249	-0.247	-0.245	-0.244
346	-0.242	-0.240	-0.239	-0.237	-0.235	-0.233	-0.232	-0.230	-0.228	-0.227
347	-0.225	-0.223	-0.222	-0.220	-0.218	-0.216	-0.215	-0.213	-0.211	-0.210
348	-0.208	-0.206	-0.204	-0.203	-0.201	-0.199	-0.198	-0.196	-0.194	-0.193
349	-0.191	-0.189	-0.187	-0.186	-0.184	-0.182	-0.181	-0.179	-0.177	-0.175
350	-0.174	-0.172	-0.170	-0.168	-0.167	-0.165	-0.163	-0.162	-0.160	-0.158
351	-0.156	-0.155	-0.153	-0.151	-0.150	-0.148	-0.146	-0.144	-0.143	-0.141
352	-0.139	-0.137	-0.136	-0.134	-0.132	-0.131	-0.129	-0.127	-0.125	-0.124
353	-0.122	-0.120	-0.118	-0.117	-0.115	-0.113	-0.111	-0.110	-0.108	-0.106
354	-0.105	-0.103	-0.101	-0.099	-0.098	-0.096	-0.094	-0.092	-0.091	-0.089
355	-0.087	-0.085	-0.084	-0.082	-0.080	-0.078	-0.077	-0.075	-0.073	-0.071
356	-0.070	-0.068	-0.066	-0.065	-0.063	-0.061	-0.059	-0.058	-0.056	-0.054
357	-0.052	-0.051	-0.049	-0.047	-0.045	-0.044	-0.042	-0.040	-0.038	-0.037
358	-0.035	-0.033	-0.031	-0.030	-0.028	-0.026	-0.024	-0.023	-0.021	-0.019
359	-0.017	-0.016	-0.014	-0.012	-0.010	-0.009	-0.007	-0.005	-0.003	-0.002

Lab 1: finite differences

Faction / name(s): _____

Babbage's Difference Engine was meant to automatically calculate and typeset numerical tables of the kind used by navigators, astronomers, and other men of science. Such tables were contained in large volumes, such as *Numerical Tables and Constants in Elementary Science* [Lup84], or Babbage's own *Table of Logarithms of the Natural Numbers from 1 to 108,000* [Bab15]. (Such books persisted well into the computer age. Consider, for instance, *Statistical Tables* [RS69], published in 1969.) The Difference Engine was to use the method of finite differences to perform the calculations. This lab will introduce you to the method of finite differences, as it would have been used to calculate numerical tables.

Where's the beef?

Let us begin with an example similar to one used by Babbage himself in his autobiography, *Passages from the Life of a Philosopher* [Bab64]. Let us suppose the owner of a butcher shop wishes to post a sign with the prices of various quantities of ground beef. Assuming the price of ground beef is \$3.19 per pound, we might envision a table like the one shown below in Table 2.3.

Table 2.3: Prices for various quantities of ground beef

Pounds	Price
1	\$3.19
2	\$6.38
3	\$9.57
4	\$12.76
5	\$15.95
6	\$19.14
7	\$22.33
8	\$25.52
9	\$28.71
10	\$31.90

Such a table is easy enough to produce, is it not? A butcher with moderate mathematical skills, even without a modern calculator, could simply do ten separate multiplications to make the table. Suppose the butcher uses paper and pencil to do the calculations, with multiplications like this:

$$\begin{array}{r}
 2 \\
 3.19 \\
 \times 3 \\
 \hline
 9.57
 \end{array}$$

Another row in the table could be computed like this:

$$\begin{array}{r}
 18 \\
 3.19 \\
 \times \quad 3 \\
 \hline
 28.71
 \end{array}$$

The butcher repeats this pencil-and-paper multiplication a total of nine times and produces the prices shown in Table 2.4:

Table 2.4: Prices computed by the butcher

Pounds	Price
1	\$3.19
2	\$6.38
3	\$9.57
4	\$12.67
5	\$15.95
6	\$19.14
7	\$22.33
8	\$25.42
9	\$28.71
10	\$31.90

How did our butcher do? Since the tables are small, it is not too difficult to see that there are two errors in the table computed by the butcher, specifically, the lines for four and eight pounds of beef. The error in the four pound line is a simple transposition (the 7 and 6 have been switched in \$12.76). The error in the eight pound line was caused by making a carry mistake during the actual calculation. These errors, if left uncorrected, would cost our butcher money! The prices for four pounds and eight pounds of ground beef are each a dime too low.

What if the table was much larger, however? How easy would it be to detect errors in a table of, say, the logarithms of the natural numbers from 1 to 108,000? Certainly detecting errors would be much harder in this case. For example, a printed multiplication table of the period was found to have no less than 40 errors on one page alone [Smi64]! Also, suppose we do detect an error somewhere in a large table. Does this help us to find any other errors that might exist? No, it does not.

Now, let us consider a second method to produce the price table. Instead of multiplying the number of pounds by \$3.19 for each row, we will start with \$3.19 for the one-pound line, and then *add \$3.19 to each successive line*. Using this method, we perform nine additions instead of nine multiplications. The first addition would be:

$$\begin{array}{r}
 1 \\
 3.19 \\
 + 3.19 \\
 \hline
 6.38
 \end{array}$$

The next would be:

$$\begin{array}{r}
 1 \\
 6.38 \\
 + 3.19 \\
 \hline
 9.57
 \end{array}$$

And so on. Using our second method to produce the table, if we make an error during a calculation, what are the implications? It is obvious that one error would cause all of the rest of the calculations to be off as well. But, that is actually a good thing. After we have completed all the additions to produce the table, we can check the last line of the table using multiplication. If our multiplication produces the same answer as the last line of the table, we can assume (with high confidence) that all of the lines are correct. If our multiplication does not agree with the last line, then we know that an error is present somewhere in the table, and we can go about searching for it. So, our second method introduces a self-checking feature to the table that was not present if the table was computed with a multiplication for each line.

There is another important benefit of the second method for producing the numbers for our price table, a special case of the method of finite differences. Babbage could conceive of a way to add numbers together by some sort of machine. So, important mathematical tables might be produced *automatically*, and with a guarantee that the tables themselves were *without error*.

Your turn

1. Suppose ground turkey costs \$2.43 per pound. Your job is to complete the following tables of prices for ground turkey. One faction member should calculate the values via ten separate multiplications, while another faction member should perform ten additions. Do you work on scratch paper, and copy each answer to the appropriate column of Table 2.5 as soon as the number has been calculated.
2. How did you do? Do your tables agree? If they do not, which table is correct? Which method of computation was easier?

Table 2.5: Prices for ground turkey

Multiplication		Addition	
Pounds	Price	Pounds	Price
1	\$2.43	1	\$2.43
2		2	
3		3	
4		4	
5		5	
6		6	
7		7	
8		8	
9		9	
10		10	

Finite differences and polynomials

Of course, Babbage was not interested in producing pricing charts for butchers. He wanted to mechanize the production of mathematical tables which could then be used by navigators, astronomers, and other men of science. So, let us turn our attention to calculating a table of numbers for a mathematical function. We will create a table that could be used to compute successive values of the function via simple addition and subtraction, like the second meat price table we created above. We will focus on a function that Babbage used himself when talking about the Difference Engine: $f(x) = x^2 + x + 41$.

Our table will have four columns. The first column will be the value of x , which in this case we will constrain to be from the non-negative integers, $\{0, 1, 2, 3, \dots\}$. Next, we will have the value of our function, $f(x)$. The last two columns will be for the first and second differences, defined respectively as $d_1(x) = f(x+1) - f(x)$ and $d_2(x) = d_1(x+1) - d_1(x)$.

To begin our table, we must compute the first three values of our function, $f(0)$, $f(1)$, and $f(2)$, so that we have enough data to compute the first row's first and second differences, $d_1(0)$ and $d_2(0)$. Here is our start:

x	$f(x)$	$d_1(x)$	$d_2(x)$
0	41		
1	43		
2	47		

Now we can compute some of the differences:

$$d_1(0) = f(1) - f(0) = 43 - 41 = 2,$$

$$d_1(1) = f(2) - f(1) = 47 - 43 = 4,$$

and

$$d_2(0) = d_1(1) - d_1(0) = 4 - 2 = 2.$$

Filling those values into our table, we have:

x	$f(x)$	$d_1(x)$	$d_2(x)$
0	41	2	2
1	43	4	
2	47		

Your turn

3. We have computed a few more values of the function $f(x) = x^2 + x + 41$ for you and inserted them as new rows into Table 2.6. Fill in as many of the differences ($d_1(x)$ and $d_2(x)$) as you can.

Table 2.6: Differences for $f(x) = x^2 + x + 41$

x	$f(x)$	$d_1(x)$	$d_2(x)$
0	41	2	2
1	43	4	
2	47		
3	53		
4	61		
5	71		

4. Do you notice anything special about the fourth column of the table, $d_2(x)$? Explain your findings.

5. Now, compare the fourth column header to the degree of our polynomial function $f(x) = x^2 + x + 41$. (The degree is the largest power of x in the function). What do you see?
6. Table 2.7 is a partially completed differences table, for a different function: $f(x) = 3x^3 - 14x^2 + 12x + 13$. The new column, for the third difference, is $d_3(x) = d_2(x+1) - d_2(x)$. Fill in as much of the table as you can.

Table 2.7: Differences for $f(x) = 3x^3 - 14x^2 + 12x + 13$

x	$f(x)$	$d_1(x)$	$d_2(x)$	$d_3(x)$
0	13			
1	14			
2	5			
3	4			
4	29			
5	98			

7. Now, what do you think about the relationship between the third difference column and the degree of the polynomial $f(x) = 3x^3 - 14x^2 + 12x + 13$? Based on your findings with these two tables, what can you generalize about the degree of the polynomial, k , and the k^{th} difference $d_k(x)$?

Working backwards

For any polynomial function of degree k , $f(x) = c_k x^k + c_{k-1} x^{k-1} + c_{k-2} x^{k-2} + \dots + c_1 x^1 + c_0$, the k^{th} difference is always a constant. If that is the generalization you came to after your work on the previous two tables, congratulations!

Now, let us employ finite differences in the way Babbage intended the Difference Engine to work. In our butcher example, the second method of producing the price table generated each successive row of the chart via repeated additions. Once we have completed the first row and have found the k^{th} constant difference, we can use a similar procedure to determine the successive values of a polynomial function of degree k , *using only addition*.

We will work with a quadratic for the following example – another polynomial with degree two. As we saw for the previous quadratic example, our formula for computing the second difference is:

$$d_2(x) = d_1(x+1) - d_1(x).$$

If we do some algebraic manipulation, we can find the next value of the first difference like this:

$$d_1(x+1) = d_2(x) + d_1(x).$$

Similar algebra gives us a way to find the next value of $f(x)$ as well:

$$f(x+1) = d_1(x) + f(x).$$

Now, consider the following partial table, with only the first row completed, for the function $f(x) = 19x^2 - 5x + 17$. Using the new formulae we just developed, we can now fill in as many lines as we choose.

x	$f(x)$	$d_1(x)$	$d_2(x)$
0	17	14	38
1			
2			
3			
4			
5			

For example, to produce the next for for $x = 1$,

$$d_2(1) = 38,$$

since the 2nd difference will always be constant for a polynomial of degree two. No we can calculate the entries for the row from right to left:

$$d_1(1) = d_2(0) + d_1(0) = 38 + 14 = 52,$$

and

$$f(1) = d_1(0) + f(0) = 14 + 17 = 31.$$

Your turn

8. Complete Table 2.8, for $f(x) = 19x^2 - 5x + 17$, up through $x = 5$, using the method of finite differences elucidated above.

Table 2.8: Differences for $f(x) = 19x^2 - 5x + 17$

x	$f(x)$	$d_1(x)$	$d_2(x)$
0	17	14	38
1	31	52	38
2			
3			
4			
5			

9. Check your table by computing $f(5)$ directly. Did your finite differences method produce the correct value? If so, what conclusions can you draw about the other numbers in the table? If your finite differences method *did not* produce the same answer as the direct calculation, what conclusions could you draw?

Estimating other functions

The method of finite differences can be used to calculate many other sorts of functions in addition to the polynomials we have examined so far – as long as we are willing to settle for very accurate approximations instead of the “real” values. We can do this via Taylor Series approximations.

For our purposes, a Taylor Series expansion is a way to determine the value of a function, by computing the value of an abbreviated form of an infinitely long polynomial. For example, if we wish to find the cosine of an angle theta,

$$\cos(\Theta) = 1 - \frac{\Theta^2}{2!} + \frac{\Theta^4}{4!} - \frac{\Theta^6}{6!} + \frac{\Theta^8}{8!} - \dots$$

Of course as it stands the Taylor Series expansion of $\cos(\Theta)$ is not very useful, since it takes a really long time to compute the value of an infinitely long polynomial. However, if we are willing to settle for values of $\cos(\Theta)$ that are “good enough,” i.e., approximations of $\cos(\Theta)$, we can use an abbreviated version of the Taylor Series expansion, along with the method of finite differences, to compute our values.

In order to do finite differences calculations, let us suppose we end our approximating Taylor Series expansion at the $\frac{\Theta^6}{6!}$ term. Then the degree of our polynomial is six, and therefore our table would have eight columns if we use the same format as our previous examples. To use finite differences, we must have a complete first row, and to do that, we must compute the Taylor Series expansion values directly for the first seven rows, so we have enough data to compute the differences $d_1(x)$ through $d_6(x)$.

Here is this process would work. Our function in this case is our truncated Taylor Series approximation,

$$\cos(\Theta) = 1 - \frac{\Theta^2}{2!} + \frac{\Theta^4}{4!} - \frac{\Theta^6}{6!}$$

Our values for x will be angles, in radians:

$$\left\{0, \frac{\pi}{32}, \frac{\pi}{16}, \frac{3\pi}{32}, \frac{\pi}{8}, \dots\right\}.$$

Here is the table, with enough values for $f(x)$ to be able to calculate the values in the first row. All the computed values are shown in scientific notation.

x	$f(x)$	$d_1(x)$	$d_2(x)$	$d_3(x)$	$d_4(x)$	$d_5(x)$	$d_6(x)$
0	1.0000E+00	-4.8153E-03	-9.5842E-03	1.3867E-04	9.0957E-05	-2.2384E-06	-8.9536E-07
$\frac{\pi}{32}$	9.9518E-01	-1.4399E-02	-9.4455E-03	2.2963E-04	8.8718E-05	-3.1338E-06	
$\frac{\pi}{16}$	9.8079E-01	-2.3845E-02	-9.2159E-03	3.1835E-04	8.5584E-05		
$\frac{3\pi}{32}$	9.5694E-01	-3.3061E-02	-8.8975E-03	4.0393E-04			
$\frac{\pi}{8}$	9.2388E-01	-4.1958E-02	-8.4936E-03				
$\frac{5\pi}{32}$	8.8192E-01	-5.0452E-02					
$\frac{3\pi}{16}$	8.3147E-01						

Once we have that start, we could produce as many entries in the table as we wish, using the addition procedure outlined above.

There are Taylor Series expansions for many other functions that men of science might be interested in. For example, another trigonometric function, $\sin(\Theta)$, can be expanded as:

$$\sin(\Theta) = 1 - \frac{\Theta^3}{3!} + \frac{\Theta^5}{5!} - \frac{\Theta^7}{7!} + \dots$$

Another example is the natural logarithm. For values of $x < 1$, the natural logarithm $\ln(x)$ can be expanded thus:

$$\ln(x+1) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Summing up (sorry about that)

We have seen two separate, but complimentary, mathematical concepts in this lab. First, the method of *finite differences* allows us to compute successive values of polynomial functions through the repeated application of addition. A bonus of the finite differences method for computing functions is that it provides a self-checking mechanism so that we can be confident that our calculations are correct.

Secondly, we have seen that *Taylor Series approximations* can give us approximate values for other functions, such as $\cos(\Theta)$, $\sin(\Theta)$, and $\ln(x)$. We can make our approximations as accurate as we want by including more of the terms in the expansion.

Charles Babbage envisioned combining these two techniques in his Difference Engine to automatically produce error-free mathematical tables.

Programming labs (optional)

Lab A: computing $\cos(\Theta)$ with Taylor Series approximations

(This programming lab is appropriate for students who have been introduced to the basics of programming, up through the concept of iteration.)

Write a program to produce approximations for $\cos(\Theta)$, for values of

$$\Theta = \left\{ 0, \frac{\pi}{32}, \frac{\pi}{16}, \frac{3\pi}{32}, \frac{\pi}{8}, \dots, \pi \right\}$$

using a Taylor Series approximation. Output three columns of numbers: Θ , the value of $\cos(\Theta)$ as computed by your programming language's built-in cosine function, and the approximation of $\cos(\Theta)$ you compute via the Taylor Series approximation. Use enough terms in your Taylor Series approximation function so that the two values of $\cos(\Theta)$ agree to four decimal places.

Lab B: computing $\cos(\Theta)$ with finite differences

(This programming lab is appropriate for students who have been introduced to two-dimensional arrays.)

Write a program to produce approximations for $\cos(\Theta)$, for values of

$$\Theta = \left\{ 0, \frac{\pi}{32}, \frac{\pi}{16}, \frac{3\pi}{32}, \frac{\pi}{8}, \dots, \pi \right\}$$

using a finite differences approximation. Output three columns of numbers: Θ , the value of $\cos(\Theta)$ as computed by your programming language's built-in cosine function, and the approximation of $\cos(\Theta)$ you compute via the finite differences approximation. Use enough terms in your finite differences approximation table so that the two values of $\cos(\Theta)$ agree to four decimal places.

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Navigation problem

Assume that Broadstairs, Kent (B) is due West of Knokke-Heist, Belgium (KH). Napoleon sails from Knokke-Heist on a heading of 320° . Simultaneously, Admiral Lord Nelson sails from Broadstairs on a heading of 50° , sailing at a speed of 12 knots (nautical miles per hour, abbreviated as kts). Nelson and Napoleon maintain constant course and speed, and their ships meet so that Nelson can apprehend Napoleon.

1. How fast was Napoleon's ship traveling?
2. The distance from Broadstairs to Knokke-Heist is 80 nautical miles (nm). How long did it take Nelson to intercept Napoleon?
3. What distance did each ship travel?

Table of tangents

Table of tangents

Θ	0	1	2	3	4	5	6	7	8	9
0	0.000	0.002	0.003	0.005	0.007	0.009	0.010	0.012	0.014	0.016
1	0.017	0.019	0.021	0.023	0.024	0.026	0.028	0.030	0.031	0.033
2	0.035	0.037	0.038	0.040	0.042	0.044	0.045	0.047	0.049	0.051
3	0.052	0.054	0.056	0.058	0.059	0.061	0.063	0.065	0.066	0.068
4	0.070	0.072	0.073	0.075	0.077	0.079	0.080	0.082	0.084	0.086
5	0.087	0.089	0.091	0.093	0.095	0.096	0.098	0.100	0.102	0.103
6	0.105	0.107	0.109	0.110	0.112	0.114	0.116	0.117	0.119	0.121
7	0.123	0.125	0.126	0.128	0.130	0.132	0.133	0.135	0.137	0.139
8	0.141	0.142	0.144	0.146	0.148	0.149	0.151	0.153	0.155	0.157
9	0.158	0.160	0.162	0.164	0.166	0.167	0.169	0.171	0.173	0.175
10	0.176	0.178	0.180	0.182	0.184	0.185	0.187	0.189	0.191	0.193
11	0.194	0.196	0.198	0.200	0.202	0.203	0.205	0.207	0.209	0.211
12	0.213	0.214	0.216	0.218	0.220	0.222	0.224	0.225	0.227	0.229
13	0.231	0.233	0.235	0.236	0.238	0.240	0.242	0.244	0.246	0.247
14	0.249	0.251	0.253	0.255	0.257	0.259	0.260	0.262	0.264	0.266
15	0.268	0.270	0.272	0.274	0.275	0.277	0.279	0.281	0.283	0.285
16	0.287	0.289	0.291	0.292	0.294	0.296	0.298	0.300	0.302	0.304
17	0.306	0.308	0.310	0.311	0.313	0.315	0.317	0.319	0.321	0.323
18	0.325	0.327	0.329	0.331	0.333	0.335	0.337	0.338	0.340	0.342
19	0.344	0.346	0.348	0.350	0.352	0.354	0.356	0.358	0.360	0.362
20	0.364	0.366	0.368	0.370	0.372	0.374	0.376	0.378	0.380	0.382
21	0.384	0.386	0.388	0.390	0.392	0.394	0.396	0.398	0.400	0.402
22	0.404	0.406	0.408	0.410	0.412	0.414	0.416	0.418	0.420	0.422
23	0.424	0.427	0.429	0.431	0.433	0.435	0.437	0.439	0.441	0.443
24	0.445	0.447	0.449	0.452	0.454	0.456	0.458	0.460	0.462	0.464
25	0.466	0.468	0.471	0.473	0.475	0.477	0.479	0.481	0.483	0.486
26	0.488	0.490	0.492	0.494	0.496	0.499	0.501	0.503	0.505	0.507
27	0.510	0.512	0.514	0.516	0.518	0.521	0.523	0.525	0.527	0.529
28	0.532	0.534	0.536	0.538	0.541	0.543	0.545	0.547	0.550	0.552
29	0.554	0.557	0.559	0.561	0.563	0.566	0.568	0.570	0.573	0.575
30	0.577	0.580	0.582	0.584	0.587	0.589	0.591	0.594	0.596	0.598
31	0.601	0.603	0.606	0.608	0.610	0.613	0.615	0.618	0.620	0.622
32	0.625	0.627	0.630	0.632	0.635	0.637	0.640	0.642	0.644	0.647
33	0.649	0.652	0.654	0.657	0.659	0.662	0.664	0.667	0.669	0.672
34	0.675	0.677	0.680	0.682	0.685	0.687	0.690	0.692	0.695	0.698
35	0.700	0.703	0.705	0.708	0.711	0.713	0.716	0.719	0.721	0.724
36	0.727	0.729	0.732	0.735	0.737	0.740	0.743	0.745	0.748	0.751
37	0.754	0.756	0.759	0.762	0.765	0.767	0.770	0.773	0.776	0.778
38	0.781	0.784	0.787	0.790	0.793	0.795	0.798	0.801	0.804	0.807
39	0.810	0.813	0.816	0.818	0.821	0.824	0.827	0.830	0.833	0.836
40	0.839	0.842	0.845	0.848	0.851	0.854	0.857	0.860	0.863	0.866
41	0.869	0.872	0.875	0.879	0.882	0.885	0.888	0.891	0.894	0.897
42	0.900	0.904	0.907	0.910	0.913	0.916	0.920	0.923	0.926	0.929
43	0.933	0.936	0.939	0.942	0.946	0.949	0.952	0.956	0.959	0.962
44	0.966	0.969	0.972	0.976	0.979	0.983	0.986	0.990	0.993	0.997
45	1.000	1.003	1.007	1.011	1.014	1.018	1.021	1.025	1.028	1.032
46	1.036	1.039	1.043	1.046	1.050	1.054	1.057	1.061	1.065	1.069
47	1.072	1.076	1.080	1.084	1.087	1.091	1.095	1.099	1.103	1.107
48	1.111	1.115	1.118	1.122	1.126	1.130	1.134	1.138	1.142	1.146
49	1.150	1.154	1.159	1.163	1.167	1.171	1.175	1.179	1.183	1.188
50	1.192	1.196	1.200	1.205	1.209	1.213	1.217	1.222	1.226	1.230
51	1.235	1.239	1.244	1.248	1.253	1.257	1.262	1.266	1.271	1.275
52	1.280	1.285	1.289	1.294	1.299	1.303	1.308	1.313	1.317	1.322
53	1.327	1.332	1.337	1.342	1.347	1.351	1.356	1.361	1.366	1.371
54	1.376	1.381	1.387	1.392	1.397	1.402	1.407	1.412	1.418	1.423
55	1.428	1.433	1.439	1.444	1.450	1.455	1.460	1.466	1.471	1.477
56	1.483	1.488	1.494	1.499	1.505	1.511	1.517	1.522	1.528	1.534
57	1.540	1.546	1.552	1.558	1.564	1.570	1.576	1.582	1.588	1.594
58	1.600	1.607	1.613	1.619	1.625	1.632	1.638	1.645	1.651	1.658
59	1.664	1.671	1.678	1.684	1.691	1.698	1.704	1.711	1.718	1.725
60	1.732	1.739	1.746	1.753	1.760	1.767	1.775	1.782	1.789	1.797

Table of tangents

Θ	0	1	2	3	4	5	6	7	8	9
61	1.804	1.811	1.819	1.827	1.834	1.842	1.849	1.857	1.865	1.873
62	1.881	1.889	1.897	1.905	1.913	1.921	1.929	1.937	1.946	1.954
63	1.963	1.971	1.980	1.988	1.997	2.006	2.014	2.023	2.032	2.041
64	2.050	2.059	2.069	2.078	2.087	2.097	2.106	2.116	2.125	2.135
65	2.145	2.154	2.164	2.174	2.184	2.194	2.204	2.215	2.225	2.236
66	2.246	2.257	2.267	2.278	2.289	2.300	2.311	2.322	2.333	2.344
67	2.356	2.367	2.379	2.391	2.402	2.414	2.426	2.438	2.450	2.463
68	2.475	2.488	2.500	2.513	2.526	2.539	2.552	2.565	2.578	2.592
69	2.605	2.619	2.633	2.646	2.660	2.675	2.689	2.703	2.718	2.733
70	2.747	2.762	2.778	2.793	2.808	2.824	2.840	2.856	2.872	2.888
71	2.904	2.921	2.937	2.954	2.971	2.989	3.006	3.024	3.042	3.060
72	3.078	3.096	3.115	3.133	3.152	3.172	3.191	3.211	3.230	3.251
73	3.271	3.291	3.312	3.333	3.354	3.376	3.398	3.420	3.442	3.465
74	3.487	3.511	3.534	3.558	3.582	3.606	3.630	3.655	3.681	3.706
75	3.732	3.758	3.785	3.812	3.839	3.867	3.895	3.923	3.952	3.981
76	4.011	4.041	4.071	4.102	4.134	4.165	4.198	4.230	4.264	4.297
77	4.331	4.366	4.402	4.437	4.474	4.511	4.548	4.586	4.625	4.665
78	4.705	4.745	4.787	4.829	4.872	4.915	4.959	5.005	5.050	5.097
79	5.145	5.193	5.242	5.292	5.343	5.396	5.449	5.503	5.558	5.614
80	5.671	5.730	5.789	5.850	5.912	5.976	6.041	6.107	6.174	6.243
81	6.314	6.386	6.460	6.535	6.612	6.691	6.772	6.855	6.940	7.026
82	7.115	7.207	7.300	7.396	7.495	7.596	7.700	7.806	7.916	8.028
83	8.144	8.264	8.386	8.513	8.643	8.777	8.915	9.058	9.205	9.357
84	9.514	9.677	9.845	10.019	10.199	10.385	10.579	10.780	10.988	11.205
85	11.430	11.664	11.909	12.163	12.429	12.706	12.996	13.300	13.617	13.951
86	14.301	14.669	15.056	15.464	15.895	16.350	16.832	17.343	17.886	18.464
87	19.081	19.740	20.446	21.205	22.022	22.904	23.859	24.898	26.031	27.271
88	28.636	30.145	31.821	33.694	35.801	38.188	40.917	44.066	47.740	52.081
89	57.290	63.657	71.615	81.847	95.489	114.589	143.237	190.984	286.478	572.957
90	-	-	-	-	-	-	-	-	-	-
91	-57.290	-52.081	-47.740	-44.066	-40.917	-38.188	-35.801	-33.694	-31.821	-30.145
92	-28.636	-27.271	-26.031	-24.898	-23.859	-22.904	-22.022	-21.205	-20.446	-19.740
93	-19.081	-18.464	-17.886	-17.343	-16.832	-16.350	-15.895	-15.464	-15.056	-14.669
94	-14.301	-13.951	-13.617	-13.300	-12.996	-12.706	-12.429	-12.163	-11.909	-11.664
95	-11.430	-11.205	-10.988	-10.780	-10.579	-10.385	-10.199	-10.019	-9.845	-9.677
96	-9.514	-9.357	-9.205	-9.058	-8.915	-8.777	-8.643	-8.513	-8.386	-8.264
97	-8.144	-8.028	-7.916	-7.806	-7.700	-7.596	-7.495	-7.396	-7.300	-7.207
98	-7.115	-7.026	-6.940	-6.855	-6.772	-6.691	-6.612	-6.535	-6.460	-6.386
99	-6.314	-6.243	-6.174	-6.107	-6.041	-5.976	-5.912	-5.850	-5.789	-5.730
100	-5.671	-5.614	-5.558	-5.503	-5.449	-5.396	-5.343	-5.292	-5.242	-5.193
101	-5.145	-5.097	-5.050	-5.005	-4.959	-4.915	-4.872	-4.829	-4.787	-4.745
102	-4.705	-4.665	-4.625	-4.586	-4.548	-4.511	-4.474	-4.437	-4.402	-4.366
103	-4.331	-4.297	-4.264	-4.230	-4.198	-4.165	-4.134	-4.102	-4.071	-4.041
104	-4.011	-3.981	-3.952	-3.923	-3.895	-3.867	-3.839	-3.812	-3.785	-3.758
105	-3.732	-3.706	-3.681	-3.655	-3.630	-3.606	-3.582	-3.558	-3.534	-3.511
106	-3.487	-3.465	-3.442	-3.420	-3.398	-3.376	-3.354	-3.333	-3.312	-3.291
107	-3.271	-3.251	-3.230	-3.211	-3.191	-3.172	-3.152	-3.133	-3.115	-3.096
108	-3.078	-3.060	-3.042	-3.024	-3.006	-2.989	-2.971	-2.954	-2.937	-2.921
109	-2.904	-2.888	-2.872	-2.856	-2.840	-2.824	-2.808	-2.793	-2.778	-2.762
110	-2.747	-2.733	-2.718	-2.703	-2.689	-2.675	-2.660	-2.646	-2.633	-2.619
111	-2.605	-2.592	-2.578	-2.565	-2.552	-2.539	-2.526	-2.513	-2.500	-2.488
112	-2.475	-2.463	-2.450	-2.438	-2.426	-2.414	-2.402	-2.391	-2.379	-2.367
113	-2.356	-2.344	-2.333	-2.322	-2.311	-2.300	-2.289	-2.278	-2.267	-2.257
114	-2.246	-2.236	-2.225	-2.215	-2.204	-2.194	-2.184	-2.174	-2.164	-2.154
115	-2.145	-2.135	-2.125	-2.116	-2.106	-2.097	-2.087	-2.078	-2.069	-2.059
116	-2.050	-2.041	-2.032	-2.023	-2.014	-2.006	-1.997	-1.988	-1.980	-1.971
117	-1.963	-1.954	-1.946	-1.937	-1.929	-1.921	-1.913	-1.905	-1.897	-1.889
118	-1.881	-1.873	-1.865	-1.857	-1.849	-1.842	-1.834	-1.827	-1.819	-1.811
119	-1.804	-1.797	-1.789	-1.782	-1.775	-1.767	-1.760	-1.753	-1.746	-1.739
120	-1.732	-1.725	-1.718	-1.711	-1.704	-1.698	-1.691	-1.684	-1.678	-1.671
121	-1.664	-1.658	-1.651	-1.645	-1.638	-1.632	-1.625	-1.619	-1.613	-1.607
122	-1.600	-1.594	-1.588	-1.582	-1.576	-1.570	-1.564	-1.558	-1.552	-1.546
123	-1.540	-1.534	-1.528	-1.522	-1.517	-1.511	-1.505	-1.499	-1.494	-1.488
124	-1.483	-1.477	-1.471	-1.466	-1.460	-1.455	-1.450	-1.444	-1.439	-1.433
125	-1.428	-1.423	-1.418	-1.412	-1.407	-1.402	-1.397	-1.392	-1.387	-1.381

Table of tangents

Θ	0	1	2	3	4	5	6	7	8	9
126	-1.376	-1.371	-1.366	-1.361	-1.356	-1.351	-1.347	-1.342	-1.337	-1.332
127	-1.327	-1.322	-1.317	-1.313	-1.308	-1.303	-1.299	-1.294	-1.289	-1.285
128	-1.280	-1.275	-1.271	-1.266	-1.262	-1.257	-1.253	-1.248	-1.244	-1.239
129	-1.235	-1.230	-1.226	-1.222	-1.217	-1.213	-1.209	-1.205	-1.200	-1.196
130	-1.192	-1.188	-1.183	-1.179	-1.175	-1.171	-1.167	-1.163	-1.159	-1.154
131	-1.150	-1.146	-1.142	-1.138	-1.134	-1.130	-1.126	-1.122	-1.118	-1.115
132	-1.111	-1.107	-1.103	-1.099	-1.095	-1.091	-1.087	-1.084	-1.080	-1.076
133	-1.072	-1.069	-1.065	-1.061	-1.057	-1.054	-1.050	-1.046	-1.043	-1.039
134	-1.036	-1.032	-1.028	-1.025	-1.021	-1.018	-1.014	-1.011	-1.007	-1.003
135	-1.000	-0.997	-0.993	-0.990	-0.986	-0.983	-0.979	-0.976	-0.972	-0.969
136	-0.966	-0.962	-0.959	-0.956	-0.952	-0.949	-0.946	-0.942	-0.939	-0.936
137	-0.933	-0.929	-0.926	-0.923	-0.920	-0.916	-0.913	-0.910	-0.907	-0.904
138	-0.900	-0.897	-0.894	-0.891	-0.888	-0.885	-0.882	-0.879	-0.875	-0.872
139	-0.869	-0.866	-0.863	-0.860	-0.857	-0.854	-0.851	-0.848	-0.845	-0.842
140	-0.839	-0.836	-0.833	-0.830	-0.827	-0.824	-0.821	-0.818	-0.816	-0.813
141	-0.810	-0.807	-0.804	-0.801	-0.798	-0.795	-0.793	-0.790	-0.787	-0.784
142	-0.781	-0.778	-0.776	-0.773	-0.770	-0.767	-0.765	-0.762	-0.759	-0.756
143	-0.754	-0.751	-0.748	-0.745	-0.743	-0.740	-0.737	-0.735	-0.732	-0.729
144	-0.727	-0.724	-0.721	-0.719	-0.716	-0.713	-0.711	-0.708	-0.705	-0.703
145	-0.700	-0.698	-0.695	-0.692	-0.690	-0.687	-0.685	-0.682	-0.680	-0.677
146	-0.675	-0.672	-0.669	-0.667	-0.664	-0.662	-0.659	-0.657	-0.654	-0.652
147	-0.649	-0.647	-0.644	-0.642	-0.640	-0.637	-0.635	-0.632	-0.630	-0.627
148	-0.625	-0.622	-0.620	-0.618	-0.615	-0.613	-0.610	-0.608	-0.606	-0.603
149	-0.601	-0.598	-0.596	-0.594	-0.591	-0.589	-0.587	-0.584	-0.582	-0.580
150	-0.577	-0.575	-0.573	-0.570	-0.568	-0.566	-0.563	-0.561	-0.559	-0.557
151	-0.554	-0.552	-0.550	-0.547	-0.545	-0.543	-0.541	-0.538	-0.536	-0.534
152	-0.532	-0.529	-0.527	-0.525	-0.523	-0.521	-0.518	-0.516	-0.514	-0.512
153	-0.510	-0.507	-0.505	-0.503	-0.501	-0.499	-0.496	-0.494	-0.492	-0.490
154	-0.488	-0.486	-0.483	-0.481	-0.479	-0.477	-0.475	-0.473	-0.471	-0.468
155	-0.466	-0.464	-0.462	-0.460	-0.458	-0.456	-0.454	-0.452	-0.449	-0.447
156	-0.445	-0.443	-0.441	-0.439	-0.437	-0.435	-0.433	-0.431	-0.429	-0.427
157	-0.424	-0.422	-0.420	-0.418	-0.416	-0.414	-0.412	-0.410	-0.408	-0.406
158	-0.404	-0.402	-0.400	-0.398	-0.396	-0.394	-0.392	-0.390	-0.388	-0.386
159	-0.384	-0.382	-0.380	-0.378	-0.376	-0.374	-0.372	-0.370	-0.368	-0.366
160	-0.364	-0.362	-0.360	-0.358	-0.356	-0.354	-0.352	-0.350	-0.348	-0.346
161	-0.344	-0.342	-0.340	-0.338	-0.337	-0.335	-0.333	-0.331	-0.329	-0.327
162	-0.325	-0.323	-0.321	-0.319	-0.317	-0.315	-0.313	-0.311	-0.310	-0.308
163	-0.306	-0.304	-0.302	-0.300	-0.298	-0.296	-0.294	-0.292	-0.291	-0.289
164	-0.287	-0.285	-0.283	-0.281	-0.279	-0.277	-0.275	-0.274	-0.272	-0.270
165	-0.268	-0.266	-0.264	-0.262	-0.260	-0.259	-0.257	-0.255	-0.253	-0.251
166	-0.249	-0.247	-0.246	-0.244	-0.242	-0.240	-0.238	-0.236	-0.235	-0.233
167	-0.231	-0.229	-0.227	-0.225	-0.224	-0.222	-0.220	-0.218	-0.216	-0.214
168	-0.213	-0.211	-0.209	-0.207	-0.205	-0.203	-0.202	-0.200	-0.198	-0.196
169	-0.194	-0.193	-0.191	-0.189	-0.187	-0.185	-0.184	-0.182	-0.180	-0.178
170	-0.176	-0.175	-0.173	-0.171	-0.169	-0.167	-0.166	-0.164	-0.162	-0.160
171	-0.158	-0.157	-0.155	-0.153	-0.151	-0.149	-0.148	-0.146	-0.144	-0.142
172	-0.141	-0.139	-0.137	-0.135	-0.133	-0.132	-0.130	-0.128	-0.126	-0.125
173	-0.123	-0.121	-0.119	-0.117	-0.116	-0.114	-0.112	-0.110	-0.109	-0.107
174	-0.105	-0.103	-0.102	-0.100	-0.098	-0.096	-0.095	-0.093	-0.091	-0.089
175	-0.087	-0.086	-0.084	-0.082	-0.080	-0.079	-0.077	-0.075	-0.073	-0.072
176	-0.070	-0.068	-0.066	-0.065	-0.063	-0.061	-0.059	-0.058	-0.056	-0.054
177	-0.052	-0.051	-0.049	-0.047	-0.045	-0.044	-0.042	-0.040	-0.038	-0.037
178	-0.035	-0.033	-0.031	-0.030	-0.028	-0.026	-0.024	-0.023	-0.021	-0.019
179	-0.017	-0.016	-0.014	-0.012	-0.010	-0.009	-0.007	-0.005	-0.003	-0.002

Table of tangents

Table of tangents

Θ	0	1	2	3	4	5	6	7	8	9
0	0.000	0.002	0.003	0.005	0.007	0.009	0.010	0.012	0.014	0.016
1	0.017	0.019	0.021	0.023	0.024	0.026	0.028	0.030	0.031	0.033
2	0.035	0.037	0.038	0.040	0.042	0.044	0.045	0.047	0.049	0.051
3	0.052	0.054	0.056	0.058	0.059	0.061	0.063	0.065	0.066	0.068
4	0.070	0.072	0.720	0.075	0.077	0.079	0.080	0.082	0.084	0.086
5	0.087	0.089	0.091	0.093	0.095	0.096	0.098	0.100	0.102	0.103
6	0.105	0.107	0.109	0.110	0.112	0.114	0.116	0.117	0.119	0.121
7	0.123	0.125	0.126	0.128	0.130	0.132	0.133	0.135	0.137	0.139
8	0.141	0.142	0.144	0.146	0.148	0.149	0.151	0.153	0.155	0.157
9	0.158	0.160	0.162	0.164	0.166	0.167	0.169	0.171	0.173	0.175
10	0.176	0.178	0.180	0.182	0.184	0.185	0.187	0.189	0.191	0.193
11	0.194	0.196	0.198	0.200	0.202	0.203	0.205	0.207	0.209	0.211
12	0.213	0.214	0.216	0.218	0.220	0.222	0.224	0.225	0.227	0.229
13	0.231	0.233	0.235	0.236	0.238	0.240	0.242	0.244	0.246	0.247
14	0.249	0.251	0.253	0.255	0.257	0.259	0.260	0.262	0.264	0.266
15	0.268	0.270	0.272	0.274	0.275	0.514	0.279	0.281	0.283	0.285
16	0.287	0.289	0.291	-0.986	0.294	0.296	0.298	0.300	0.302	0.304
17	0.306	0.308	0.310	0.311	0.313	0.315	0.317	0.319	0.321	0.323
18	0.325	0.327	0.329	0.331	0.333	0.335	0.337	0.338	0.340	0.342
19	0.344	0.346	0.348	0.350	0.352	0.354	0.356	0.358	0.360	0.362
20	0.364	0.366	0.368	0.370	0.372	0.374	0.376	0.378	0.380	0.382
21	0.384	0.386	0.388	0.390	0.392	0.394	0.396	0.398	0.400	0.402
22	0.404	0.406	0.408	0.410	0.412	0.414	0.416	0.418	0.420	0.422
23	0.424	0.427	0.429	0.431	0.433	0.435	0.437	0.439	0.441	0.443
24	0.445	0.447	0.449	0.452	0.454	0.456	0.458	0.460	0.462	0.464
25	0.466	0.468	0.471	0.473	0.475	0.477	0.479	0.481	0.483	0.486
26	0.488	0.490	0.492	0.494	0.496	0.499	0.501	0.503	0.505	0.507
27	0.510	0.512	0.514	0.516	0.518	0.521	0.523	0.525	0.527	0.529
28	0.532	0.534	0.536	0.538	0.541	0.543	0.545	0.547	0.550	0.552
29	0.554	0.557	0.559	0.561	0.563	0.566	0.568	0.570	0.573	0.575
30	0.577	0.580	0.582	0.584	0.587	0.589	0.591	0.594	0.596	0.598
31	0.347	0.603	0.606	0.608	5.242	0.613	0.615	0.618	0.620	0.622
32	0.625	0.627	0.630	0.632	0.635	0.637	0.640	0.642	0.644	0.647
33	0.649	0.652	0.654	0.657	0.659	0.662	0.664	0.667	0.669	0.672
34	0.675	0.677	0.680	0.682	0.685	0.687	0.690	0.692	0.695	0.698
35	0.700	0.703	0.705	0.708	0.711	0.713	0.716	0.719	0.721	0.724
36	0.727	0.729	0.732	0.735	0.737	0.740	0.743	0.745	0.748	0.751
37	0.754	0.756	0.759	0.762	0.765	0.767	0.770	0.773	0.776	0.778
38	0.781	0.784	0.787	0.790	0.793	0.795	0.798	0.801	0.804	0.807
39	0.810	0.813	0.816	0.818	0.821	0.824	0.827	0.830	0.833	0.836
40	0.839	0.842	0.845	0.848	0.851	0.854	0.857	0.860	0.863	0.866
41	0.869	0.872	0.875	0.879	0.882	0.885	0.888	0.891	0.894	0.897
42	0.900	0.904	0.907	0.910	0.913	0.916	0.920	0.923	0.926	0.929
43	0.933	0.936	0.939	0.942	0.946	0.949	0.952	0.956	0.959	0.962
44	0.966	0.969	0.972	0.976	0.979	0.983	0.986	0.990	0.993	0.997
45	1.000	1.003	1.007	1.011	1.014	1.018	1.021	1.025	1.028	1.032
46	1.036	1.039	1.043	1.046	1.050	1.054	1.057	1.061	1.065	1.069
47	1.072	1.076	1.080	1.084	1.087	1.091	1.095	1.099	1.103	1.107
48	1.111	1.115	1.118	1.122	1.126	1.130	1.134	1.138	1.142	1.146
49	1.150	1.154	1.159	1.163	1.167	1.171	1.175	1.179	1.183	1.188
50	1.192	1.196	1.200	1.205	1.209	1.213	1.217	1.222	1.226	1.230
51	1.235	1.239	1.244	1.248	1.253	1.257	1.262	1.266	1.271	1.275
52	1.280	1.285	1.289	1.294	1.299	1.303	1.308	1.313	1.317	1.322
53	1.327	1.332	1.337	1.342	1.347	1.351	1.356	1.361	1.366	1.371
54	1.376	1.381	1.387	1.392	1.397	1.402	1.407	1.412	1.418	1.423
55	1.428	1.433	1.439	1.444	1.450	1.455	1.460	1.466	1.471	1.477
56	1.483	1.488	1.494	1.499	1.505	1.511	1.517	1.522	1.528	1.534
57	1.540	1.546	1.552	1.558	1.564	1.570	1.576	1.582	1.588	1.594
58	1.600	1.607	1.613	1.619	1.625	1.632	1.638	1.645	1.651	1.658
59	1.664	1.671	1.678	1.684	1.691	1.698	1.704	1.711	1.718	2.235
60	1.732	1.739	1.746	1.753	1.760	1.767	1.775	1.782	1.789	1.797

Table of tangents

Θ	0	1	2	3	4	5	6	7	8	9
61	1.804	1.811	1.819	1.827	1.834	1.842	1.849	1.857	1.865	1.873
62	1.881	1.889	1.897	1.905	1.913	1.921	1.929	1.937	1.946	1.954
63	1.963	1.971	1.980	1.988	1.997	2.006	2.014	2.023	2.032	2.041
64	2.050	2.059	2.069	2.078	2.087	2.097	2.106	2.116	2.125	2.135
65	2.145	2.154	2.164	2.174	2.184	2.194	2.204	2.215	2.225	2.236
66	2.246	2.257	2.267	2.278	2.289	2.300	2.311	2.322	2.333	2.344
67	2.356	2.367	2.379	2.391	2.402	2.414	2.426	2.438	2.450	2.463
68	2.475	2.488	3.383	2.513	2.526	2.539	2.552	2.565	2.578	2.592
69	2.605	2.619	2.633	2.646	2.660	2.675	2.689	2.703	2.718	2.733
70	2.747	2.762	2.778	2.793	2.808	2.824	2.840	2.856	2.872	2.888
71	2.904	2.921	2.937	2.954	2.971	2.989	3.006	3.024	3.042	3.060
72	3.078	3.096	3.115	3.133	3.152	3.172	3.191	3.211	3.230	3.251
73	3.271	3.291	3.312	3.333	3.354	3.376	3.398	3.420	3.442	3.465
74	3.487	3.511	3.534	3.558	3.582	3.606	3.630	3.655	3.681	3.706
75	3.732	4.178	3.785	3.812	3.839	3.867	3.895	3.923	3.952	3.981
76	4.011	4.041	4.071	4.102	4.134	4.165	4.198	4.230	4.264	4.297
77	4.331	4.366	4.402	4.437	4.474	4.511	4.548	4.586	4.625	4.665
78	4.705	4.745	4.787	4.829	4.872	4.915	4.959	5.005	5.050	5.097
79	5.145	5.365	5.242	5.292	5.343	5.396	5.449	6.231	5.558	5.614
80	4.829	5.730	5.789	5.850	5.912	5.976	6.041	6.107	6.174	6.243
81	6.314	6.386	6.460	6.535	6.612	6.691	6.772	6.855	6.940	6.806
82	7.115	7.207	7.300	7.396	7.495	7.596	7.700	7.806	7.916	8.028
83	8.144	8.264	8.386	8.513	8.643	8.777	8.915	9.058	9.205	9.357
84	9.514	9.677	9.845	10.019	10.199	10.385	10.579	10.780	10.988	11.205
85	11.430	11.664	11.909	12.163	12.429	12.706	12.996	13.300	13.617	13.951
86	14.301	14.669	15.056	15.464	15.895	16.350	16.832	17.343	17.886	18.464
87	19.081	19.740	20.446	21.205	22.022	22.904	23.859	24.898	26.031	27.271
88	28.636	30.145	31.821	33.694	35.801	38.188	40.917	44.066	47.740	52.081
89	57.290	63.657	71.615	81.847	95.489	114.589	143.237	190.984	286.478	572.957
90	-	-	-	-	-	-	-	-	-	-
91	-57.290	-52.081	-47.740	-44.066	-40.917	-38.188	-35.801	-33.694	-31.821	-30.145
92	-28.636	-27.271	-26.031	-24.898	-23.859	-22.904	-22.022	-21.205	-20.446	-19.740
93	-19.081	-18.464	-17.886	-17.343	-16.832	-16.350	-15.895	-15.464	-15.056	-14.669
94	-14.301	-13.951	-13.617	-13.300	-12.996	-12.706	-12.429	-12.163	-11.909	-11.664
95	-11.430	-11.205	-10.988	-10.780	-10.579	-10.385	-10.199	-10.019	-9.845	-9.677
96	-9.514	-9.357	-9.205	-9.058	-8.915	-8.777	-8.643	-8.513	-8.386	-8.264
97	-8.144	-8.028	-7.916	-7.806	-7.700	-7.596	-7.495	-7.396	-7.300	-7.207
98	-7.115	-7.026	-6.940	-6.855	-6.772	-6.691	-6.612	-6.535	-6.460	-6.386
99	-6.314	-6.243	-6.174	-6.107	-6.041	-5.976	-5.912	-5.850	-5.789	-5.730
100	-5.671	-5.614	-5.558	-5.503	-5.449	-5.396	-5.343	-5.292	-5.242	-5.193
101	-5.145	-5.097	-5.050	-5.005	-4.959	-4.915	-4.872	-4.829	-4.787	-4.745
102	-4.705	-4.665	-4.625	-4.586	-4.548	-4.511	-4.474	-4.437	-4.402	-4.366
103	-4.331	-4.297	-4.264	-4.230	-4.198	-4.165	-4.134	-4.102	-4.071	-4.041
104	-4.011	-3.981	-3.952	-3.923	-3.895	-3.867	-3.839	-3.812	-3.785	-3.758
105	-3.732	-3.706	-3.681	-3.655	-3.630	-3.606	-3.582	-3.558	-3.534	-3.511
106	-3.487	-3.465	-3.442	-3.420	-3.398	-3.376	-3.354	-3.333	-3.312	-3.291
107	-3.271	-3.251	-3.230	-3.211	-3.191	-3.172	-3.152	-3.133	-3.115	-3.096
108	-3.078	-3.060	-3.042	-3.024	-3.006	-2.989	-2.971	-2.954	-2.937	-2.921
109	-2.904	-2.888	-2.872	-2.856	-2.840	-2.824	-2.808	-2.793	-2.778	-2.762
110	-2.747	-2.733	-2.718	-2.703	-2.689	-2.675	-2.660	-2.646	-2.633	-2.619
111	-2.605	-2.592	-2.578	-2.565	-2.552	-2.539	-2.526	-2.513	-2.500	-2.488
112	-2.475	-2.463	-2.450	-2.438	-2.426	-2.414	-2.402	-2.391	-2.379	-2.367
113	-2.356	-2.344	-2.333	-2.322	-2.311	-2.300	-2.289	-2.278	-2.267	-2.257
114	-2.246	-2.236	-2.225	-2.215	0.517	-2.194	-2.184	-2.174	-2.164	-2.154
115	-2.145	-2.135	-2.125	-2.116	-2.106	-2.097	-2.087	-2.078	-2.069	-2.059
116	-2.050	-2.041	-2.032	-2.023	-2.014	-2.006	-1.997	-1.988	-1.980	-1.971
117	-1.963	-1.954	-1.946	-1.937	-1.929	-1.921	-1.913	-1.905	-1.897	-1.889
118	-1.881	-1.873	-1.865	-1.857	-1.849	-1.842	-1.834	-1.827	-1.819	-1.811
119	-1.804	-1.797	-1.789	-1.782	-1.775	-1.767	-1.760	-1.753	-1.746	-1.739
120	-1.732	-1.725	-1.718	-1.711	-1.704	-1.698	-1.691	-1.684	-1.678	-1.671
121	-1.664	-1.658	-1.651	-1.645	-1.638	-1.632	-1.625	-1.619	-1.613	-1.607
122	-1.600	-1.594	-1.588	-1.582	-1.576	-1.570	-1.564	-1.558	-1.552	-1.546
123	-1.540	-1.534	-1.528	-1.522	-1.517	-1.511	-1.505	-1.499	-1.494	-1.488
124	-1.483	-1.477	-1.471	-1.466	-1.460	-1.455	-1.450	-1.444	-1.439	-1.433
125	-1.428	-1.423	-1.418	-1.412	-1.407	-1.402	-1.397	-1.392	-1.387	-1.381

Table of tangents

Θ	0	1	2	3	4	5	6	7	8	9
126	-1.376	-1.371	-1.366	-1.361	-1.356	-1.351	-1.347	-1.342	-1.337	-1.332
127	-1.327	-1.322	-1.317	-1.313	-1.308	-1.303	-1.299	-1.294	-1.289	-1.285
128	-1.280	-1.275	-1.271	-1.266	-1.262	-1.257	-1.253	-1.248	-1.244	-1.239
129	-1.235	-1.230	-1.226	-1.222	-1.217	-1.213	-1.209	-1.205	-1.200	-1.196
130	-1.192	-1.188	-1.183	-1.179	-1.175	-1.171	-1.167	-1.163	-1.159	-1.154
131	-1.150	-1.146	-1.142	-1.138	-1.134	-1.130	-1.126	-1.122	-1.118	-1.115
132	-1.111	-1.107	-1.103	-1.099	-1.095	-1.091	-1.087	-1.084	-1.080	-1.076
133	-1.072	-1.069	-1.065	-1.061	-1.057	-1.054	-1.050	-1.046	-1.043	-1.039
134	-1.036	-1.032	-1.028	-1.025	-1.021	-1.018	-1.014	-1.011	-1.007	-1.003
135	-1.000	-3.422	-0.993	-0.990	-0.986	-0.983	-0.979	-0.976	-0.972	-0.969
136	-0.966	-0.962	-0.959	-0.956	-0.952	-0.949	-0.946	-0.942	-0.939	-0.936
137	-0.933	-0.929	-0.926	-0.923	-0.920	-0.916	-0.913	-0.910	-0.907	-0.904
138	-0.900	-0.897	-0.894	-0.891	-0.888	-0.885	-0.882	-0.879	-0.875	-0.872
139	-0.869	-0.866	-0.863	-0.860	-0.857	-0.854	-0.851	-0.848	-0.845	-0.842
140	-0.839	-0.836	-0.833	-0.830	-0.827	-0.824	-0.821	-0.818	-0.816	-0.813
141	-0.810	-0.807	-0.804	-0.801	-0.798	-0.795	-0.793	-0.790	-0.787	-0.784
142	-0.781	-0.778	-0.776	-0.773	-0.770	-0.767	-0.765	-0.762	-0.759	-0.756
143	-0.754	-0.751	-0.748	-0.745	-0.743	-0.740	-0.737	-0.735	-0.732	-0.729
144	-0.727	-0.724	-0.721	-0.719	-0.716	-0.713	-0.711	-0.708	-0.705	-0.703
145	-0.700	-0.698	-0.695	-0.692	-0.690	-0.687	-0.685	-0.682	-0.680	-0.677
146	-0.675	-0.672	-0.669	-0.667	-0.664	-0.662	-0.659	-0.657	-0.654	-0.652
147	-0.649	-0.647	-0.644	-0.642	-0.640	-0.637	-0.635	-0.632	-0.630	-0.627
148	-0.625	-0.622	-0.620	-0.618	-0.615	-0.613	-0.610	-0.608	-0.606	-0.603
149	-0.601	-0.598	-0.596	-0.804	-0.591	-0.589	-0.587	-0.584	-0.582	0.915
150	-0.577	-0.575	-0.573	-0.570	-0.568	-0.566	-0.563	-0.561	-0.559	-0.557
151	-0.554	-0.552	-0.550	-0.547	-0.545	-0.543	-0.541	-0.538	-0.536	-0.534
152	-0.532	-0.529	-0.527	-0.525	-0.523	-0.521	-0.518	-0.516	-0.514	-0.512
153	-0.510	-0.507	-0.505	-0.503	-0.501	-0.499	-0.496	-0.494	-0.492	-0.490
154	-0.488	-0.486	-0.483	-0.481	-0.479	-0.477	-0.475	-0.473	-0.471	-0.468
155	-0.466	-0.464	-0.462	-0.460	-0.458	-0.456	-0.454	-0.452	-0.449	-0.447
156	-0.445	-0.443	-0.441	-0.439	-0.437	-0.435	-0.433	-0.431	-0.429	-0.427
157	-0.424	-0.422	-0.420	-0.418	-0.416	-0.414	-0.412	-0.410	-0.408	-0.406
158	-0.404	-0.402	-0.400	-0.398	-0.396	-0.394	-0.392	-0.390	-0.388	-0.386
159	-0.384	-0.382	-0.380	-0.378	-0.376	-0.374	-0.372	-0.370	-0.368	-0.366
160	-0.364	-0.362	-0.360	-0.358	-0.356	-0.354	-0.352	-0.350	-0.348	-0.346
161	-0.344	-0.342	-0.340	-0.338	-0.337	-0.335	-0.333	-0.331	-0.329	-0.327
162	-0.325	-0.323	-0.321	-0.319	-0.317	-0.315	-0.313	-0.311	-0.310	-0.308
163	-0.306	-0.304	-0.302	-0.300	-0.298	-0.296	-0.294	-0.292	-0.291	-0.289
164	-0.287	-0.285	-0.283	-0.281	-0.279	-0.277	-0.275	-0.274	-0.272	-0.270
165	-0.268	-0.266	-0.264	-0.262	-0.260	-0.259	-0.257	-0.255	-0.253	-0.251
166	-0.249	-0.247	-0.246	-0.244	-0.242	-0.240	-0.238	-0.236	-0.235	-0.233
167	-0.231	-0.229	-0.227	-0.225	-0.224	-0.222	-0.220	-0.218	-0.216	-0.214
168	-0.213	-0.211	-0.209	-0.207	-0.205	-0.203	-0.202	-0.200	-0.198	-0.196
169	-0.194	-0.193	-0.191	-0.189	-0.187	-0.185	-0.184	-0.182	-0.180	-0.178
170	-0.176	-0.175	-0.173	-0.171	-0.169	-0.167	-0.166	-0.164	-0.162	-0.160
171	-0.158	-0.157	-0.155	-0.153	-0.151	-0.149	-0.148	-0.146	-0.144	-0.142
172	-0.141	-0.139	-0.137	-0.135	-0.133	-0.132	-0.130	-0.128	-0.126	-0.125
173	-0.123	-0.121	-0.119	-0.117	-0.116	-0.114	-0.112	-0.110	-0.109	-0.107
174	-0.105	-0.103	-0.102	-0.100	-0.098	-0.096	-0.095	-0.093	-0.091	-0.089
175	-0.087	-0.086	-0.084	-0.215	-0.080	-0.079	-0.077	-0.075	-0.073	-0.072
176	-0.070	-0.068	-0.066	-0.065	-0.063	-0.061	-0.059	-0.058	-0.056	-0.054
177	-0.052	-0.051	-0.049	-0.047	-0.045	-0.044	-0.042	-0.040	-0.038	-0.037
178	-0.035	-0.033	-0.031	-0.030	-0.028	-0.026	-0.024	-0.023	-0.021	-0.019
179	-0.017	-0.016	-0.014	-0.012	-0.010	-0.009	-0.007	-0.005	-0.003	-0.002

Table of square roots

Table of square roots

x	0	1	2	3	4	5	6	7	8	9
1	1.000	1.049	1.095	1.140	1.183	1.225	1.265	1.304	1.342	1.378
2	1.414	1.449	1.483	1.517	1.549	1.581	1.612	1.643	1.673	1.703
3	1.732	1.761	1.789	1.817	1.844	1.871	1.897	1.924	1.949	1.975
4	2.000	2.025	2.049	2.074	2.098	2.121	2.145	2.168	2.191	2.214
5	2.236	2.258	2.280	2.302	2.324	2.345	2.366	2.387	2.408	2.429
6	2.449	2.470	2.490	2.510	2.530	2.550	2.569	2.588	2.608	2.627
7	2.646	2.665	2.683	2.702	2.720	2.739	2.757	2.775	2.793	2.811
8	2.828	2.846	2.864	2.881	2.898	2.915	2.933	2.950	2.966	2.983
9	3.000	3.017	3.033	3.050	3.066	3.082	3.098	3.114	3.130	3.146
10	3.162	3.178	3.194	3.209	3.225	3.240	3.256	3.271	3.286	3.302
11	3.317	3.332	3.347	3.362	3.376	3.391	3.406	3.421	3.435	3.450
12	3.464	3.479	3.493	3.507	3.521	3.536	3.550	3.564	3.578	3.592
13	3.606	3.619	3.633	3.647	3.661	3.674	3.688	3.701	3.715	3.728
14	3.742	3.755	3.768	3.782	3.795	3.808	3.821	3.834	3.847	3.860
15	3.873	3.886	3.899	3.912	3.924	3.937	3.950	3.962	3.975	3.987
16	4.000	4.012	4.025	4.037	4.050	4.062	4.074	4.087	4.099	4.111
17	4.123	4.135	4.147	4.159	4.171	4.183	4.195	4.207	4.219	4.231
18	4.243	4.254	4.266	4.278	4.290	4.301	4.313	4.324	4.336	4.347
19	4.359	4.370	4.382	4.393	4.405	4.416	4.427	4.438	4.450	4.461
20	4.472	4.483	4.494	4.506	4.517	4.528	4.539	4.550	4.561	4.572
21	4.583	4.593	4.604	4.615	4.626	4.637	4.648	4.658	4.669	4.680
22	4.690	4.701	4.712	4.722	4.733	4.743	4.754	4.764	4.775	4.785
23	4.796	4.806	4.817	4.827	4.837	4.848	4.858	4.868	4.879	4.889
24	4.899	4.909	4.919	4.930	4.940	4.950	4.960	4.970	4.980	4.990
25	5.000	5.010	5.020	5.030	5.040	5.050	5.060	5.070	5.079	5.089
26	5.099	5.109	5.119	5.128	5.138	5.148	5.158	5.167	5.177	5.187
27	5.196	5.206	5.215	5.225	5.235	5.244	5.254	5.263	5.273	5.282
28	5.292	5.301	5.310	5.320	5.329	5.339	5.348	5.357	5.367	5.376
29	5.385	5.394	5.404	5.413	5.422	5.431	5.441	5.450	5.459	5.468
30	5.477	5.486	5.495	5.505	5.514	5.523	5.532	5.541	5.550	5.559
31	5.568	5.577	5.586	5.595	5.604	5.612	5.621	5.630	5.639	5.648
32	5.657	5.666	5.675	5.683	5.692	5.701	5.710	5.718	5.727	5.736
33	5.745	5.753	5.762	5.771	5.779	5.788	5.797	5.805	5.814	5.822
34	5.831	5.840	5.848	5.857	5.865	5.874	5.882	5.891	5.899	5.908
35	5.916	5.925	5.933	5.941	5.950	5.958	5.967	5.975	5.983	5.992
36	6.000	6.008	6.017	6.025	6.033	6.042	6.050	6.058	6.066	6.075
37	6.083	6.091	6.099	6.107	6.116	6.124	6.132	6.140	6.148	6.156
38	6.164	6.173	6.181	6.189	6.197	6.205	6.213	6.221	6.229	6.237
39	6.245	6.253	6.261	6.269	6.277	6.285	6.293	6.301	6.309	6.317
40	6.325	6.332	6.340	6.348	6.356	6.364	6.372	6.380	6.387	6.395
41	6.403	6.411	6.419	6.427	6.434	6.442	6.450	6.458	6.465	6.473
42	6.481	6.488	6.496	6.504	6.512	6.519	6.527	6.535	6.542	6.550
43	6.557	6.565	6.573	6.580	6.588	6.595	6.603	6.611	6.618	6.626
44	6.633	6.641	6.648	6.656	6.663	6.671	6.678	6.686	6.693	6.701
45	6.708	6.716	6.723	6.731	6.738	6.745	6.753	6.760	6.768	6.775
46	6.782	6.790	6.797	6.804	6.812	6.819	6.826	6.834	6.841	6.848
47	6.856	6.863	6.870	6.877	6.885	6.892	6.899	6.907	6.914	6.921
48	6.928	6.935	6.943	6.950	6.957	6.964	6.971	6.979	6.986	6.993
49	7.000	7.007	7.014	7.021	7.029	7.036	7.043	7.050	7.057	7.064
50	7.071	7.078	7.085	7.092	7.099	7.106	7.113	7.120	7.127	7.134
51	7.141	7.148	7.155	7.162	7.169	7.176	7.183	7.190	7.197	7.204
52	7.211	7.218	7.225	7.232	7.239	7.246	7.253	7.259	7.266	7.273
53	7.280	7.287	7.294	7.301	7.308	7.314	7.321	7.328	7.335	7.342
54	7.348	7.355	7.362	7.369	7.376	7.382	7.389	7.396	7.403	7.409
55	7.416	7.423	7.430	7.436	7.443	7.450	7.457	7.463	7.470	7.477
56	7.483	7.490	7.497	7.503	7.510	7.517	7.523	7.530	7.537	7.543
57	7.550	7.556	7.563	7.570	7.576	7.583	7.589	7.596	7.603	7.609
58	7.616	7.622	7.629	7.635	7.642	7.649	7.655	7.662	7.668	7.675
59	7.681	7.688	7.694	7.701	7.707	7.714	7.720	7.727	7.733	7.740
60	7.746	7.752	7.759	7.765	7.772	7.778	7.785	7.791	7.797	7.804
61	7.810	7.817	7.823	7.829	7.836	7.842	7.849	7.855	7.861	7.868

Table of square roots

x	0	1	2	3	4	5	6	7	8	9
62	7.874	7.880	7.887	7.893	7.899	7.906	7.912	7.918	7.925	7.931
63	7.937	7.944	7.950	7.956	7.962	7.969	7.975	7.981	7.987	7.994
64	8.000	8.006	8.012	8.019	8.025	8.031	8.037	8.044	8.050	8.056
65	8.062	8.068	8.075	8.081	8.087	8.093	8.099	8.106	8.112	8.118
66	8.124	8.130	8.136	8.142	8.149	8.155	8.161	8.167	8.173	8.179
67	8.185	8.191	8.198	8.204	8.210	8.216	8.222	8.228	8.234	8.240
68	8.246	8.252	8.258	8.264	8.270	8.276	8.283	8.289	8.295	8.301
69	8.307	8.313	8.319	8.325	8.331	8.337	8.343	8.349	8.355	8.361
70	8.367	8.373	8.379	8.385	8.390	8.396	8.402	8.408	8.414	8.420
71	8.426	8.432	8.438	8.444	8.450	8.456	8.462	8.468	8.473	8.479
72	8.485	8.491	8.497	8.503	8.509	8.515	8.521	8.526	8.532	8.538
73	8.544	8.550	8.556	8.562	8.567	8.573	8.579	8.585	8.591	8.597
74	8.602	8.608	8.614	8.620	8.626	8.631	8.637	8.643	8.649	8.654
75	8.660	8.666	8.672	8.678	8.683	8.689	8.695	8.701	8.706	8.712
76	8.718	8.724	8.729	8.735	8.741	8.746	8.752	8.758	8.764	8.769
77	8.775	8.781	8.786	8.792	8.798	8.803	8.809	8.815	8.820	8.826
78	8.832	8.837	8.843	8.849	8.854	8.860	8.866	8.871	8.877	8.883
79	8.888	8.894	8.899	8.905	8.911	8.916	8.922	8.927	8.933	8.939
80	8.944	8.950	8.955	8.961	8.967	8.972	8.978	8.983	8.989	8.994
81	9.000	9.006	9.011	9.017	9.022	9.028	9.033	9.039	9.044	9.050
82	9.055	9.061	9.066	9.072	9.077	9.083	9.088	9.094	9.099	9.105
83	9.110	9.116	9.121	9.127	9.132	9.138	9.143	9.149	9.154	9.160
84	9.165	9.171	9.176	9.182	9.187	9.192	9.198	9.203	9.209	9.214
85	9.220	9.225	9.230	9.236	9.241	9.247	9.252	9.257	9.263	9.268
86	9.274	9.279	9.284	9.290	9.295	9.301	9.306	9.311	9.317	9.322
87	9.327	9.333	9.338	9.343	9.349	9.354	9.359	9.365	9.370	9.375
88	9.381	9.386	9.391	9.397	9.402	9.407	9.413	9.418	9.423	9.429
89	9.434	9.439	9.445	9.450	9.455	9.460	9.466	9.471	9.476	9.482
90	9.487	9.492	9.497	9.503	9.508	9.513	9.518	9.524	9.529	9.534
91	9.539	9.545	9.550	9.555	9.560	9.566	9.571	9.576	9.581	9.586
92	9.592	9.597	9.602	9.607	9.612	9.618	9.623	9.628	9.633	9.638
93	9.644	9.649	9.654	9.659	9.664	9.670	9.675	9.680	9.685	9.690
94	9.695	9.701	9.706	9.711	9.716	9.721	9.726	9.731	9.737	9.742
95	9.747	9.752	9.757	9.762	9.767	9.772	9.778	9.783	9.788	9.793
96	9.798	9.803	9.808	9.813	9.818	9.823	9.829	9.834	9.839	9.844
97	9.849	9.854	9.859	9.864	9.869	9.874	9.879	9.884	9.889	9.894
98	9.899	9.905	9.910	9.915	9.920	9.925	9.930	9.935	9.940	9.945
99	9.950	9.955	9.960	9.965	9.970	9.975	9.980	9.985	9.990	9.995
100	10.000	10.005	10.010	10.015	10.020	10.025	10.030	10.035	10.040	10.045

Table of square roots

Table of square roots

x	0	1	2	3	4	5	6	7	8	9
1	1.000	1.049	1.095	1.140	1.183	1.225	1.265	1.304	1.342	1.378
2	1.414	1.449	1.483	1.517	1.549	1.581	1.612	1.643	1.673	1.703
3	1.732	1.761	1.789	1.817	1.844	1.871	1.897	1.924	1.949	1.975
4	2.000	2.025	2.049	2.074	2.098	2.121	2.145	2.168	2.191	2.214
5	2.236	2.258	2.280	2.302	2.324	2.345	2.366	2.387	2.408	1.824
6	2.449	2.470	2.490	2.510	2.530	2.550	2.569	2.588	2.608	2.627
7	2.646	2.665	2.683	2.702	2.720	2.739	2.757	2.775	2.793	2.811
8	2.828	2.846	2.864	2.881	2.898	2.915	2.933	2.950	2.966	2.983
9	3.000	3.017	3.033	3.050	3.066	3.082	3.098	3.114	3.130	3.146
10	3.162	3.178	3.194	3.209	3.225	3.240	3.256	3.271	3.286	3.302
11	3.317	3.332	3.347	3.362	3.376	3.391	3.406	3.421	3.435	3.450
12	3.464	3.479	3.493	3.507	3.521	3.536	3.550	3.564	3.578	3.592
13	3.606	3.619	3.633	3.647	3.661	3.674	3.688	3.701	3.715	3.728
14	3.742	3.755	3.768	3.782	3.795	3.808	3.821	3.834	3.847	3.860
15	3.873	3.886	3.899	3.912	3.924	3.937	3.950	3.962	3.975	3.987
16	4.000	4.012	4.025	4.037	4.050	4.062	4.074	4.087	4.099	4.111
17	4.123	4.135	4.147	4.159	4.171	4.183	4.195	4.207	4.219	4.231
18	4.243	4.254	4.266	4.278	4.290	4.301	4.313	4.324	4.336	4.347
19	4.359	4.370	4.382	4.393	4.405	4.416	4.427	4.438	4.450	4.461
20	4.472	4.483	4.494	4.506	4.517	4.528	4.539	4.550	4.561	4.572
21	4.583	4.593	4.604	4.615	4.626	4.637	4.648	4.658	4.669	4.680
22	4.690	4.701	4.712	4.722	4.733	4.743	4.754	4.764	4.775	4.785
23	4.796	4.806	4.817	4.827	4.837	4.848	4.858	4.868	4.879	4.889
24	4.899	5.811	4.919	4.930	4.940	4.950	4.960	4.970	4.980	4.990
25	5.000	5.010	5.020	5.030	5.040	5.550	5.060	5.070	5.079	5.089
26	5.099	5.109	5.119	5.128	5.138	5.148	5.158	5.167	5.177	5.187
27	5.196	5.206	5.215	5.225	5.235	5.244	5.254	5.263	5.273	5.282
28	5.292	5.301	5.310	5.320	5.329	5.339	5.348	5.357	5.367	5.376
29	5.385	5.394	5.404	5.413	5.422	5.431	5.441	5.450	5.459	5.468
30	5.477	5.486	5.495	5.505	5.514	5.523	5.532	5.541	5.550	5.559
31	5.568	5.577	5.586	5.595	5.604	5.612	5.621	5.630	5.639	5.648
32	5.651	5.666	5.675	5.683	5.692	5.701	5.710	5.718	5.727	5.736
33	5.745	5.753	5.762	5.771	5.779	5.788	5.797	5.805	5.814	5.822
34	5.831	5.840	5.848	5.857	5.865	5.874	5.882	5.892	5.899	5.908
35	5.916	5.925	5.933	5.941	5.950	5.958	5.967	5.975	5.983	5.992
36	6.000	6.008	6.017	6.025	6.033	6.042	6.050	6.058	6.066	6.075
37	6.083	6.091	6.099	6.107	6.116	6.124	6.132	6.140	6.148	6.156
38	6.164	6.173	6.181	6.189	6.197	6.205	6.213	6.221	6.229	6.237
39	6.245	6.253	6.261	6.269	6.277	6.285	6.293	6.301	6.309	6.317
40	6.325	6.332	6.340	6.348	6.356	6.364	6.372	6.380	6.387	6.395
41	6.403	6.411	6.419	6.427	6.434	6.442	6.450	6.458	6.465	6.473
42	6.481	6.488	6.496	6.504	6.512	6.519	6.527	6.535	6.542	6.550
43	6.557	6.565	6.573	6.580	6.588	6.595	6.603	6.611	6.618	6.626
44	6.633	6.641	6.648	6.656	6.663	6.671	6.678	6.686	6.693	6.701
45	6.708	6.716	6.723	6.731	6.738	6.745	6.753	6.760	6.768	6.775
46	6.782	6.790	6.797	6.804	6.812	6.819	6.826	6.834	6.841	6.848
47	6.856	6.863	6.870	6.877	4.214	6.892	6.899	6.907	6.914	6.921
48	6.928	6.935	6.943	6.950	6.957	6.964	6.971	6.979	6.986	6.993
49	7.000	7.007	7.014	7.021	7.029	7.036	7.043	7.050	7.057	7.064
50	7.071	7.078	7.085	7.092	7.099	7.106	7.113	7.120	7.127	7.134
51	7.141	7.148	7.155	7.162	7.169	7.176	7.183	7.190	7.197	7.204
52	7.211	7.589	7.225	7.232	7.239	7.246	7.253	7.259	7.266	7.273
53	7.280	7.287	7.294	7.301	7.308	7.314	7.321	7.328	7.335	7.342
54	7.348	7.355	7.362	7.369	7.376	7.382	7.389	7.396	7.403	7.409
55	7.416	7.423	7.430	7.436	7.443	7.450	7.457	7.463	7.470	7.477
56	7.483	7.490	7.497	7.503	7.510	7.517	7.523	7.530	7.537	7.543
57	7.550	7.556	7.563	7.570	7.576	7.583	7.589	7.596	7.603	7.609
58	7.616	7.622	7.629	7.635	7.642	7.649	7.655	7.662	7.855	7.675
59	7.681	7.688	7.694	7.701	7.707	7.714	7.720	7.727	7.733	7.740
60	7.746	7.752	7.759	7.765	7.772	7.778	7.785	7.791	7.797	7.804
61	7.810	7.817	7.823	7.829	7.836	7.842	7.849	7.855	7.861	7.868

Table of square roots

x	0	1	2	3	4	5	6	7	8	9
62	7.874	7.880	7.887	7.893	7.899	7.906	7.912	7.918	7.925	7.931
63	7.937	7.944	7.950	7.956	8.666	7.969	7.975	7.981	7.987	7.994
64	8.000	8.006	8.012	8.019	8.025	8.031	8.037	8.044	8.050	8.056
65	8.062	8.068	8.075	8.081	8.087	8.093	8.099	8.106	8.112	8.118
66	8.124	8.130	8.136	8.142	8.149	8.155	8.161	8.167	8.173	8.179
67	8.185	8.191	8.198	8.204	8.210	8.216	8.222	8.228	8.234	8.240
68	8.246	8.252	8.258	8.264	8.270	8.276	8.283	8.289	8.295	8.301
69	8.307	8.313	8.319	8.325	8.331	8.337	8.343	8.349	8.355	8.361
70	8.367	8.373	8.379	8.385	8.390	8.396	8.402	8.408	8.414	8.420
71	8.426	8.432	8.438	8.444	8.450	8.456	8.462	8.468	8.473	8.479
72	8.485	8.491	8.497	8.503	8.509	8.515	8.521	8.526	8.532	8.538
73	8.544	8.550	8.556	8.562	8.567	8.573	8.579	8.585	8.591	8.597
74	8.602	8.608	8.614	8.620	8.626	8.631	8.637	8.643	8.649	8.654
75	8.660	8.666	8.672	8.678	8.683	8.689	8.695	8.701	8.706	8.712
76	8.718	8.724	8.729	8.735	8.741	8.746	8.752	8.758	8.764	8.769
77	8.775	8.781	8.786	8.792	8.798	8.803	8.809	8.815	8.820	8.826
78	8.832	8.837	8.843	8.849	8.854	8.860	8.866	8.871	8.877	8.883
79	8.888	8.894	8.899	8.905	8.911	8.916	8.922	8.927	8.933	8.939
80	8.944	8.950	8.955	8.961	8.967	8.972	8.978	8.983	8.989	8.994
81	9.000	9.006	9.011	9.017	9.022	9.028	9.033	9.039	9.044	9.050
82	9.055	9.061	9.066	9.072	9.077	9.083	9.088	9.094	9.099	9.105
83	9.110	9.116	9.121	9.127	9.132	9.138	9.143	9.149	9.154	9.160
84	9.165	9.171	9.176	9.182	9.187	9.192	9.198	9.203	9.209	9.214
85	9.220	9.225	9.230	8.833	9.241	9.247	9.252	9.257	9.263	9.268
86	9.274	9.279	9.284	9.290	9.295	9.301	9.306	9.311	9.317	9.322
87	9.327	9.333	9.338	9.343	9.349	9.354	9.359	9.365	9.370	7.936
88	9.381	9.386	9.391	9.397	9.402	9.407	9.413	9.418	9.423	9.429
89	9.434	9.439	9.445	9.450	9.455	9.460	9.466	9.471	9.476	9.482
90	9.487	9.492	9.497	9.503	9.508	9.513	9.518	9.524	9.529	9.534
91	9.501	9.545	9.550	9.555	9.560	9.566	9.571	9.576	9.581	9.586
92	9.592	9.597	9.602	9.607	9.612	9.618	9.623	9.628	9.633	9.638
93	9.644	9.649	9.654	9.659	9.664	9.670	9.675	9.680	9.685	9.690
94	9.695	9.701	9.706	9.711	9.716	9.721	9.726	9.731	9.737	9.742
95	9.747	9.752	9.757	9.762	9.767	9.772	9.778	9.783	9.788	9.793
96	9.798	9.803	9.808	9.813	9.818	9.823	9.829	9.834	9.839	9.844
97	9.849	9.854	9.859	10.651	9.869	9.874	9.879	9.884	9.889	9.894
98	9.899	9.905	9.910	9.915	9.920	9.925	9.930	9.935	9.940	9.945
99	9.950	9.955	9.960	9.965	9.970	9.975	9.980	9.985	9.990	9.995
100	10.000	10.005	10.010	10.015	10.020	10.025	10.030	10.035	10.040	10.045

Lab 2: the Difference Engine

Faction / name(s): _____

This lab should help you understand more about how the Difference Engine would be used to make calculations. This lab assumes that you have completed the preceding lab on finite differences. Rather than using a large, heavy, expensive mechanism made from hundreds of mechanical parts, the lab uses a simulation that runs in a Web site. The site we will use simulates a machine using five orders of difference, and approximately 17 digits of accuracy.

This lab focuses on what the Difference Engine would have done, rather than on how it would have worked. Our simulation will produce a table of numbers, but it will not illustrate the movement of the gears, levers, and cams that the Difference Engine would have had. Rather, it uses a modern computer program to move through multiple iterations of a finite differences calculation, given the initial conditions for a particular function.

Also, rather than the more complicated Taylor series approximations that were introduced in the previous lab, this lab will continue to work with simple polynomial equations.

Your turn

Before we can start the Difference Engine simulation, we need to have a function for it to calculate. Since the simulator supports five orders of difference, let us use a polynomial of degree five, namely:

$$f(x) = y = 6x^5 + 10x^4 - 8x^3 - 6x^2 - 8x - 7$$

We will also need to calculate the first row of a table of finite differences, as this is what the Difference Engine simulator will require to calculate successive values of the function. For this first example, we have performed the calculations for you, in this table:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
0	-7	-6	260	1212	1680	720
1	-13	254	1472	2892	2400	
2	241	1726	4364	5292		
3	1967	6090	9656			
4	8057	15746				
5	23803					

In order to completely fill out the first row, we had to first calculate the first six values of y , for $x = 0, 1, 2, 3, 4$, and 5 . Then, we calculated five values for the first difference, here labeled Δy , using the formula $\Delta y(x) = f(x+1) - f(x)$. Next, we calculated the second difference column, using $\Delta^2 y(x) = \Delta y(x+1) - \Delta y(x)$. In a similar way, we calculated values for the third, fourth, and fifth differences. Recall that since the polynomial is of

degree five, the fifth difference column is a constant value, i.e., the entire column would have the value 720.

Now we can go to the simulation and see how the Difference Engine could take the $x = 0$ line from the table above, and from that calculate as many values of $f(x)$ as we desire.

On your computer, open a Web browser, and navigate to the following site to find the Difference Engine simulator:

<http://dunx1.irt.drexel.edu/~cp676/DiffEngine.html>

You should see something like this:

Difference Engine Simulator

The following program is a simulation of Charles Babbage's renowned Difference Engine

Number of Rows:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$

On the site, fill in the columns for x , y , Δy , $\Delta^2 y$, $\Delta^3 y$, $\Delta^4 y$, and $\Delta^5 y$, based on the values in the table. The simulator should look like this:

Number of Rows:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
0	-7	-6	260	1212	1680	720

Now it is time to set the Difference Engine in motion! Enter some number of rows – perhaps 20 – and click on the start button. The table will expand from one line to many, and the y column of the simulation shows the values of the function $f(x)$ for the various values of x .

Use the simulation to write down the values of $f(x)$ in the Table 2.13.

Of course, there would be no need for someone in Babbage's day to write down the results of a Difference Engine calculation, because the engine itself would typeset the results on copper plates that could be used to print the tables.

Table 2.13: Difference Engine output for $f(x) = y = 6x^5 + 10x^4 - 8x^3 - 6x^2 - 8x - 7$

x	$f(x)$
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	

Another example

Now, use the Difference Engine simulator to find values for the function:

$$f(x) = y = x^4 + 11x^3 - 2x^2 + 9x - 5$$

This time, determine the initial values for the Difference Engine yourself, by completing Table 2.14. Perform the smallest number of calculations you can in order to complete the full first line.

Table 2.14: Table for calculating Difference Engine input

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
0						
1						
2						
3						
4						

Now, reset the Difference Engine simulator by clicking the “Reset” button, and then enter the first line values. Create 100 lines of the table, and then write down the function values for the five values of x shown in Table 2.15.

Table 2.15: Difference Engine output for $f(x) = y = x^4 + 11x^3 - 2x^2 + 9x - 5$

x	$f(x)$
37	
49	
53	
88	
100	

Reflection

1. For the function you just worked with, evaluate $f(100)$ by hand and compare the results with the result produced by the Difference Engine simulator. Do your results agree? If so, what does that tell you? If not, what does that tell you?

2. One of the benefits claimed by proponents of the Difference Engine was that it would eliminate the chance of human error in calculation. Given your experience in this lab, what do you think of this claim? To what extent would the Difference Engine have eliminated the chance for human error?

Resolution to Recommend Continued Funding of the Difference Engine

Whereas the Chancellor of the Exchequer has inquired of the Royal Society to answer the following question, related to the Difference Engine being constructed by Mr. C. Babbage FRS: “Whether the progress of the Machine confirms them in their former opinion, that it will ultimately prove adequate to the important object it was intended to attain;”

Whereas the Government has spent some £1,500, and Mr. Babbage a considerable amount of his own funds, to produce a working fragment of the Difference Engine, thus demonstrating the feasibility of the machine;

Whereas the construction of the Difference Engine requires funding far beyond the capability of any one individual to bear;

Whereas there is ample evidence, provided by Mr. Dionysius Lardner FRS and others, that the perfectly error-free mathematical tables produced by the Difference Engine will have demonstrable economic benefit to the Nation, not the least through improved maritime navigation;

Whereas reliable and readily-available mathematical tables are cornerstone requirements for the advancement of scientific knowledge; and

Whereas the Royal Society must be ardent and faithful in supporting those natural philosophers with the demonstrated capability to make great strides forward in natural philosophy; now, therefore, be it

Resolved, that the Royal Society of London

1. fully supports the efforts of Mr. C. Babbage FRS to bring construction of his Difference Engine to completion; and
2. therefore, strongly urges His Majesty’s Government to provide further funding to Mr. Babbage, to finish the Difference Engine, so that it may be effectively employed by the Government.

Resolution to Recommend Discontinuation of Funding for the Difference Engine

Whereas the Chancellor of the Exchequer has inquired of the Royal Society to answer the following question, related to the Difference Engine being constructed by Mr. C. Babbage FRS: “Whether the progress of the Machine confirms them in their former opinion, that it will ultimately prove adequate to the important object it was intended to attain;”

Whereas there is no concrete evidence that erroneous navigational tables have ever caused the loss of even a single vessel from the Royal Navy or the nations’ mercantile fleet;

Whereas there is no concrete evidence that erroneous tables of any sort have had any measurable detrimental impact on the pursuit of scientific study, or upon the economic well-being of the nation;

Whereas it is not the place of the British Government to fund such expensive, speculative projects that have not been shown to have direct, tangible benefits to the Government; and

Whereas Mr. Babbage’s machines do not, in fact, remove the possibility of human error from the production of mathematical, navigational, or astronomical tables; now, therefore, be it

Resolved, that the Royal Society of London

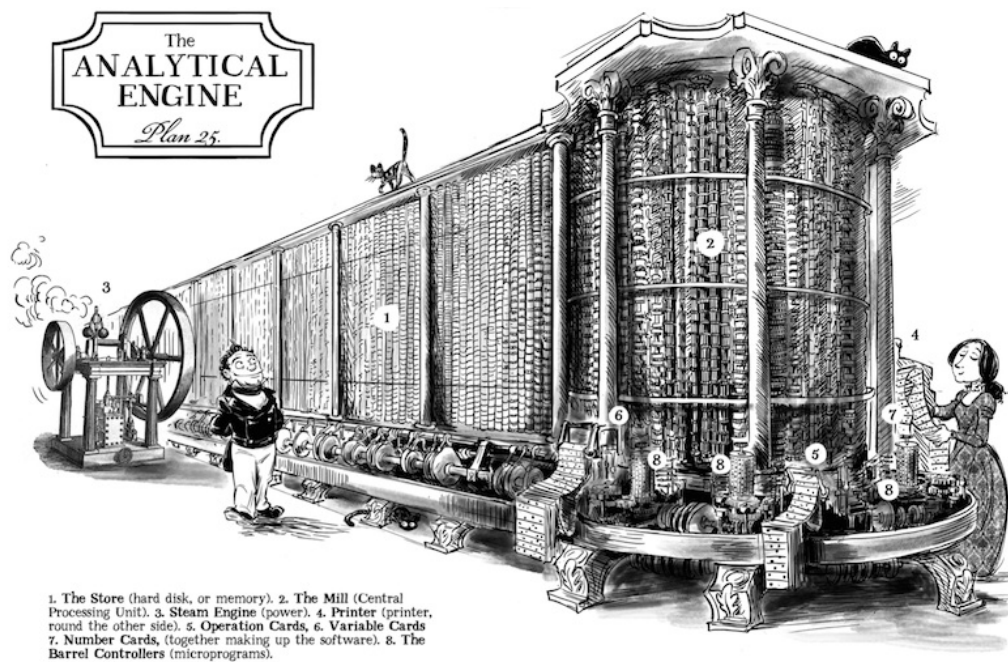
1. believes that Mr. Babbage’s Difference Engine, while it may be useful in some distant future application, at present serves no demonstrable practical use;
2. believes that scientific and engineering undertakings of this scale are not the domain of the Government of the United Kingdom; and
3. therefore, strongly urges His Majesty’s Government to decline to provide further funding to Mr. Babbage for his Difference Engine project.

Lab 3: the Analytical Engine

Faction / name(s): _____

In this lab, you will experiment a bit with programming the Analytical Engine. As in the previous lab, rather than working with a huge, steam-powered brass machine, you will work with a program designed to simulate the Analytical Engine.

The architecture of the Analytical Engine



This image, from the excellent graphic novel by Sydney Padua, *The Thrilling Adventures of Lovelace and Babbage: With Interesting & Curious Anecdotes of Celebrated and Distinguished Characters, Fully Illustrating a Variety of Instructive and Amusing Scenes; as Performed within and without the Remarkable Difference Engine, Embellished with Portraits and Scientifick Diagrams* [Pad15] illustrates the major components of the Analytical Engine. The components that are important to the simulator you will use in this lab are the store, the mill, and the program (operation, variable, and number) cards.

- The store holds numeric data that may be needed later during a computation. It corresponds to the memory of a modern computer. The store can hold 1,000 50-digit numbers, but no number has a decimal point. To work with fractional numbers, the person using the computer would have to keep track of where the decimal point is. Each storage location, called a column by Babbage, is assigned an address, from zero to 999.
- The mill performs operations on numbers, or on the sequence of program cards controlling the operation of the Engine. These operations include addition, subtraction,

multiplication, and division, as well as sequence operations that you will learn about in the next section.

- The program cards instruct the mill to perform operations (operation cards), input numbers into the store (number cards), or direct the transfer of a number from the store to the mill or vice versa (variable cards). Since the Web simulator we use is not capable of reading in Jacquard Loom cards, lines of text will simulate these cards, where each line represents one card.

There are some pieces of the Engine that are not visible in the diagram that will be referenced in the background materials associated with the Web simulator. The arguments for mathematical operations are on the two *ingress axes*, while the results of the operation are placed on the *egress axis*. These are similar to the *registers* that are used inside modern CPUs. There is also a *run-up lever* which is raised in certain circumstances, like when the result of a subtraction has a different sign from the first argument in the calculation (say, $4 - 5 = -1$).

Now, some research . . .

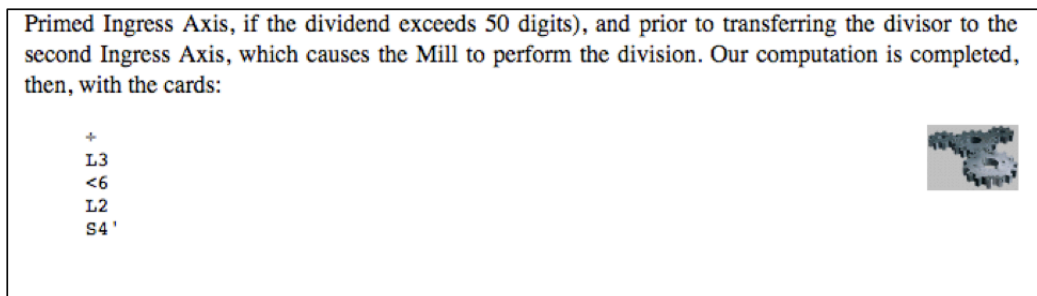
Before you can do some programming with the Analytical Engine simulator, you will need to do some reading about what those programs are like. To begin to understand about Analytical Engine programming, visit this Web site:

<https://www.fourmilab.ch/babbage/cards.html>

Read the *Introduction* and *Program Cards* sections, up through and including the *Stepping Up and Down Cards* subsection.

. . . and some experimentation

When you get to the part about computing $(4000 \times 2.5) \div 28$, you will see a little gear symbol to the right of the last part of the program. This is a button that allows you to run the program on the Analytical Engine simulator. You are looking for something like this:



Right-click on the gears, and choose to open the link in a new tab or new window. This will load the program to compute $(4000 \times 2.5) \div 28$ into the virtual Analytical Engine. There are several areas in the simulator, but the ones we are interested in are shown below:

Program

```

N0 4000000000
N1 2500000
N2 28000000

×
L0
L1
>6
S3

÷
L3
<6
L2
S4'

```

☒ **Keep comments**

No file chosen

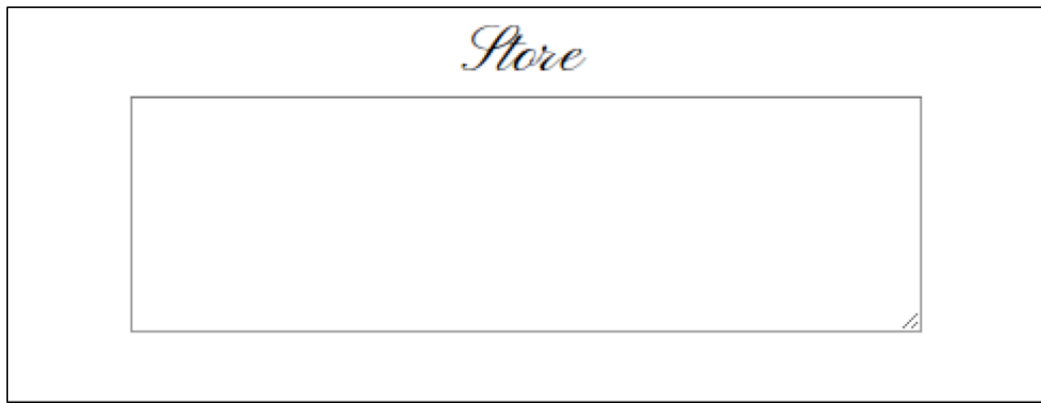
The Program area represents the three card readers that would be used to read in the operation, number and variable cards. In the simulation, all three types of card are grouped here, in the order that they are needed. Each line represents one card.

Attendant's Log

Controls

☒ **Panel**
☒ **Trace**
☒ **Animate**

The Attendant's Log contains a sort of narration of what happens when the program runs. The Controls area is where you start the Analytical Engine working. You should make sure that the Trace box is checked so that you can see as much detail as you can during the run.



The Store area will show the values of the numbers in the store during the computation and after it is complete.

1. Now, run the program to compute $(4000 \times 2.5) \div 28$, by clicking on the Start button in the Controls area. Write down the contents of the store after the computation is complete:
2. There should be five numbers in the store. Which column of the store contains the answer to the computation? How do you know where to place the decimal point in the answer?

Significant events between 1830 (GS 4) and 1846 (GS 5)

His Majesty, King George IV, died on June 26, 1830. He was succeeded by his brother, His Majesty King William IV, who reigned until his death on June 20, 1837. After his death, William IV was succeeded by his niece, Her Majesty Queen Victoria. Only 18 and unmarried when she became queen, Victoria was exceptionally influential. She reigned for the rest of the century, until her death on January 22, 1901. In 1840, Victoria married her first cousin, Prince Albert.

Prince Albert was a very influential consort for Queen Victoria. He supported educational reform in the kingdom, the abolishment of slavery, and a more constitutional monarchy.

Most significant for our game, Prince Albert was also an enthusiastic supporter of industry, engineering, and science. As such, shortly after his marriage to Queen Victoria, he persuaded the sitting Prime Minister, William Lamb, the 2nd Viscount Melbourne, to provide funding for construction of Charles Babbage's Difference Engine.

Babbage used a significant portion of the funds he received to convert his stables into a new workshop adjacent to his house on Dorset Street. To date, Babbage has received approximately £17,000 of government funds for his Difference Engine, and has contributed a significant portion of his own fortune to the project.

But, the Difference Engine remains unfinished.

In other happenings, Mr. Airy has been named a Fellow of the Royal Society. Congratulations, Mr. Airy!

Voting slips

Grey: Vote for most persuasive GS 1 presentation. ☐ The Fury ☐ Frend / Somerville ☐ King ☐ Herschel	Peel: Vote for most persuasive GS 1 presentation. ☐ The Fury ☐ Frend / Somerville ☐ King ☐ Herschel	Robinson: Vote for most persuasive GS 1 presentation. ☐ The Fury ☐ Frend / Somerville ☐ King ☐ Herschel
Russell: Vote for most persuasive GS 1 presentation. ☐ The Fury ☐ Frend / Somerville ☐ King ☐ Herschel	Wellesley: Vote for most persuasive GS 1 presentation. ☐ The Fury ☐ Frend / Somerville ☐ King ☐ Herschel	GM: Vote for best GS 1 presentation ☐ The Fury ☐ Frend / Somerville ☐ King ☐ Herschel
TA: Vote for best GS 1 presentation ☐ The Fury ☐ Frend / Somerville ☐ King ☐ Herschel	Vote for most persuasive GS 1 presentation. ☐ The Fury ☐ Frend / Somerville ☐ King ☐ Herschel	Vote for best GS 1 presentation ☐ The Fury ☐ Frend / Somerville ☐ King ☐ Herschel

Grey: Vote for most persuasive GS 2 presentation. ☐ Sussex ☐ Sabine ☐ Lardner ☐ Swing ☐ Ghost of Byron	Peel: Vote for most persuasive GS 2 presentation. ☐ Sussex ☐ Sabine ☐ Lardner ☐ Swing ☐ Ghost of Byron	Robinson: Vote for most persuasive GS 2 presentation. ☐ Sussex ☐ Sabine ☐ Lardner ☐ Swing ☐ Ghost of Byron
Russell: Vote for most persuasive GS 2 presentation. ☐ Sussex ☐ Sabine ☐ Lardner ☐ Swing ☐ Ghost of Byron	Wellesley: Vote for most persuasive GS 2 presentation. ☐ Sussex ☐ Sabine ☐ Lardner ☐ Swing ☐ Ghost of Byron	GM: Vote for best GS 2 presentation ☐ Sussex ☐ Sabine ☐ Lardner ☐ Swing ☐ Ghost of Byron
TA: Vote for best GS 2 presentation ☐ Sussex ☐ Sabine ☐ Lardner ☐ Swing ☐ Ghost of Byron	Vote for most persuasive GS 2 presentation. ☐ Sussex ☐ Sabine ☐ Lardner ☐ Swing ☐ Ghost of Byron	Vote for best GS 2 presentation ☐ Sussex ☐ Sabine ☐ Lardner ☐ Swing ☐ Ghost of Byron

Grey: Vote for most persuasive GS 3 presentation. ___ Faraday ___ Buckland	Peel: Vote for most persuasive GS 3 presentation. ___ Faraday ___ Buckland	Robinson: Vote for most persuasive GS 3 presentation. ___ Faraday ___ Buckland
Russell: Vote for most persuasive GS 3 presentation. ___ Faraday ___ Buckland	Wellesley: Vote for most persuasive GS 3 presentation. ___ Faraday ___ Buckland	GM: Vote for best GS 3 presentation ___ Faraday ___ Buckland
TA: Vote for best GS 3 presentation ___ Faraday ___ Buckland	Vote for most persuasive GS 3 presentation. ___ Faraday ___ Buckland	Vote for best GS 3 presentation ___ Faraday ___ Buckland

Grey: Vote for most persuasive GS 4 presentation. ___ Gilbert ___ Babbage	Peel: Vote for most persuasive GS 4 presentation. ___ Gilbert ___ Babbage	Robinson: Vote for most persuasive GS 4 presentation. ___ Gilbert ___ Babbage
Russell: Vote for most persuasive GS 4 presentation. ___ Gilbert ___ Babbage	Wellesley: Vote for most persuasive GS 4 presentation. ___ Gilbert ___ Babbage	GM: Vote for best GS 4 presentation ___ Gilbert ___ Babbage
TA: Vote for best GS 4 presentation ___ Gilbert ___ Babbage	Vote for most persuasive GS 4 presentation. ___ Gilbert ___ Babbage	Vote for best GS 4 presentation ___ Gilbert ___ Babbage

Grey: Vote for most persuasive GS 5 presentation. ☐ Clement ☐ Brunel ☐ Lovelace ☐ Crosse	Peel: Vote for most persuasive GS 5 presentation. ☐ Clement ☐ Brunel ☐ Lovelace ☐ Crosse	Robinson: Vote for most persuasive GS 5 presentation. ☐ Clement ☐ Brunel ☐ Lovelace ☐ Crosse
Russell: Vote for most persuasive GS 5 presentation. ☐ Clement ☐ Brunel ☐ Lovelace ☐ Crosse	Wellesley: Vote for most persuasive GS 5 presentation. ☐ Clement ☐ Brunel ☐ Lovelace ☐ Crosse	GM: Vote for best GS 5 presentation ☐ Clement ☐ Brunel ☐ Lovelace ☐ Crosse
TA: Vote for best GS 5 presentation ☐ Clement ☐ Brunel ☐ Lovelace ☐ Crosse	Vote for most persuasive GS 5 presentation. ☐ Clement ☐ Brunel ☐ Lovelace ☐ Crosse	Vote for best GS 5 presentation ☐ Clement ☐ Brunel ☐ Lovelace ☐ Crosse

Grey: Vote for most persuasive GS 6 presentation. ☐ Adams ☐ Airy	Peel: Vote for most persuasive GS 6 presentation. ☐ Adams ☐ Airy	Robinson: Vote for most persuasive GS 6 presentation. ☐ Adams ☐ Airy
Russell: Vote for most persuasive GS 6 presentation. ☐ Adams ☐ Airy	Wellesley: Vote for most persuasive GS 6 presentation. ☐ Adams ☐ Airy	GM: Vote for best GS 6 presentation ☐ Adams ☐ Airy
TA: Vote for best GS 6 presentation ☐ Adams ☐ Airy	Vote for most persuasive GS 6 presentation. ☐ Adams ☐ Airy	Vote for best GS 6 presentation ☐ Adams ☐ Airy

Adams: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Airy: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Babbage: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley
Brunel: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Buckland: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Crosse: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley
Faraday: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Frend: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Gilbert: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley

Grey: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Herschel: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Peel: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley
Robinson: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Russell: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Sabine: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley
Wellesley: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	King: Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley	Vote for the best PM speech ☐ Grey ☐ Peel ☐ Robinson ☐ Russell ☐ Wellesley

Which should be preferred: ☐ Imagination ☐ Reason	Which should be preferred: ☐ Imagination ☐ Reason	Which should be preferred: ☐ Imagination ☐ Reason
Which should be preferred: ☐ Imagination ☐ Reason	Which should be preferred: ☐ Imagination ☐ Reason	Which should be preferred: ☐ Imagination ☐ Reason
Which should be preferred: ☐ Imagination ☐ Reason	Which should be preferred: ☐ Imagination ☐ Reason	Which should be preferred: ☐ Imagination ☐ Reason
Which should be preferred: ☐ Imagination ☐ Reason	Which should be preferred: ☐ Imagination ☐ Reason	Which should be preferred: ☐ Imagination ☐ Reason
Which should be preferred: ☐ Imagination ☐ Reason	Which should be preferred: ☐ Imagination ☐ Reason	Which should be preferred: ☐ Imagination ☐ Reason

Make enough copies of this sheet for each character in the game to get a vote.